

Stanford Education Data Archive

Technical Documentation

Version 2025.1

May 2026

Erin M. Fahle, Stanford University

Jim Saliba, Stanford University

Jie Min, Stanford University

Sean F. Reardon, Stanford University

Benjamin R. Shear, University of Colorado Boulder

Jiyeon Shim, Stanford University

Demetra Kalogrides, Stanford University

Andrew D. Ho, Harvard University

Suggested citation:

Fahle, E. M., Saliba, J., Min, J., Reardon, S. F., Shear, B. R., Shim, J., Kalogrides, D., & Ho, A. D. (2026). Stanford Education Data Archive Technical Documentation: SEDA 2025.1. Retrieved from <https://purl.stanford.edu/hm970gr1371>.

Contents

What is SEDA?	3
I.A. Overview of Test Score Data Files	4
I.B. Covariate Data	5
I.C. Data Use Agreement	6
II. Achievement Data Construction	8
II.A. Source Data	8
II.B. Definitions	10
II.C. Construction Overview	11
II.D. Notation	13
II.E. Detailed Construction Overview	14
Step 1. Creating the Crosswalk	14
Step 2. Data Cleaning	18
Step 3. Cutscore Estimation	24
Step 4. Selecting Data for Mean Estimation	27
Step 5. Estimating Test Score Means	30
Step 6. Linking Mean Estimates	34
Step 7. Creating Reporting Scales	35
Step 8. Annual Estimates	38
Step 9. Pooled Estimates	44
Step 10. Aggregating Cross-State Unit Components	51
Step 11. Calculating Achievement Gaps	51
Step 12. Suppressing and Flagging Data for Release	52
II.F. Additional Notes	54
II.G. How to Obtain Values Shown on edopportunity.org	57
III. Covariate Data Construction	57
III.A. ACS Data and SES Composite Construction	57
III.B. Common Core of Data Imputation	60
IV. Website-Specific Variables	61
V. Versioning and Publication	62
References	63
Tables	65
Figures	99
Appendices	100
Appendix A: Additional Detail on Statistical Methods	100
1. Estimating Means and Standard Deviations for Units that Cross State Lines	100
2. NAEP 2024 Adjustment and Estimated NAEP 2025 Values	102
Appendix B: Covariates	108
1. List of Raw ACS Tables Used for SES Composite	108
2. Measurement Error, Attenuation Bias and Solutions	111
3. Computing the Sampling Variance of Sums of ACS Variables	113
4. Estimating Sampling Variance of Composite SES Measures	119

What is SEDA?

The Stanford Education Data Archive (SEDA) is created by the Educational Opportunity Project (EOP) at Stanford University (<https://edopportunity.org>). The EOP harnesses data to help scholars, policymakers, educators, and parents learn how to improve educational opportunities. SEDA includes detailed data on educational conditions, contexts, and outcomes in schools, school districts, counties, commuting zones, metropolitan statistical areas, and states across the United States. Available measures differ by aggregation; see Sections I.A. and I.B. for a complete list of files and data.

By making the data files available to the public, we hope that anyone who is interested can obtain detailed information about educational opportunity in U.S. schools and communities, and that researchers will use these data to generate evidence about how educational policies and contexts can increase educational opportunity.

SEDA 2025.1 includes trend data for the 2008-09 through 2024-25 school years to the Stanford Education Data Archive for schools, administrative school districts, and states with sufficient publicly available data.

Acknowledgements. The construction of SEDA has been supported by grants from the Gates Foundation, the Institute of Education Sciences, the Spencer Foundation, the William T. Grant Foundation, the Overdeck Family Foundation, and by a visiting scholar fellowship from the Russell Sage Foundation. Some of the data used in constructing the SEDA files were provided by the National Center for Education Statistics (NCES), the Education Data Center (EDC), and the National Assessment Governing Board. The findings and opinions expressed herein are those of the authors alone; they do not represent the views of the U.S. Department of Education, NCES, EDC, or any of the funding agencies.

We would like to thank: Emily Oster, Clare Halloran, and the EDC team for providing state assessment data files; Sadie Richardson, Sofia Wilson, Julia Paris, Ishita Panda, Amelia Bloom, Nahian Haque, Jackson Kinsella, and Aditi Oak for their assistance in producing SEDA.

I.A. Overview of Test Score Data Files

SEDA 2025.1 contains test score data files for schools, administrative school districts, and states. Test score data files contain information about the average academic achievement as measured by standardized test scores administered in 3rd through 8th grade in mathematics and Reading/English Language Arts (RLA) over the 2008-09 through 2018-19 and 2021-22 through 2024-25 school years. The measures contained in the data files are detailed below. For a list of key changes from SEDA 2024.3 see [Section V. Versioning and Publication](#).

Administrative School District and State Files. Forty test score files correspond to the two units (administrative school districts and states) by four scales (the cohort standardized scale, CS; the grade cohort standardized scale, GCS; the year standardized scale, YS; and the grade year standardized scale, GYS) by five pooling levels (long, annual by subject, annual, pooled by subject, and pooled). “Long” files contain estimates for each grade and year separately; “annual by subject” (or annualsub) files contain estimates for each year pooled over grades within subject; “annual” files contain estimates for each year pooled over grades and subjects; “pooled by subject” (or poolsub) files contain estimates that are averaged across grades and years within subjects; and “pooled” (or pool) files contain estimates that are averaged across grades, years, and subjects.

In the long files there are variables corresponding to test score means by subgroup and their respective standard errors in each grade, year, and subject. In the annualsub files, there are variables corresponding to the average test score mean by year in math and in RLA (averaged over grades). In the annual files, there are variables corresponding to the average test score mean by year (averaged over grades and subjects). In the poolsub files, there are variables corresponding to the average test score mean in math and in RLA (averaged across grades and years), the average “learning rate” across grades in math and in RLA, the average “trend” in test scores across cohorts in math and in RLA, and the average trend in learning rates in math and in RLA, along with their standard errors. In the pooled files, there are variables corresponding to the average test score mean (averaged across grades, years, and subjects), the average learning rate across grades, the trend in average test scores across cohorts, the average trend in learning

rates, and the average difference between math and RLA test scores, along with their standard errors. Estimates are reported for all students and by demographic subgroups.

School Files. We release eight school-level test score data files. These include two types of files described above—the “pooled” (or pool) files and the “pooled by subject” (or poolsub) files—in each of the four scales: CS, GCS, YS, and GYS. The file contents are as defined above for other geographic units, except that school estimates are only reported for all students; no estimates are provided by demographic subgroup.

The [Achievement Data Construction](#) section of the documentation provides more detail about the construction of the test score estimates. [Table 1](#) lists the files and file structures. Lists of variables can be found in the codebooks that accompany this documentation.

I.B. Covariate Data

SEDA 2025.1 also includes estimates of socioeconomic, demographic, and segregation¹ characteristics of schools, administrative districts, and states. The measures we report come from two primary sources:

- The Common Core of Data (CCD). The CCD is an annual survey of all public elementary and secondary schools and school districts in the United States. The CCD data include basic descriptive information on schools and school districts, including demographic characteristics.² We aggregate school-level CCD data to larger units (e.g., geographic districts, counties).³
- The American Community Survey (ACS). We obtain detailed tables from the National Historical Geographic Information System (NHGIS) web portal,⁴ which include data on the demographic and socioeconomic characteristics of individuals and households residing in each unit. The measures from the ACS are not available at the school level.

¹ For more detail on the segregation measures, see the Segregation Tracking Project at <https://edopportunity.org/segregation/>.

² The CCD is available for download from the NCES website: <https://nces.ed.gov/ccd/>.

³ The exception is the finance data (total instructional expenditures and per pupil expenditures) which are only available at the district level. These measures are aggregated from the district level to the county and metropolitan area levels and are not available at the school level.

⁴ The ACS data is available for download from the IPUMS-NHGIS website at: <https://www.nhgis.org/>. Full citation: Manson, S., Schroeder, J., Van Riper, D., & Ruggles, S. (2019). *IPUMS National Historical Geographic Information System: Version 14.0* [Data set]. IPUMS. <https://doi.org/10.18128/D050.V14.0>.

Administrative School District and State Files. We release six covariate data files for these units (three per unit). The three covariate files we construct for each unit contain the same variables but at different aggregations: separately for each grade and year, averaged across grades (providing a single value per unit per year), or averaged across grades and years (providing a single value per unit).

School Files. We provide one covariate data file for schools, which reports a single record for each school that is the average across years.

The [Covariate Data Construction](#) section of the documentation provides more detail about the construction of these data files and the computation of derived variables. [Table 2](#) lists the names and file structures of the covariate data files. Lists of variables can be found in the codebooks that accompanies this documentation.

I.C. Data Use Agreement

Prior to downloading the data, users must sign the data use agreement, shown below.

You agree not to use the data sets for commercial advantage, or in the course of for-profit activities. Commercial entities wishing to use this Service should contact Stanford University's Office of Technology Licensing (info@otlmail.stanford.edu).

You agree that you will not use these data to identify or to otherwise infringe the privacy or confidentiality rights of individuals.

You agree not to publish the data files, in full or in part, without explicit permission of the Educational Opportunity Project at Stanford.

THE DATA SETS ARE PROVIDED "AS IS" AND STANFORD MAKES NO REPRESENTATIONS AND EXTENDS NO WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED. STANFORD SHALL NOT BE LIABLE FOR ANY CLAIMS OR DAMAGES WITH RESPECT TO ANY LOSS OR OTHER CLAIM BY YOU OR ANY THIRD PARTY ON ACCOUNT OF, OR ARISING FROM THE USE OF THE DATA SETS.

You agree that this Agreement and any dispute arising under it is governed by the laws of the State of California of the United States of America, applicable to agreements negotiated, executed, and performed within California.

You agree to acknowledge the Stanford Education Data Archive as the source of these data. In publications, please cite the data as:

Reardon, S. F., Fahle, E. M., Ho, A. D., Shear, B. R., saliba, j., Min, J., Shim, J., & Kalogrides, D.
(2026). Stanford Education Data Archive (Version SEDA 2025.1). Retrieved from:
<https://purl.stanford.edu/hm970qr1371>.

Subject to your compliance with the terms and conditions set forth in this Agreement,
Stanford grants you a revocable, non-exclusive, non-transferable right to access and make use of
the Data Sets.

II. Achievement Data Construction

II.A. Source Data

SEDA 2025.1 test scores estimates are constructed using data from: (a) the *EDFacts* data system, (b) the Education Data Center (EDC), and (c) the National Assessment of Educational Progress (NAEP).

EDFacts. The *EDFacts* data system, housed by the U.S. Department of Education, collects aggregated test score data from each state’s standardized testing program as required by federal law. Specifically, each state is required to test every student in grades 3 through 8 in math and Reading Language Arts (RLA) each year.⁵ States have the flexibility to select or design a test of their choice that measures student achievement relative to the state’s standards. Additionally, states set their own benchmarks or thresholds for the levels of performance, e.g., “proficient,” in each grade and subject. Typically, states select 2 to 5 performance levels, where one or more levels represent “proficient” grade-level achievement or above.

The *EDFacts* data system includes state-reported numbers of students in each school, subgroup, subject, grade, and year scoring at each of the state-defined performance levels. No individual student-level data is reported. SEDA uses these school assessment outcomes for eleven consecutive school years from the 2008-09 school year to the 2018-19 school year in grades 3 to 8 in RLA and math.⁶ Student subgroup scores in SEDA include disaggregations by race/ethnicity, gender, and socioeconomic disadvantage. Other subgroup data in *EDFacts* is not used in SEDA.

The raw *EDFacts* data used to construct SEDA include no suppressed cells, nor do they have a minimum cell size for reporting. Each row of data corresponds to a school-subgroup-

⁵ Federal law also requires states to report data for one high school grade; however, that data is not currently used in SEDA.

⁶ Data are also available for the 2020-21 and 2021-22; however, they are not reported in sufficient detail for our use. Data for 2022-23 and 2023-24 are not yet available. From 2013-2019, the data is further broken out by the assessment type administered to students: regular assessments, regular assessments with accommodations, and alternate assessments with grade-level standards, modified standards and alternate standards. However, from 2009-2012, *EDFacts* does not distinguish students taking regular from alternate assessments. Therefore, for consistency in all years, we use all performance data reported in *EDFacts*, including results of students taking both regular and alternate assessments.

subject-grade-year cell. [Table 3](#) illustrates the structure of the raw data from *EDFacts* prior to use in constructing SEDA.

EDC. We use state-reported data for 2021-22 through 2024-25 from EDC's [State Assessment Data Repository \(SADR\)](#) Version 3.0 (Education Data Center, 2025). SADR includes publicly-available assessment data from all 50 states and D.C. for students in grades 3-8, reported at the state, administrative school district, and school levels, and disaggregated by subject, grade level and student subgroups. Because these are public data, data for small places or small subgroups are often suppressed to maintain student privacy. There are also variations in data that states report (e.g., some states may not report tested counts of students or data for subgroups). We have taken measures to clean these data, so that we retain the maximum number of states, districts, and schools in our sample. Our data cleaning process is described in II.E.

NAEP. NAEP is a congressionally mandated biennial assessment administered by the National Center for Education Statistics (NCES). NAEP reports national, state, and select district means and standard deviations, as well as their standard errors, in 4th and 8th grade math and reading.⁷ Data are available from the NAEP Data Explorer.⁸ To account for NAEP initiatives to expand and standardize inclusion of English learners and students with disabilities over this time period, we rely on the Expanded Population Estimates (EPE) of means and standard deviations provided by the National Center of Education Statistics in all years in which they are available.⁹ Because we did not have EPEs for 2024 at the time of this release, we created adjustments based on historical NAEP data to apply to the 2024 regular NAEP means and SDs; we use these 'adjusted' NAEP values in SEDA. The construction of these adjustments is described in **Appendix A2.1, NAEP 2024 Adjustment**. Additionally, we need to estimate NAEP state means and SDs in untested years (e.g., 2010, 2012, 2014, 2016, 2018, 2023, and 2025) and grades (3, 5, 6, and 7). In most untested years and grades, we use interpolation to get NAEP estimates. This is described

⁷ The data described here only reflect a subset of the NAEP data. For additional details on NAEP, please see: <https://nces.ed.gov/nationsreportcard>.

⁸ <https://nces.ed.gov/nationsreportcard/data/>

⁹ Where unavailable, we use the NAEP regular estimates and apply an adjustment to produce an EPE-like NAEP estimate. The correlation of Expanded Population Estimates and regular NAEP estimates are near unity. Generally, substantive conclusions do not depend upon whether EPE or regular NAEP estimates are used.

in **Section 5A. State Estimates for the All Student Group**. For 2025, we produce NAEP estimates using a different process that is described in **Appendix A2.2 Estimated NAEP 2025 Values**.

II.B. Definitions

Administrative School District: Administrative school districts operate sets of public and charter schools. The schools operated by each school district are identified using the NCES school and district identifiers. Most commonly, administrative school districts operate local public schools within a given physical boundary; these are what we refer to as “traditional public school districts.” There are also specialized administrative districts that do not have a physical boundary, like charter school, virtual school, and state agency school districts.

Geographic School District: The aggregate of all public schools in SEDA (except for special education¹⁰, virtual¹¹, and other non-geographic¹² schools) that are physically located within a geographically defined public Elementary or Unified school district. We use the 2019 Elementary and Unified School District Boundaries

(<https://nces.ed.gov/programs/edge/Geographic/DistrictBoundaries>) to define these districts. In this document, the terms “geographic district,” “geographic school district,” and “geographically-defined school district” are used interchangeably.

Group: The term “group” refers to a unit-subgroup. For schools, subgroup sample sizes are too small to support reliable subgroup estimates, so the only available subgroup is “all students.” For administrative school districts and states, data for other subgroups are available when estimates are sufficiently precise.

State: States are identified by their FIPS state code. We include all 50 states plus Washington, DC.

Subcategory: The term “subcategory” refers to the subcategory to which a subgroup belongs. In addition to data for all students, we have data for the following subcategories: gender, race, and

¹⁰ As defined by school type in the CCD Public Elementary/Secondary School Universe Survey Data.

¹¹ As defined by virtual text in the CCD Public Elementary/Secondary School Universe Survey Data.

¹² Other non-geographic schools are those that are not identified as charter or magnet schools CCD Public Elementary/Secondary School Universe Survey Data and whose administrative district is absent from the 2019 Elementary and Unified School District Boundaries.

economic status. The gender subcategory contains two subgroups, male and female. The race subcategory includes the Asian, Black, Hispanic, Native American, and White subgroups. The economic status subcategory includes the economically disadvantaged (ECD) and not economically disadvantaged (nonECD) subgroups.

Subgroup: The term “subgroup” refers to the group of students to which an estimate pertains. Subgroups include all, Asian, Black, Hispanic, Native American, White, female, male, economically disadvantaged (ECD), and not economically disadvantaged students (nonECD).

Unit: The term “unit” refers to the aggregation or the geographic level of the data. This may be a school, administrative school district, or state.

II.C. Construction Overview

The construction process produces mean test score estimates for schools, geographic districts, administrative districts, counties, metropolitan areas, commuting zones, and states on four nationally comparable scales in a series of twelve steps, outlined in [Figure 1](#). We provide a brief conceptual description of each step here and the full technical details about each step in [Section II.E](#).

Step 1: Creating the Crosswalk. This step links each public school to a unique stable school, geographic school district, administrative school district, county, commuting zone, metropolitan area, and state, and links each administrative school district to its district id in the most recent year.

Step 2: Data Cleaning. This step removes data not used in SEDA.

Step 3: Estimating Cutscores. This step uses Heteroskedastic Ordered Probit (HETOP) models or the inverse cumulative standard normal distribution function to estimate the state-grade-subject-year cutscores.

Step 4: Selecting Data for Mean Estimation. This step selects data for *unit-subgroup-subject-grade-year* cases that will be used in estimation. We exclude cases with low participation in the assessment or high percentages of students taking alternate

assessments. We also exclude cases for which we have insufficient data to produce an estimate.

Step 5: Estimating Means. This step first uses the NAEP data to estimate state-level means and SDs for all grades and years. We then use the pooled HETOP model to estimate unit-subgroup-subject-grade-year means and standard deviations, along with their standard errors, based on the cutscores from Step 3 and the data prepared in Step 4.

Step 6: Linking Means. This step uses the state NAEP data to put the estimates from Step 5 onto a common scale.

Step 7: Creating Additional Reporting Scales. This step creates Cohort Standardized (CS), Grade Cohort Standardized (GCS), Year Standardized (YS), and Grade Year Standardized (GYS) estimates. From this point onward, we have four scales of data for all units. Subsequent steps are equivalent for all scales unless otherwise noted.

Step 8: Creating Annual Estimates. This step estimates the average achievement by subject and overall for each unit and scale by year.

Step 9: Creating Pooled Estimates. This step estimates the average achievement, learning rate, trend in test scores, and trend in learning rates by subject and overall for each unit and scale across all years.

At the end of Step 9, we have five types of data for administrative districts and states: long (not pooled by grade, year, or subject), annual by subject (annualsub; pooled over grades by years and subjects), annual (pooled over grades and subjects by year), pooled by subject (poolsub; pooled over grades and years by subject), and pooled overall (pool; pooled over grades, years, and subjects). For schools, we only report the pooled overall (pool) and pooled by subject (poolsub) estimates.

Step 10: Aggregating Unit Components. In this step, we aggregate components of units that took different state tests. Examples of these are units that contain BIE schools that take a different state test and Metropolitan Statistical Areas that cross state lines.

Step 11: Calculating Achievement Gaps. We do not calculate achievement gaps for SEDA 2025.1.

Step 12: Preparing Data for Release. For long, annual, and pooled files for all units and scales, this step suppresses estimates that are too imprecise to be useful or do not reflect the performance of at least 20 unique students. For 2009-2019 estimates reported in the long files, this step adds a small amount of random noise to meet the reporting requirements of the US Department of Education.

II.D. Notation

In the remainder of the documentation, we use the following mathematical notation:

- Mean estimates are denoted by $\hat{\mu}$ and standard deviation estimates by $\hat{\sigma}$.
- The cutscore estimates are denoted as $\hat{c}_1, \dots, \hat{c}_{K-1}$. There are $K - 1$ total cutscores in each state-subject-grade-year.
- A subscript indicates the aggregation of the estimate. We use the following subscripts:

u = unit (generic)
 n = school
 d = geographic school district
 a = administrative school district
 c = county
 z = commuting zone
 m = metropolitan area
 s = state
 N = national

r = subgroup

all = all students
 asn = Asian
 blk = Black
 hsp = Hispanic
 nam = Native American
 wht = White
 fem = female
 mal = male
 ecd = economically disadvantaged
 nec = not economically disadvantaged
 wag = White-Asian gap
 wbg = White-Black gap
 whg = White-Hispanic gap

wng = White-Native American gap

mfg = male-female gap

neg =not economically disadvantaged-economically disadvantaged gap

y = year

b = subject

g = grade

- A superscript indicates the scale of the estimate. The metric is generically designated as *x*. There are six scales. The first two scales are only used in construction. The latter four scales are reported:

state = state-standardized metric

naep = NAEP test score scale metric

cs = cohort scale metric

gcs = grade within cohort scale metric

ys = year scale metric

gys = grade within year scale metric

II.E. Detailed Construction Overview

Step 1. Creating the Crosswalk

The primary purpose of the crosswalk is to create stable school identifiers and assign schools to larger units such as geographic school districts, administrative school districts, and states. We use the CCD's *Public Elementary/Secondary School Universe Survey Data* (Directory and School Characteristics files) and the *Local Education Agency (School District) Universe Survey Data* (Directory files) as the basis for the crosswalk.

Stable School IDs. Since we want to be able to track schools as they change districts (district changes could be due to districts splitting, merging or some other administrative changes), we create stable school IDs using the CCD's *Longitudinal ID Crosswalks*. According to the CCD documentation, "Schools are uniquely identified in CCD by the 12-digit variable **ncessch**. This variable is a combination of the state code (the first two digits or **FIPST**), the Local Education Agency (LEA) ID (the first seven digits or **leaid**) and the last five digits (**schid**). It was always intended that the **schid** should be unique within the state so that a school could be tracked from year-to-year even if a re-organization caused it to change LEAs. However, a system error created some duplicate **schids** within some states."¹³

¹³ See Page 1 in the NCES *School Crosswalk (SY 2014-15 to 2015-16)*. Retrieved from: https://nces.ed.gov/ccd/doc/3_Changes_to_NCES_School_ID_2015_16.docx. Bolding added for emphasis.

Because some schools changed school IDs during the 2008-09 to 2018-19 time period, we use the CCD's longitudinal ID crosswalks¹⁴ from the CCD's Reference Library (https://nces.ed.gov/ccd/reference_library.asp) to uniquely identify schools. These stable school IDs became the last 5 digits of the **sedasch** IDs. The final **sedasch** ID is comprised of 12 digits in the same format as the NCES school ID (ncessch). The **sedasch** ID's first 2 digits correspond to the state FIPS code, first 7 digits correspond to a stable administrative district ID (**sedaadmin**), and the last 5 digits correspond to the school ID within the state.

Stable Administrative District IDs. In order to track administrative districts over time, including accounting for district mergers, we align administrative districts in all years with their administrative district ID in the most recent year available, as much as possible.

School Assignment to Administrative Districts. With the following exceptions, we use the NCES district ID (leaid) provided in the CCD in the most recent year the school was observed as the administrative district ID (sedaadmin). Exceptions to this rule:

1. Schools operated by supervisory unions: We place all schools in Vermont and New York that are part of supervisory unions in their supervisory union LEAs. For example, New York City School District (LEA 3620580) is a supervisory union comprised of 33 subordinate school districts.
2. Non-charter schools with district ID's that no longer exist in the CCD in 2025 (extinct district ID's): For all extinct administrative district IDs, we first identify the schools that used to belong to the extinct district ID. If any of those schools belong to a district ID that exists in the CCD in 2025, we use that ID as the district ID for all schools that belonged to the extinct district ID. This ensures that most schools that closed prior to their district merging into or with another district are consolidated into the surviving or new administrative district ID. If none of the schools that belonged to an extinct district ID can be linked to a district in the 2024 CCD data, we use news reports or other historical sources to identify what happened to the extinct district and assign each of its schools a current administrative district ID.

¹⁴ School Crosswalk (SY 2014-15 to 2015-16), School Crosswalk (SY 2015-16 to 2016-17)

3. Charter and recovery district schools geographically located in Orleans Parish: We use the Orleans Parish geographic district ID as the administrative ID to more closely align with public state reporting.

You can identify a given school's administrative district by looking at the first 7 digits of the **sedasch** ID, which will be the **sedaadmin** ID.

Assignment of Schools to Geographic Districts. Most public school districts in the U.S. are geographically defined.¹⁵ In SEDA we use the 2019 EDGE Unified and Elementary School District Boundaries to define geographic school districts. Commonly, public school districts have administrative control over the traditional public schools that fall within their specific geographic boundaries. However, there may be some schools physically located within the geographic boundary of a school district that are not under its administrative control. For example, there may be charter schools located within the boundaries of a school district that are operated by a different school district or a charter school network (which may have no geographic boundary).

In SEDA we have several rules around what schools are placed or excluded from geographic school districts based on location (latitude and longitude coordinates), school type information, and school status information. Our goal is for geographic district test score estimates in SEDA to reflect most of the public school students living within the geographic boundaries of the school district. One motivation for this assignment is to better align the average test score estimates with the demographic and socioeconomic data from ACS, which are reported for all families living within geographic school district boundaries.

We use a school's most recently observed CCD information on school operating status, charter status, coordinates, and county ID to create time-invariant information for schools in SEDA. Below are the geographic district assignment rules in SEDA based on these time-invariant characteristics:

¹⁵ According to NCES, "The US has more than 13,000 geographically defined school districts. These include districts that are administratively and fiscally independent of any other government, as well as public school systems that lack sufficient autonomy to be counted as separate governments and are classified as a dependent agency of some other government—a county, municipal, township, or state. Most public school systems are Unified districts that operate regular, special, and/or vocational programs for children in Prekindergarten through 12th grade." Retrieved from: <https://nces.ed.gov/programs/edge/Geographic/DistrictBoundaries>

Charter schools: All (except for virtual or special education) charter schools are geolocated and reassigned to the Elementary or Unified District in which they physically reside.

Schools operated by secondary districts: All schools with NCES LEAIDs corresponding to secondary school districts in the Secondary School District Boundary file are geolocated to the Elementary or Unified geographic district in which they physically reside.

Virtual schools: By their nature, many virtual schools do not draw students from within geographic district boundaries. We identify schools as virtual using CCD data from 2013-14 through 2024-25 Public Elementary/Secondary School Universe Survey Data. The virtual school identifier did not exist in earlier years of data, so we identify virtual schools based on their names. We flag schools as virtual in all years if they were identified as virtual in the last year in which they were observed in the data.¹⁶ We exclude virtual schools from geographic districts.

Special education schools: We classify schools as special education schools if they are ever classified as “special education” in the school-type variable in the CCD data between 2009 and 2025.¹⁷ We exclude these schools from their geographic districts. We do not report test scores for schools with high proportions of students taking alternative tests because their school-average test scores are not comparable to those in other schools. This restriction affects many special education schools.

BIE controlled schools: Schools controlled by the Bureau of Indian Education (BIE) are placed in the Elementary or Unified District in which they are physically located.

Schools operated by supervisory unions: Except for special education and virtual schools, we place all schools in Vermont and New York that are part of supervisory unions in their supervisory union LEAs. For example, New York City School District (LEAID 3620580) is a supervisory union comprised of 33 subordinate school districts.

District of Columbia schools: All schools within Washington, DC are given DC’s geographic district ID (LEAID 1100030).

¹⁶ In 2013-2015, we identified 12 non-virtual schools in Alabama identified as “virtual” by the CCD indicator. We treat these as regular schools in all subsequent steps.

¹⁷ Special education is defined by School Type in CCD Public Elementary/Secondary School Universe Survey Data.

Hawaii schools: All schools within Hawaii are given Hawaii’s geographic district ID (LEAID 1500030), because Hawaii operates as a single statewide school district.

Other non-geographic schools: We classify schools as non-geographic if they are not administered by a geographic elementary or unified school district, not charter schools, not BIE schools, not operated by a geographic secondary school district, and not in DC or Hawaii. We exclude these schools from their geographic districts, counties, commuting zones, and metropolitan areas. These schools include state agency and regional schools that likely draw students from multiple geographic districts.

Geographic school districts are identifiable by their **sedalea**. In SEDA 2025.1, we do not report estimates for geographic school districts; we do use geographic school districts in the process of estimating cutscores for 2009-2019.

Use of Crosswalk for 2021-22 through 2024-25 School Years. We do not need to assign schools to administrative districts after the 2018-19 school year as we use proficiency data reported separately at the school, administrative district, and state levels. However, we do use the crosswalk to align administrative district IDs with their ID in the most recent year available.

Merge with ED*Facts* and EDC Data. After creating the crosswalk, we merge the ED*Facts* and EDC data (described under II.A. **Source Data**) with the crosswalk developed in **Step 1**, which matches the data to the stable unit IDs (**sedasch**, **sedaadmin**, **sedalea**, and **sedafips**) used throughout the SEDA process. For ED*Facts*, we merge by NCES school ID (ncessch) and year; for EDC we merge by NCES school ID and year for schools and by NCES district ID (leaid) and year for administrative school districts.

Step 2. Data Cleaning

First, we preprocess the EDC data to allow it to be used in the SEDA process. Then we implement state, district, and school level exclusions for both EDC and ED*Facts* data.

EDC Data Preprocessing. The state-reported data the EDC SADR provides follow three common reporting patterns:

- a. Number of students scoring in each proficiency category. These are the necessary student-level data and were used “as is.”

- b. Total number of students tested and the percent scoring in each proficiency category. From these data, we derived the counts in each category (with some rounding error).
- c. Only the percent scoring in each proficiency category. We estimate the total number of test takers (in each school district, subgroup, subject, grade, and year) using the Common Core of Data (CCD).¹⁸ We then derive an estimated count scoring in each category. In the downloadable data files, we flag all cells where we estimated counts.

There are a handful of additional data issues that are common in the state-reported data. These include:

Partial suppression. States used different suppression rules to protect student privacy. Some states suppress entire rows of data that do not meet reporting thresholds typically based on sample size, while others use partial suppression (suppressing some cells of data within a row). The extent and type of suppression affected our methodology for estimating district average scores and sometimes prohibited our ability to produce estimates for individual districts. For partially suppressed rows, we first identify whether we know both the number of students tested and the number of students considered to be “proficient or above.” There are four common cases:

- a. The number of students tested and the number of students “proficient or above” are known.
- b. The number of students tested is known, but the number of students “proficient or above” is missing.
- c. The number of students tested is missing but the number of students “proficient or above” is known.
- d. Both the number of students tested and the number of students “proficient or above” are missing.

¹⁸ We use the number of enrolled students in the district-subgroup-subject-grade from the CCD in the fall of the school year as the starting point to estimate counts when missing. For example, we use 2023 CCD data to estimate the tested count in Spring 2024. If the number of enrolled students is missing for a given district-subgroup-grade, then we use imputed values. These numbers serve as the starting point; we then estimate a sample size that is consistent with the reported proficiency category percentages and the original count estimate from CCD.

In cases a, b, c, and we can fill in some or all of the missing data in the row. In case d, no steps to fill in the data can be taken. After filling in the missing data, we drop unit-subject-year-grades that match either of these conditions: (1) when the reported student counts do not yield percentages, within acceptable rounding tolerance, that match the reported percentages across all proficiency categories; or (2) when the percentages sum to less than 98% or greater than 102%.

Inconsistent number of students tested across subgroups. When the combined number of students tested across subgroup categories (race, economic status, and gender) exceeds the total number of students tested for a given school or school district-subject-grade-year, we exclude the subgroup data within that category.

Missing data for one gender and economic subgroup. Not all states reported data for all subgroups. When subgroup data are unavailable, but data for both all students and the complementary group are available, we infer the number of students tested and the number of students in each proficiency category for the missing subgroup using the complementary group. However, when all-student data is unavailable, we exclude all subgroup data for that school or school district-subject-grade-year observation.

Subgroup data quality concerns. We dropped all race subgroup data for Alabama, New Mexico, and Vermont across all years due to evidence that the reported racial categories are not mutually exclusive.

State Exclusions. Table 4a shows the state-grade-year-subjects for which we produce estimates. Table 4b shows why estimates are not available when they are not. We drop state-subject-grade-years that meet the following criteria:

Data not reported in ED*Facts* (2009-2019) or in EDC (2022-2025). In some years, our sources may include no school or school district data for a particular state-subject-grade-year. While we do not actively remove these data, they are noted in Table 4b for completeness.

State participation is less than 94% or more than 105%. The ED*Facts* data includes participation information for all state-subject-grade-year cases in the 2012-13 through

2018-19 school years. We calculate participation as the ratio of the number of test scores reported to the number enrolled students in a given state-subject-grade-year:

$$participation_{sygb} = \frac{numscores_{sygb}}{numenrl_{sygb}} \quad (2.1)$$

for each state s , year y , grade g , and subject b . If the participation rate falls outside the accepted bounds, we remove all data for the state-subject-grade-year. This participation data file is not available prior to the 2012-13 school year, and therefore we cannot calculate participation rates prior to 2012-13 and do not remove any cases for this rule. The EDC data include state-level participation rates, when available from states, for the 2021-22 through 2024-25 school years. When available, we again remove any case where the state participation falls outside acceptable bounds. When unavailable, we make no removals for this rule.

These state-level drops are important because both the quality of the estimates and the linkage process depend on having the full population of student test scores for that state-subject-grade-year. State participation may be low due to several factors, including student opt out or pilot testing. Note that we do not drop any entire state-subject-grade-year cases prior to the 2012-13 school year based on participation as enrollment data are not available in *EDFacts*. However, opt out and non-participation are low in 2012-13 (no state was excluded based on this threshold), which suggests states met 94% threshold in prior years when participation data are not available.

Not all students took comparable tests within the state-subject-grade-year. There are three ways this appears within the data. First, there are cases where districts were permitted to administer locally selected assessments. This occurred in Nebraska during SY 2008-2009 (RLA and Math) and SY 2009-2010 (Math). Second, in some cases students take end-of-course rather than end-of-grade assessments. This is most common in 8th grade mathematics, where many students may take an Algebra I course in place of the standard math course. Third, a Spanish-language version of an assessment may be used in some subjects and grades. Spanish- and English-language versions of the assessments are not always equivalent, particularly when assessing reading and language skills. When

test scores measure different content and are reported on different scales using different cutscores, proficiency counts cannot be compared across districts or schools within these state-subject-grade-year cases. Our estimation methodology relies on the fact that all students took a common test (within a state-subject-grade-year).

Different student counts. As an extra check for the completeness of data, we (a) compare the number of test scores in each subject to each other and (b) compare the tested counts to the number of enrolled students reported in the CCD. We drop state-year-grade-subjects when:

- a. The state-subject-grade-year tested count is lower than 90% (for 2009-2019) or lower than 95% (for 2022-2025) of the number of test scores reported for the other subject in the same state-grade-year (unless participation in the other subject is higher than 105%); or,
- b. The state-subject-grade-year test count is less than 90% or more than 110% of the number of enrolled students in the CCD.

For 2022-2025, we also drop state-subject-grade-years where the number of test scores reported is less than 90% of the number reported in the prior grade in the same subject-year.

Insufficient data was reported to ED Facts or EDC. We remove state data from which we cannot recover reliable estimates. For example, in the 2008-09, 2009-10, and 2010-11 school years, Colorado reported data in only two proficiency categories, and a large majority of the data (88% across subjects, grades, and years) fall into a single category. These data do not provide sufficient information to estimate means and/or standard deviations in most regions. As a second example, in 2016, Tennessee testing was largely canceled after technical and logistical failures, making their data unreliable.

Test changes between 2023-24 and 2024-25. We exclude the entire state-subject data for 2025 if the state changed the assessment or cutscores from the prior year.

School Exclusions. We remove two sets of schools at this stage in our process. First, we remove a handful of schools from the data that report results under both BIE school IDs and regular school IDs. These were identified by the NCES. According to the CCD documentation,

“There is a possibility that some schools are reported in CCD by both the BIE and the state in which the schools are located, leading to a double counting of students and staff. (NCES allows for the possibility of co-located schools, so a double counting of schools is not an issue.) This arises from situations where both the state and BIE share operational or financial responsibilities for a school.”¹⁹ In order for SEDA to also avoid double counting, we remove the schools from the list and retain their counterparts listed in [Table 5](#). Second, we remove schools that appear in *EDFacts* but never appear in the CCD from 2009 through 2019. This affects 47 school-years: 11 in 2009, 7 in 2010, 19 in 2013, and no more than 2 in any other year. Most have fewer than 10 assessments across all grades; none have more than 30.

For 2022-2025, we exclude virtual schools, vocational schools, and special education schools.

Alternate Assessments. In 2008-09 through 2011-12, *EDFacts* does not distinguish students taking regular from alternate assessments; these counts were combined in the reported data. Therefore, for consistency in all years, we combine the performance data for regular and alternate assessments as reported in *EDFacts*. In some states, alternate assessments have different performance categories relative to the regular assessment.²⁰ To ensure that the alternate assessment’s proficiency levels match the regular assessment’s proficiency levels, we collapse the top categories for any places that report students scoring in one higher proficiency level than exists for the regular assessment. The affected state, subject, grade, and year cases include: Arkansas, math and RLA, grades 3-8, years 2012, 2013, and 2014; Colorado, math and RLA, grades 3-8, years 2012, 2013, and 2014; Colorado, RLA, grade 6 in 2018 and 2019; DC, RLA, grade 6 in 2011; Iowa, math and RLA, grades 3 through 8, years 2015 through 2019; and South Carolina, math and RLA, grades 3-8, years 2012-2014. In practice, alternate assessments make up a small share of tested students in most units. We flag and remove cases in which more than 20% of students take alternate assessments to ensure that our estimates are less affected when combining data from potentially different test score scales.

¹⁹ See Page 1 in the NCES [Double Counting of BIE Reported Schools](https://nces.ed.gov/ccd/doc/5%20Double%20Counting%20of%20Bureau%20of%20Indian%20Education%20Schools%203.4.2020.docx) documentation. Retrieved from: <https://nces.ed.gov/ccd/doc/5 Double Counting of Bureau of Indian Education Schools 3.4.2020.docx>

²⁰ The *EDFacts* documentation notes proficiency levels by assessment type in years after 2011-12.

For 2022-2025, we use data as reported, which may or may not include alternative assessments. The inclusion of alternative assessments in publicly reported counts varies by state.

Step 3. Cutscore Estimation

School years 2008-09 through 2018-19. In this step, we use HETOP models and the all-student geographic school district proficiency count data to estimate state-subject-grade-year cutscores on a state standardized scale. We exclude virtual schools from cutscore estimation.

To address practical challenges that can arise in HETOP cutscore estimation for a specific state-subject-grade-year, we:

Rearrange geographic school districts. We reconfigure geographic school districts that meet certain criteria within a state-subject-grade-year in order to improve the HETOP estimation process. First, we combine vectors of counts that have fewer than 20 students into “overflow” groups because estimates based on small sample sizes can be inaccurate. Second, in some vectors with more than 20 students the pattern of counts does not provide enough information to estimate a mean or a standard deviation; we also place these count vectors into the “overflow” group. If the resulting overflow groups have parameters that cannot be estimated via maximum likelihood, they are removed from the data. This reconfiguration allows us to retain the maximum possible number of test scores in the estimation sample for the cutscores. This is important as the linking methods we use later in this step rely on having information about the same full population that NAEP represents in each state-grade-year-subject.

Constrain geographic school districts. For groups not in the “overflow” group, we always estimate a unique mean. But we can sometimes obtain more precise and identifiable estimates by placing additional constraints on group standard deviation parameters in the HETOP model. We constrain standard deviation parameter estimates for groups that meet the following conditions during estimation:

- There are fewer than 50 student assessment outcomes in a geographic school district.

- There are not sufficient data to estimate both a mean and standard deviation (all student assessment outcomes fall in only two adjacent performance level categories; all student assessment outcomes fall in the top and bottom performance categories; or all student assessment outcomes fall in a single performance level category).

After these data processing steps, we estimate a separate HETOP model for each state-subject-grade-year and save the cutscore estimates. For state-grade-year-subjects with only two proficiency categories, we cannot estimate unique geographic school district standard deviations and instead we use the model with a single, fixed standard deviation parameter (the homoscedastic ordered probit, or HOMOP model). We denote the estimated cutscores as $\widehat{c}_{1sygb}^{state}, \dots, \widehat{c}_{K-1sygb}^{state}$, for a state s , year y , grade g , and subject b , where the proficiency data are reported in K categories. These cutscores are expressed in units of their respective state-year-grade-subject student-level standardized distribution. The HETOP model estimation procedure also provides standard errors of these cutscore estimates, denoted $se(\widehat{c}_{ksygb}^{state})$ for $k = 1, \dots, K - 1$, respectively (Reardon, et al., 2017). Note that we do not use the group-specific means or standard deviations that are simultaneously estimated along with the cutscores. See Reardon et al. (2017) for additional details about the HETOP model.

We adjust cutscores based on the state reliability to account for measurement error in state accountability test scores. Note that cutscores on the state scale are expressed in terms of standard deviation units of the state score distribution. Because measurement error inflates the observed standard deviation, these state-scale cutscores are biased toward zero. We disattenuate the cutscores by dividing them by the square root of the state test score reliability estimate, where r_s is the reliability of the state test in the given subject-grade-year:

$$c_{ks}^{state'} = \frac{c_{ks}}{\sqrt{r_s}}$$

The initial reliability data used to disattenuate the estimates come from Reardon and Ho (2015); these have been supplemented with publicly available information from state technical reports from recent years. For cases where no information was available, test reliabilities were imputed using data from other grades and years in the same state.

PARCC & SBAC cutscores for BIE waiver schools. Once we have disattenuated cutscores, we take all states, years, and subjects that took the PARCC and average their cutscores together to get a set of average PARCC cutscores. We do the same for the SBAC cutscores. These average cutscores are then used for BIE waiver schools taking the respective exams. We do not use these averaged cutscores for the overall state data. Table 6a and Table 6b show the states-years-subjects averaged for this cutscore creation. Applying Cuts to BIE schools: Some BIE schools submitted data in different proficiency categories than the proficiency categories reported by the states in which the BIEs reside. In addition, the Navajo Nation had test waivers for the PARCC beginning in SY 2015-16 and Miccosukee Indian School which had a waiver for the SBAC starting in SY 2014-15. For these waiver schools, we use the averaged waiver cuts discussed above. For non-waiver BIE schools, we realign the BIE cuts to match the cuts for the states in which they were located. A few schools whose cuts we could not determine were omitted from SEDA. See Table 7.

School years 2021-22 through 2024-25. Because of suppression in the public district level data from EDC, we do not have the full state population of students reflected in the sum of the district counts for each state-subject-grade-year. Therefore, we estimate state-grade-subject-year cutscores using the inverse cumulative standard normal distribution function using state-level data. Specifically, using the state-subject-grade-year counts for all students, we estimate the k th cutscore in the state-subject-grade-year as (subject, grade, year subscripts omitted):

$$c_{ks} = \Phi^{-1}(p_{ks}) \quad (3.1)$$

where p_{ks} is the proportion of students scoring below the k th cutscore in the state-grade-subject-year.

Again, we disattenuate the cutscores by dividing them by the square root of the state test score reliability estimate in the given subject-grade-year. This is the same process as described in the section on the 2009-2019 cutscores above. The disattenuated cut scores will be merged with the district-level proficiency counts to fit HETOP models at the district level in **Step 5**.

Applying cuts to partially suppressed data. Because state-level data reflect the full population of tested students, suppression does not affect state-level cut score estimation. At the school and district level, however, small cell sizes frequently trigger suppression of one or more proficiency level counts in the public EDC data. When one or more district-level counts are suppressed, we identify the unsuppressed adjacent levels for each district and retain only the cut scores corresponding to boundaries between those levels. For example, if a district has counts for levels 1, 2, and 3 but level 4 is suppressed, we retain cut scores c_1 and c_2 , drop c_3 , and treat the district as having three proficiency levels. This procedure requires that the observed levels form a contiguous (adjacent) subset; districts where suppression creates a gap in the interior of the proficiency scale (e.g., level 2 suppressed but levels 1, 3, and 4 observed) cannot be salvaged by realignment and are dropped from estimation.

Step 4. Selecting Data for Mean Estimation

In Step 5, we estimate a model separately for each unit-subgroup using only on the subject-grade-year data for that unit-subgroup. For some subjects, grades, and years, we are less confident in the quality of the unit-subgroup data and do not want to include these in the estimation as it may bias the parameter estimates.²¹ We create flags for dropping these cases, which are described below:

There are fewer than 2 test scores for the unit-year-subject-grade. There is insufficient data to construct test score means and standard deviations for these cells.

The participation rate is less than 94%. In these cases, the population of tested students may not be representative of the population of students in that school. Therefore, we flag and remove all unit-subgroup-subject-grade-year cases where participation was lower than 94%. Participation is defined as:

²¹ This logic of this data selection differs from the cleaning done in Step 2 to support cutscore estimation. For the cutscore estimation, we wanted to keep as much data as possible in the estimation process because the linking procedure at the beginning of Step 6 requires population-based data. Moreover, the cutscores are not particularly sensitive to low-quality data for individual geographic school districts. In contrast, the individual school/geographic district mean/SD estimates will be strongly affected by low quality data (which is defined and described in this section).

$$participation_{urygb} = \frac{numscores_{urygb}}{numenrl_{urygb}}. \quad (4.1)$$

This measure can be constructed starting in the 2012-13 school year; we do not remove data based on this rule in earlier years. If the participation rate for “all students” is less than 94%, we do not report any estimates for demographic subgroups regardless of whether the subgroup-specific participation rate was greater than 94% because we are concerned about data quality in cases with low overall participation. We set a slightly lower threshold (94%) for overall participation to avoid dropping units due to minor enrollment or reporting discrepancies, while requiring at least 95% representation (in the next step) for race and gender subgroups so that subgroup estimates are based on nearly all reported test scores. For 2022-2025, we use participation rate data when it is available. If it is unavailable, we assume that the participation rate is acceptable

Incomplete data reported by student demographic subgroups. There are a small number of cases where the total number of test scores reported by race or gender is less than 95% of the total reported test scores for all students. For example, there may be 50 test scores reported for all students, but only 20 test scores for male students and 20 test scores for female students. In this case, we would not report the male or female test score means because insufficient test scores were reported by gender. We calculate the reported percentage as:

$$representation_{urygb} = \frac{\sum_r numscores_{urygb}}{numscores_{u,all,ygb}}. \quad (4.2)$$

This measure can be constructed in all years. We flag and remove unit-subgroup-subject-grade-year data from SEDA where there is not at least a 95% representation rate for the gender or race subcategories. For 2022-2025, we cannot enforce this rule as we often do not have full subgroup data.

More than 20% of students take alternate assessments. Measurement error may affect unit-subgroup-subjects where over 20% of the students take alternate assessments.

These assessments typically differ from the regular assessment and have different proficiency thresholds. For 2009-2019, we base this flag on the percentage of students

taking alternative assessments in the unit-subgroup-subject from all grade-years for which it is available, starting in the 2012-13 school year. We flag and remove unit-subgroup-subjects that meet this criterion from SEDA at this step. We do not have sufficient data to enforce this rule for 2022-2025.

Insufficient data. In some cases, the data we have does not meet the minimum statistical estimation requirements. First, we flag and remove all unit-subgroup-subject-grade-years where students scored only in the top or only in the bottom proficiency category. We cannot obtain maximum likelihood estimates of unique means for these cases. Second, we flag unit-subgroup-subject-grade-year cells as insufficient if:

1. All observations are in a single, middle proficiency category.
2. All observations are in only 2 adjacent proficiency categories.
3. All observations are only in the top and bottom proficiency categories.

We remove all unit-subgroup-subjects where *all* grade-year cells are insufficient. We will be unable to produce estimates in these cases as there is insufficient data to freely estimate a mean and standard deviation in any cell.

Table 8 shows how many unit-subgroup-subject-grade-years cells these drops affect and what is excluded from mean estimation.

Flags for estimation. We create two flags for use in estimation: an “insufficient data” flag and a “size” flag. During estimation, the insufficient data and size flags are used to place constraints on the standard deviation estimates for individual grade-year cells.

Insufficient data flag. Unit-subgroup-subjects may contain one or more “insufficient” grade-year cells which do not have sufficient data to freely estimate both a mean and a standard deviation. A grade-year cell will be insufficient if:

1. All observations are in a single, middle proficiency category.
2. All observations are in only 2 adjacent proficiency categories.
3. All observations are only in the top and bottom proficiency categories.

Size flag. We flag cells as “small” if they have fewer than 100 test scores; otherwise, cells are flagged as “large”.

Step 5. Estimating Test Score Means

The goal of this step is to estimate the mean and standard deviation of test scores for each subgroup in each unit across subjects, grades, and years.

5A. State Estimates for the All Student Group

The state estimates we report for the all student group come from the main NAEP state data, not state accountability data. NAEP data provide estimates of 4th and 8th grade test score means and standard deviations for each state on a common scale, denoted $\hat{\mu}_{sygb}^{naep}$ and $\hat{\sigma}_{sygb}^{naep}$, respectively, as well as their standard errors.²² NAEP was administered only in 4th and 8th grades in odd-numbered years from 2009 to 2019 and in even-numbered years from 2022 to 2024. We use the Expanded Population Estimates (or “EPE” estimates). These were not yet available for 2024 at the time of data construction, so we create an adjustment to construct a 2024 EPE from the regular NAEP estimate. See **Appendix A2.1 NAEP 2024 Adjustment** for more details.

We need to obtain NAEP values in non-tested years (2010, 2012, 2014, 2016, 2018, 2023, and 2025)²³ and grades (3, 5, 6, and 7). First, within each NAEP year (2009, 2011, 2013, 2015, 2017, 2019, 2022, 2024), we linearly interpolate between grades 4 and 8 to grades 5, 6, and 7 and linearly extrapolate to grade 3. This provides complete grade level data in each NAEP-tested year. Second, we focus on the years falling between each NAEP-tested year (all non-tested years, except 2025). For all grades 3-8, we linearly interpolate between the NAEP-tested years to estimate parameters in 2010, 2012, 2014, 2016, 2018, and 2023 using the formulas here:

$$\begin{aligned}\hat{\mu}_{sygb}^{naep} &= \hat{\mu}_{sy4b}^{naep} + \frac{g-4}{4} (\hat{\mu}_{sy8b}^{naep} - \hat{\mu}_{sy4b}^{naep}), \\ &\text{for } g \in \{3, 5, 6, 7\} \\ \hat{\mu}_{sygb}^{naep} &= \frac{1}{2} (\hat{\mu}_{s(y-1)gb}^{naep} + \hat{\mu}_{s(y+1)gb}^{naep}), \\ &\text{for } y \\ &\in \{2010, 2012, 2014, 2016, 2018, 2023\}\end{aligned}\tag{5.1}$$

²² Note that the NAEP scales are not comparable across math and reading, but they are comparable across states, grades and years within each subject.

²³ Note: We do not attempt to produce estimates of the NAEP means/SDs in 2020 or 2021.

We use the same process and formulas to interpolate and extrapolate the standard deviations for untested grades and years (excluding 2025).

Because producing an estimate for 2025 involves extrapolating the year trend, we take a different approach. We examine the historical differences in trends between the state tests and the NAEP tests. Using this information, we estimate what the NAEP score would be in 2025 for each state based on what we observe for the state test trend from 2024 to 2025. For more details on the production of the 2025 estimates, see Reardon et al. (2026) and **Appendix A2.2 Estimated NAEP 2025 Values**. For 2025, the grade 4 and grade 8 inputs are obtained using the procedure in Appendix A2.2. We then linearly interpolate/extrapolate inputs for grades 3, 5, 6, 7 using the first equation from Equation 5.1 above.

At the end of this step, we have estimates, on the NAEP scale, for all states plus the District of Columbia in all grades and years. For 2025, NAEP estimates are available for a subset of states where state test score changes between 2024 and 2025 could be estimated. Note, however, that the count of students tested associated with each estimate we report do not come from NAEP. We use the counts of test takers from *EDFacts* in 2009 to 2019 or EDC in 2022 to 2025 when available. For state-years when *EDFacts* and EDC data are not available, we use enrollment counts from the CCD.

5B. HETOP Estimates for Other Units and for State Subgroups

Pooled HETOP estimation. In the prior steps, we prepared two pieces of information that we use in estimation: the observed proficiency counts for each unit-subgroup-grade-year-subject from **Step 4** and the estimated cutscores separating the proficiency categories in the associated state-grade-year-subject from **Step 3**. All schools are affiliated with a single state and, thus, a single test and a single set of cutscores. While larger units (geographic districts, administrative districts, metropolitan areas, commuting zones, and states) are also typically affiliated with a single state, test, and set of cutscores, there are a few notable exceptions:

Units that contain BIE schools: As noted above, BIE schools often have different cutscores than the other schools assigned to a unit. In these cases, the unit has two or more components where the schools assigned to each component take the same test and use

the same cutscores. For example, we might have a unit with two pieces: unit-regular schools and unit-BIE waiver schools.

Metropolitan Areas or Commuting Zones that cross state lines: A subset of metropolitan areas and commuting zones cross state lines and therefore can be affiliated with several state’s cutscores. These units will have metropolitan area-by-state or commuting zone-by-state components, where the schools assigned to each component took the same test and used the same cutscores.

For both cases, we estimate pooled HETOP using data for each unit component and the appropriate cutscores, we then aggregate the components after estimation into overall unit estimates (see **Step 10**).

We use a pooled HETOP model (Shear & Reardon, 2021; Reardon et al., 2017) to estimate μ_{urygb}^{state} and σ_{urygb}^{state} , the mean and standard deviation of achievement on the state scale for unit u , subgroup r , year y , grade g , and subject b . As described below, the pooled HETOP model is estimated separately for each unit-subgroup-subject but combines data across grades and years when estimating these parameters. Combining data across grades and years allows us to get better estimates of σ_{urygb}^{state} for years and grades in which sample sizes are small or where the proficiency count data are limited.

We use a pooled HETOP model to overcome three practical challenges. The challenges are: 1) in some states, years, and grades, the number of proficiency categories $K = 2$ and there is not sufficient information to estimate both a mean and a standard deviation for each unit-subgroup-grade-year-subject; 2) if $K \geq 3$ but there are sampling zeros because test scores were not observed in all K categories for a particular grade and year, there may not be sufficient information to estimate both a mean and a standard deviation; and 3) when the sample size n_{kurygb} is small, prior simulations (e.g., Reardon et al., 2017; Shear & Reardon, 2021) have shown that estimates of standard deviations can be biased and contain excessive sampling error.

We estimate a pooled HETOP model (Shear & Reardon, 2021) for each unit, separately for each subject and subgroup, by “pooling” data across all available grades and years, and maximizing the joint log likelihood function given by:

$$\begin{aligned}
L = \ln[P(\mathbf{N}_{urb} | \mathbf{M}_{urb}^{state}, \mathbf{H}_{urb}^{state}, \mathbf{C}_{sb}^{state})] &= \sum_{y=1}^Y \sum_{g=1}^G \sum_{k=1}^K n_{kurygb} \ln(\pi_{kurygb}) \\
&= \sum_{y=1}^Y \sum_{g=1}^G \sum_{k=1}^{K_{gy}} n_{kurygb} \ln \left(\Phi \left(\frac{\mu_{urygb}^{state} - c_{k-1_{sygb}}^{state}}{\exp(h_{urb}(g, y))} \right) - \Phi \left(\frac{\mu_{urygb}^{state} - c_{k_{sygb}}^{state}}{\exp(h_{urb}(g, y))} \right) \right),
\end{aligned}$$

where $\mathbf{N}_{urb}$ is a matrix of proficiency counts across all available grades (G) and years (Y) for unit u , subgroup r and subject b , $\mathbf{M}_{urb}^{state}$ is a vector of estimated means across grades and years, $\mathbf{H}_{urb}^{state}$ is a vector of estimated parameters for the function $h(\cdot)$ described below, and $\mathbf{C}_{sb}^{state}$ is a matrix of cutscores across grades and years. The cutscores are treated as fixed here, using the values estimated in **Step 3**. We have replaced σ_{urygb}^{state} in the above equation with $\exp(h_{urb}(g, y))$, where $h_{urb}(g, y)$ is a unit-specific function used to model the natural log of the standard deviations as a function of grade and year:

$$h_{urb}(g, y) = \ln(\sigma_{urygb}^{state}) = \gamma_{urygb}^{state}.$$

We do this for two reasons. First, estimating $\gamma_{urygb}^{state} = \ln(\sigma_{urygb}^{state})$ rather than σ_{urygb}^{state} directly ensures that the ML estimate will be positive. Second, defining γ_{urygb}^{state} as a function of grade and year, rather than allowing this value to be unique in each grade and year, defines the pooled HETOP model. If we place no constraints on the model and allow $h_{urb}(g, y) = \gamma_{urbgy}$ to take on a unique value in each grade and year, maximization of this likelihood will result in identical estimates to those obtained by maximizing the likelihood separately for each grade and year.

To leverage the data available across multiple grades and years and overcome the limitations noted above, we define $h_{urb}(g, y)$ in the following way. First, we allow γ_{urygb} to be freely estimated in each grade-year cell that is both “sufficient” and “large”, by the flags defined above. For all other grade-year cells, we constrain $h_{urb}(g, y)$ such that the estimate of γ_{urygb} is equal to the mean of the $\hat{\gamma}_{urygb}$ estimates across the freely estimated cells. That is, we estimate a common “pooled” standard deviation across the grades and years in which there are “insufficient data” and/or “small” cell sizes.

More formally, for all years and grades in which $n_{urygb} < 100$ and/or in which there are insufficient data to estimate both a mean and a standard deviation, we constrain $h_{urb}(g, y) =$

γ_{urb}^{state} to be equal, while allowing $h_{urb}(g, y) = \gamma_{urygb}^{state}$ to be freely estimated in grades and years with at least 100 test scores and sufficient data to estimate both a mean and standard deviation. We further constrain the model such that the “pooled” log standard deviation is equal to the (unweighted) mean of the unconstrained log standard deviations by defining the constraint:

$$\gamma_{urb}^{state} = \frac{\sum_{g=1}^G \sum_{y=1}^Y (I_{urygb}^{100} \cdot I_{urygb}^S \cdot \gamma_{urygb}^{state})}{\sum_{g=1}^G \sum_{y=1}^Y (I_{urygb}^{100} \cdot I_{urygb}^S)},$$

where I_{urygb}^{100} is the size indicator flag (equal to 1 if size is “large”) and I_{urygb}^S is the sufficient data indicator flag (equal to 1 if there are sufficient data). If I_{urygb}^{100} and I_{urygb}^S are equal to 1 for all cells in a unit, then we estimate a unique mean and standard deviation for each cell. For all other units, there will be a mix of freely estimated and constrained standard deviation parameters. Recall that in **Step 4** that we removed unit-subgroups where $I_{urygb}^S = 0$ for all cells because we are unable to estimate a standard deviation parameter.

In sum, the models described here are used to produce ML estimates of μ_{urygb}^{state} and σ_{urygb}^{state} (where $\hat{\sigma}_{urygb}^{state}$ may be constrained to be equal in some grades and years), as well as estimated standard errors $se(\hat{\mu}_{urygb}^{state})$ and $se(\hat{\sigma}_{urygb}^{state})$ and the estimated sampling covariances $cov(\hat{\mu}_{urygb}^{state}, \hat{\sigma}_{urygb}^{state})$. The estimates are produced on the state standardized scale that is disattenuated for measurement error, and can be transformed to other scales, as described in **Step 6**.

Step 6. Linking Mean Estimates

We link the estimated means and their standard errors created in Step 5 to a common scale using the interpolated NAEP state data, described in Step 5A.²⁴ The methods we use—as well as a set of empirical analyses demonstrating the validity of this approach—are described in more detail by Reardon, Kalogrides, and Ho (2021). We provide a brief summary here.

²⁴ Note that the state all student estimates are already on a common scale and those estimates will be used in the linkage equations for all other unit-subgroups.

The linking equations are shown below (subgroup, subject, grade, year subscripts omitted for clarity).

$$\begin{aligned}\hat{\mu}_u^{naep} &= \hat{\mu}_u^{state} \cdot \hat{\sigma}_s^{naep} + \hat{\mu}_s^{naep} \\ se(\hat{\mu}_u^{naep}) &= \hat{\sigma}_s^{naep} \cdot se(\hat{\mu}_u^{state})\end{aligned}\tag{6.1}$$

The resulting $\hat{\mu}_{uygb}^{naep}$ and $se(\hat{\mu}_{uygb}^{naep})$ are on the NAEP scale. However, the linked $se(\hat{\mu}_{dygb}^{naep})$ does not yet account for the uncertainty in NAEP. We create a second standard error $se'(\hat{\mu}_{dygb}^{naep})$ that incorporates the NAEP uncertainty, using the following formula (subgroup subscript omitted for clarity):

$$\begin{aligned}se^{adj}(\hat{\mu}_{uygb}^{naep}) &= \sqrt{var(\hat{\mu}_{uygb}^{naep}) + var(\hat{\mu}_{sygb}^{naep})} \\ &\quad - \left(\frac{1}{\hat{\sigma}_{sygb}^{naep}} \right)^2 var(\hat{\mu}_{uygb}^{naep}) var(\hat{\sigma}_{sygb}^{naep}) + var(\hat{\mu}_{sygb}^{naep})\end{aligned}\tag{6.2}$$

We report both standard errors in our data. The unadjusted standard error is appropriate for within-state comparisons, while the NAEP adjusted standard error is appropriate for cross-state comparisons.

Step 7. Creating Reporting Scales

We first interpolate national NAEP data, using the same process described under Step 5 for the state NAEP data. The reported national NAEP means and standard deviations, along with interpolated values, by year and grade, are shown in [Table 9](#). Then we standardize the NAEP-linked means and SEs to our four reporting scales: the Cohort Standardized (CS); the Year Standardized (YS); Grade Cohort Standardized (GCS); and the Grade Year Standardized (GYS) scales.

Cohort Standardized (CS) Scale: We standardize relative to the average of the four national cohorts that were in 4th grade in 2009, 2011, 2013, and 2015. This standardization is accomplished by subtracting the national grade-subject-specific mean and dividing by the

national grade-subject-specific standard deviation for the average reference cohort. For each grade, year, and subject we calculate:

$$\begin{aligned}\mu_{N(avg)gb}^{naep} &= \sum_{Y \in \{2005, 2007, 2009, 2011\}} \frac{1}{4} \hat{\mu}_N^{naep} (y=Y+g)gb \\ \sigma_{N(avg)gb}^{naep} &= \sum_{Y \in \{2005, 2007, 2009, 2011\}} \frac{1}{4} \hat{\sigma}_N^{naep} (y=Y+g)gb\end{aligned}\tag{7.1}$$

In Equation (7.1), Y refers to the year in which the cohort was in the spring of kindergarten. For the 2009 4th grade cohort, this is equal to 2005 (or 2009 minus 4).

Then we standardize the means on the NAEP scale to the CS scale as follows (equations shown for districts, subgroup subscript omitted):

$$\begin{aligned}\hat{\mu}_{uygb}^{cs} &= \frac{(\hat{\mu}_{uygb}^{naep} - \hat{\mu}_{N(avg)gb}^{naep})}{\sigma_{N(avg)gb}^{naep}} \\ se(\hat{\mu}_{uygb}^{cs}) &= \frac{se(\hat{\mu}_{uygb}^{naep})}{\sigma_{N(avg)gb}^{naep}}\end{aligned}\tag{7.2}$$

All means in this scale are interpretable as an effect size, relative to the grade-subject-specific standard deviation of student-level scores in the four national cohorts that were in 4th grade in 2009, 2011, 2013, and 2015. For example, a district mean of 0.5 on the CS scale indicates that the average student scored approximately one half of a standard deviation higher than the average national reference cohort scored in that same grade.

Means reported on the CS scale have an overall average near 0 as expected. Note that this scale retains information about absolute changes over time by relying on the stability of the NAEP scale over time. This scale does not enable absolute comparisons across grades, however.

Year Standardized (YS) Scale: To create the YS scale, we standardize the NAEP-scaled estimates using the 2019 national average in each grade and subject. The equations to standardize to the YS scale are equivalent to those in 7.2 but using the scaling parameters are: $\hat{\mu}_{N(2019)gb}^{naep}$ and $\hat{\sigma}_{N(2019)gb}^{naep}$. In this scale, one unit is equivalent to a 2019 national standard deviation in the same subject and grade.

Grade Cohort Standardized (GCS): We approximate the average amount student test scores grow per grade on NAEP using the 4th and 8th grade estimates by subject for the average

of the four national cohorts that were in 4th grade in 2009, 2011, 2013, and 2015. We calculate the amount the test scores changed between 4th and 8th grade as the average score in 8th grade minus the average score in 4th grade for each cohort; then average those across the four cohorts. Then, to get an estimate of per-grade differences, we divide that value by 4.

$$\hat{\gamma}_{N(avg)b} = \frac{\hat{\mu}_{N(avg)(8)b}^{naep} - \hat{\mu}_{N(avg)(4)b}^{naep}}{4} \quad (7.3)$$

We scale the data using these parameters, such that in the GCS scale each unit is interpretable as 1 grade level referenced to the four national cohorts that were in 4th grade in 2009, 2011, 2013, and 2015 (equations shown for districts, subgroup subscript omitted):

$$\begin{aligned} \hat{\mu}_{dygb}^{gcs} &= 4 + \frac{(\hat{\mu}_{dygb}^{naep} - \hat{\mu}_{N(avg)(4)b}^{naep})}{\hat{\gamma}_{N(avg)b}} \\ se(\hat{\mu}_{dygb}^{gcs}) &= \frac{se(\hat{\mu}_{dygb}^{naep})}{\hat{\gamma}_{N(avg)b}} \end{aligned} \quad (7.4)$$

Then, $\hat{\mu}_{urygb}^{gcs}$ can be interpreted as the estimated average national “grade-level performance” of students in unit u , subgroup r , year y , grade g , and subject b . For example, if $\hat{\mu}_{ury4b}^{gcs} = 5$, 4th-grade students in unit u , subgroup r , and year y are one grade level ($\hat{\gamma}_b$) above the 4th grade 2009-2015 national average ($\hat{\mu}_{N(avg)(4)b}^{naep}$) in performance on the tested subject b .

Grade Year Standardized (GYS): We approximate the average amount student test scores grow per grade on NAEP using the 4th and 8th grade estimates by subject in 2019. We calculate the amount the test scores changed between 4th and 8th grade as the average score in 8th grade in 2019 minus the average score in 4th grade in 2019. Then, to get an estimate of per-grade differences, we divide that value by 4 and denote it as $\hat{\gamma}_{N(2019)b}$. We scale the data using these parameters and equations similar to those in Equation 7.4, such that in the GYS scale each unit is interpretable as 1 grade level referenced to the 2019 national population.

The two grade-level metrics (GCS and GYS) enable absolute comparisons across grades and over time. They do so by relying not only on the assumption that the NAEP scale is stable over time but also that it is vertically linked across grades 4 and 8 and linear between grades. These metrics are simple linear transformations of the NAEP scale, intended to render the NAEP scale more interpretable. For reference, 1 CS scale standard deviation is approximately 3 grade

levels. As such, the grade-level metrics are useful for presenting descriptive research to broad audiences not familiar with interpreting standard deviation units. However, we do not advise using them for analyses where the vertical linking across grades and the linear interpolation assumptions are not required nor defensible. We recommend CS- or YS-scaled estimates for research purposes and GCS- or GYS-scaled estimates for low-stakes reporting to non-research audiences.

Step 8. Annual Estimates

For each unit-subgroup-year, we have up to 12 subject-grade mean estimates. We pool the estimates using generalized least squares (GLS) and precision-weighted random-coefficient models to produce: (1) subject-specific annual estimates, pooled over grades within subject (“annualsub”); and (2) overall annual estimates, pooled over grades and subjects (“annual”). We produce annual estimates for all units except schools. We produce two types of estimates: ordinary least squares (“OL”) estimates and Empirical Bayes (“EB”) shrunken estimates. We describe the uses of each type of estimate in the Additional Notes section of the documentation.

The models differ by geography (state vs. other units) and by group (all students vs. subgroups). We address each in turn below. For simplicity, we only show the models for the subject-specific annual estimates. All models shown below are estimated separately for each unit, subgroup, subject, and scale. Subject subscripts and scale superscripts are omitted for parsimony throughout. For overall annual estimates, we use data for both subjects and include an indicator that the subject is math, centered at 0.5. Otherwise, the process is identical.

Because the models used to produce EB annual estimates rely on trend information for shrinkage, we estimate models separately for the 2009-2019 and 2022-2025 periods. By separating the two periods, we use only the trend information from the relevant pre- or post-pandemic period. The structure of the estimated models is the same regardless of the period; the difference is the year centering which is noted below. Prior to estimation, we drop cases that have insufficient data for estimation. For subject specific annual estimates, we flag unit-subgroup-subject-year cases as insufficient if they contain fewer than two grades. For overall, pooled across subject annual estimates, we flag unit-subgroup-year cases as insufficient if they

contain fewer than two grades or fewer than two subjects. [Table 10](#) shows how many cases these drops affect.

8.A. State All-Student Annual Estimates

We first use NAEP data to construct state annual estimates. Specifically, we use the state-level data for grades 4 and 8 from each year, including the interpolated years and the estimated 2025 value constructed in **Step 5A** (note, however, we do not use interpolated grade data for any year).

We recover least squares (denoted “OL estimates” in the data files) annual estimates and variance-covariance (VCV) matrices for state-years via GLS. We perform the following calculations separately for each state-year-subject-scale:

$$\begin{aligned}\hat{\mathbf{B}}_{(y)s} &= (\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{W}^{-1}\mathbf{Y}) \\ \hat{\mathbf{V}}_{(y)s} &= (\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1}\end{aligned}\tag{8A.1}$$

Where $\mathbf{X}$ is a matrix of two predictors, an intercept and grade term centered at 5.5; $\mathbf{Y}$ is the vector of grade 4 and 8 NAEP means in the given state, subject, and year, $\hat{\mu}_{(y)gs}$; and $\mathbf{W}$ is a diagonal matrix of their variances, $var(\hat{\mu}_{(y)gs})$.

Using the same data, we also fit an HLM model separately by year-subject-scale with state-grade observations, nested in states:

$$\begin{aligned}\hat{\mu}_{(y)gs} &= \beta_{(y)0s} + \beta_{(y)1s}(G - 5.5) + \epsilon_{(y)gs} \\ \beta_{(y)0s} &= \gamma_{(y)00} + r_{(y)0s} \\ \beta_{(y)1s} &= \gamma_{(y)10} + r_{(y)1s} \\ \epsilon_{(y)gs} &\sim (0, \zeta_{(y)gs}); \quad \mathbf{r}_{(y)s} \sim (0, \boldsymbol{\tau}_{(y)})\end{aligned}\tag{8A.2}$$

Where G is the grade, $\epsilon_{(y)gs}$ is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean estimates, $var(\hat{\mu}_{(y)gs})$. The residual variance in each year-subject-scale $\boldsymbol{\tau}_{(y)}$ is estimated. From this model, we produce EB state annual-subject means and posterior variances for all students: $\hat{\mathbf{B}}_{(y)s}^*$ and $\hat{\mathbf{V}}_{(y)s}^*$.

At the end of this step, we have both OL and EB state-level all student annual-subject estimates ($\hat{\mathbf{B}}_{(y)s}$, $\hat{\mathbf{V}}_{(y)s}$, $\hat{\mathbf{B}}_{(y)s}^*$ and $\hat{\mathbf{V}}_{(y)s}^*$) and their variances on all scales.

8.B. Other Unit All Student Annual Estimates

We then use unit-subject-grade-year data for all other units, along with the state annual-subject estimates from **Step 8A**, to construct unit annual-subject estimates for all students. Note that we use the unit data *without* the adjusted standard errors that account for the error in NAEP linking from **Step 5B** as input for these models. We will create a second adjusted standard error after model estimation.

We get OL estimates and VCV matrices ($\hat{\mathbf{B}}_{(y)u}$ and $\hat{\mathbf{V}}_{(y)u}$) for other unit-year-subjects via GLS. The process is the same as that for states (Equation 8A.1) except that we use the state accountability (not NAEP) data for all grades. We then construct an adjusted VCV matrix (for use in comparisons across states) by adding the VCV matrix from the state-year-subject model (created in step 8A, Equation 8A.1) to the adjusted unit-year-subject VCV matrix:

$$\hat{\mathbf{V}}_{(y)u}^{adj} = \hat{\mathbf{V}}_{(y)u} + \hat{\mathbf{V}}_{(y)s} \quad (8B.1)$$

To construct the EB estimates, we fit two HLM models. The logic behind the two-model structure is to first estimate a within-state linear trend in annual estimates across years, so we can then shrink the annuals toward this unit-specific annual trend rather than solely shrinking toward a state annual. We then shrink these toward the shrunken state annual trends.

Specifically, we first fit a model using all unit data within a state-subject-scale where grades are nested in years, nested in units:

$$\begin{aligned} \hat{\mu}_{(s)gyu} &= \beta_{(s)0yu} + \beta_{(s)1yu}(G - 5.5) + \epsilon_{(s)gyu} \\ \beta_{(s)0yu} &= \gamma_{(s)00u} + \gamma_{(s)01u}(Y - my) + r_{(s)0yu} \\ \beta_{(s)1yu} &= \gamma_{(s)10u} + r_{(s)1yu} \\ \gamma_{(s)00u} &= \delta_{(s)000} + u_{(s)00u} \\ \gamma_{(s)01u} &= \delta_{(s)010} + u_{(s)01u} \\ \gamma_{(s)10u} &= \delta_{(s)100} + u_{(s)10u} \\ \epsilon_{(s)gyu} &\sim (0, \zeta_{(s)gyu}); \quad \mathbf{r}_{(s)yu} \sim (0, \boldsymbol{\eta}_{(s)}); \quad \mathbf{u}_{(s)u} \sim (0, \boldsymbol{\tau}_{(s)}) \end{aligned} \quad (8B.2)$$

Where G is defined as above, Y is the spring of the test year centered at my , $\epsilon_{(y)gs}$ is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean. The residual variances $\boldsymbol{\eta}_{(s)}$ and $\boldsymbol{\tau}_{(s)}$ are estimated. my is the middle year of the panel; when estimating models for 2009-2019, we use $my = 2014$; when estimating models for 2022-2025, we use $my = 2023.5$.

From this model, $\hat{\mathbf{u}}_{(s)u}$ is the average difference between the district parameters and the state s average parameters across years. We include $\hat{\mathbf{u}}_{(s)u}$ as a predictor in the second model, estimated separately for each state-year using data for all units within a state-year-subject-scale, with grade observations nested within units:

$$\hat{\mu}_{(sy)gu} = \beta_{(sy)0u} + \beta_{(sy)1u}(G - 5.5) + \epsilon_{(sy)gd} \quad (8B.3)$$

$$\beta_{(sy)0u} = \gamma_{(sy)00} + \gamma_{(sy)01} \left(\hat{u}_{(s)00u} + \hat{u}_{(s)01u}(Y - my) \right) + r_{(sy)0u}$$

$$\beta_{(sy)1u} = \gamma_{(sy)10} + \gamma_{(sy)11} \left(\hat{u}_{(s)10u} \right) + r_{(sy)1u}$$

$$\epsilon_{(sy)gu} \sim (0, \zeta_{(sy)gu}); \quad \mathbf{r}_{(sy)u} \sim (0, \boldsymbol{\eta}_{(sy)})$$

Where all terms are defined as above, $\epsilon_{(sy)gu}$ is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean. The residual variance $\boldsymbol{\eta}_{sy}$ is estimated.

From this model, we get the EB posterior means and variances: $\hat{\mathbf{B}}^*_{(sy)u}$ and $\hat{\mathbf{V}}^*_{(sy)u}$. These EB estimates will be centered at the state-subject-scale mean for each year. However, we want the estimates to be centered at the overall state-subject-scale mean across all years to account for shrinkage of the state mean estimate to the national mean. Thus, we adjust the posterior means. To do this, we subtract the state OL estimate (created in Step 8A, Equation 8A.1) and add the state EB estimate. We also construct an adjusted posterior variance matrix by adding the posterior VCV matrix from the state-year model (created in **Step 8A**, Equation 8A.2) to the district-year posterior VCV matrix.

$$\hat{\mathbf{B}}^*_{(y)u} = (\hat{\mathbf{B}}^*_{(sy)u} - \hat{\mathbf{B}}_{(y)s}) + (\hat{\mathbf{B}}^*_{(y)s}) \quad (8B.4)$$

$$\hat{\mathbf{V}}^*_{(y)u} = \hat{\mathbf{V}}^*_{(sy)u}$$

$$\hat{\mathbf{V}}^{*adj}_{(y)u} = \hat{\mathbf{V}}^*_{(sy)u} + \hat{\mathbf{V}}^*_{(y)s}$$

8.C. Subgroup Annual Estimates for All Units

To construct unit estimates for each subgroup, denoted h , we use the data from the state assessments both from subgroups and from the all-student group. For clarity, we show the subgroup as a superscript in the below equations.

We get OL estimates, $\hat{\mathbf{B}}^h_{(y)u}$, VCV matrices, $\hat{\mathbf{V}}^h_{(y)u}$, adjusted VCV matrices, $\hat{\mathbf{V}}^{hadj}_{(y)u}$, for each unit-subgroup-year-subject via GLS following the process described in Step 8B.

To get EB estimates, we first compute the difference in mean scores between group h and the *all* mean in each unit-grade-year (where p_{gyu}^h is the proportion of group h in the unit-grade-year):

$$\begin{aligned}\Delta\hat{\mu}_{gyu}^h &= \hat{\mu}_{gyu}^h - \hat{\mu}_{gyu}^{all} \\ \zeta_{gyu} &= \left(\frac{1 - p_{gyu}^h}{p_{gyu}^h} \right) var(\hat{\mu}_{gyu}^{all})\end{aligned}\tag{8C.1}$$

We will use these differences as the outcome in our HLM models, such that the resulting estimates will be for the *differences* between the subgroup and the all group.

We fit two HLM models. Like the logic outlined above for the all group, we estimate two models to improve the shrinkage of the subgroup estimates. The first model allows us to recover a district-specific annual trend in the differences between the subgroup and the all group, which we use to inform shrinkage of the annual estimates for the subgroup-all differences.

The first model has grades, nested in years, nested in units (data for all states and years are included in one model; models are estimated separately by subject):

$$\begin{aligned}\Delta\hat{\mu}_{gyu}^h &= \beta_{0yu} + \beta_{1yu}(G - 5.5) + \epsilon_{gyu} \\ \beta_{0yu} &= \gamma_{00u} + \gamma_{01u}(Y - 2014) + r_{0yu} \\ \beta_{1yu} &= \gamma_{10u} + r_{1yu} \\ \gamma_{00u} &= \delta_{000} + u_{00u} \\ \gamma_{01u} &= \delta_{010} + u_{01u} \\ \gamma_{10u} &= \delta_{100} + u_{10u} \\ \epsilon_{gyu} &\sim (0, \zeta_{gys}); \quad \mathbf{r}_{yu} \sim (0, \boldsymbol{\eta}); \quad \mathbf{u}_u \sim (0, \boldsymbol{\tau})\end{aligned}\tag{8C.2}$$

Where grade is defined as above, and ϵ_{gyu} is a normally distributed error term with mean zero and known variance equal to the sampling variance of $\Delta\hat{\mu}_{gyu}^h$, ζ_{gys} . The residual variances $\boldsymbol{\eta}$ and $\boldsymbol{\tau}$ are estimated.

From this model, $\hat{\mathbf{u}}_u$ is the average difference between the unit subgroup-all difference parameters and the national subgroup-all difference parameters, across years. We then use $\hat{\mathbf{u}}_u$ to predict the value of the unit parameters in each year in a second HLM model (estimated separately for each year-subject, but including data for all states):

$$\Delta\hat{\mu}_{(y)gu}^h = \beta_{(y)0u} + \beta_{(y)1u}(G - 5.5) + \epsilon_{(y)gu} \quad (8C.3)$$

$$\beta_{(y)0u} = \gamma_{(y)00} + \gamma_{(y)01}(\hat{u}_{00u} + \hat{u}_{01u}(Y - my)) + r_{(y)0u}$$

$$\beta_{(y)1u} = \gamma_{(y)10} + \gamma_{(y)11}(\hat{u}_{10u}) + r_{(y)1u}$$

$$\epsilon_{(y)gu} \sim (0, \zeta_{(y)gu}); \quad \mathbf{r}_u \sim (0, \boldsymbol{\eta}_{(y)})$$

Where all terms are defined as above, $\epsilon_{(y)gu}$ is a normally distributed error term with mean zero and known variance equal to the sampling variance of $\Delta\hat{\mu}_{(y)gu}^h$. The residual variance $\boldsymbol{\eta}_y$ is estimated.

From this model, we get EB estimates of the difference between the subgroup h and the all group, $\hat{\mathbf{B}}_{(y)u}^{\Delta h*}$ and $\hat{\mathbf{V}}_{(y)u}^{\Delta h*}$. To get the state annual-subject subgroup mean estimates, we need to add the resulting EB estimates from the model above to those for all students:

$$\hat{\mathbf{B}}_{(y)u}^{h*} = \hat{\mathbf{B}}_{(y)u}^{\Delta h*} + \hat{\mathbf{B}}_{(y)u}^{all*} \quad (8C.4)$$

$$\hat{\mathbf{V}}_{(y)u}^{h*} = \hat{\mathbf{V}}_{(y)u}^{\Delta h*} + \hat{\mathbf{V}}_{(y)u}^{all*}$$

$$\hat{\mathbf{V}}_{(y)u}^{h*adj} = \hat{\mathbf{V}}_{(y)u}^{\Delta h*} + \hat{\mathbf{V}}_{(y)u}^*$$

8D. National Variance and Reliability Parameters

To get national variance and reliability estimates, we estimate one additional model for each subgroup-year-subject-scale, with grade observations nested within units within states, of the form:

$$\hat{\mu}_{(y)gu} = \beta_{(y)0u} + \beta_{(y)1u}(G - 5.5) + \epsilon_{(y)gu} \quad (8D.1)$$

$$\beta_{(y)0us} = \gamma_{(y)00s} + r_{(y)0us}$$

$$\beta_{(y)1us} = \gamma_{(y)10s} + r_{(y)1us}$$

$$\gamma_{(y)00s} = \delta_{(y)000} + u_{(y)0s}$$

$$\gamma_{(y)10s} = \delta_{(y)100} + u_{(y)1s}$$

$$\epsilon_{(y)gu} \sim (0, \zeta_{(y)gu}); \quad \mathbf{r}_{(y)us} \sim (0, \boldsymbol{\eta}_{(y)}); \quad \mathbf{u}_{(y)s} \sim (0, \boldsymbol{\tau}_{(y)})$$

Where all terms are defined as above, $\epsilon_{(s)gyu}$ is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean. The residual variances $\boldsymbol{\eta}_{(y)s}$ and $\boldsymbol{\tau}_{(y)}$ are estimated.

We take the $\boldsymbol{\tau}_{(y)}$ from Model 8A.2 (the state-level variance from the NAEP data) and add that to the $\boldsymbol{\eta}_{(y)}$ from 8D.1 (the within-state, unit-level variance from the state accountability

data). We report this sum as the national tau for that unit-subgroup-subject-year-scale case. The national reliabilities reported come from model 8D.1.

We use the national tau for the all student group to determine suppression based on imprecision; we suppress $\hat{\beta}_{(y)0u}^h$ if the $se(\hat{\beta}_{(y)0u}^h) < \sqrt{\frac{3}{7}\tau_{(y)00}}$. In other words, we report estimates with reliability greater than 0.7. We do all suppression based on the reliabilities in the CS scale to ensure consistency across scales. We use the standard error threshold determined for all students to censor estimates for every subgroup rather than calculate a subgroup-specific threshold.

Table 11 reports the variance terms from the $\hat{\tau}^2$ matrices (averaged over years) and **Table 12** reports the estimated reliabilities of estimates from the annual models (averaged over years) for all units.

Step 9. Pooled Estimates

We construct pooled estimates separately for the 2008-09 through 2019-19 school years and the 2021-22 through 2024-25 school years. For each unit-subgroup, we have up to 132 subject-grade-year mean estimates (11 years, 6 grades, 2 subjects) for the 2009-2019 period and up to 48 subject-grade-year mean estimates (4 years, 6 grades, 2 subjects) for the 2022-2025 period. We pool these estimates using GLS and precision-weighted random-coefficient models to produce: (1) subject-specific unit-subgroup pooled estimates, pooled over grades and cohorts within subject (“poolsub”); and (2) overall unit-subgroup pooled estimates, pooled over grades, cohorts, and subjects (“pool”). We produce both sets of pooled estimates for all units.

The models differ by geography (state vs. other units) and by group (all students vs. subgroups). We address each in turn below. For simplicity, we only show the models for the subject-specific pooled estimates. In all models shown below, we omit the subject subscript and scale superscript for clarity. For overall pooled estimates, we use data for both subjects and include an indicator that the subject is math, centered at 0.5. Otherwise, the process is identical.

Prior to estimation, we drop cases that have insufficient data for estimation. For subject specific pooled estimates, we flag unit-subgroup-subject cases as insufficient if they contain

fewer than three grades, fewer than two cohorts, or fewer than five cells total. For overall, pooled across subject annual estimates, we flag unit-subgroup cases as insufficient if they contain fewer than three grades, fewer than two cohorts, fewer than two subjects, or fewer than six cases total. [Table 13](#) shows how many cases these drops affect.

9A. State All-Student Pooled Estimates

We use state NAEP data to construct state pooled estimates. Specifically, we use data for grades 4 and 8 from NAEP tested years (e.g., 2009, 2011, ..., 2019, 2022, 2024) and the NAEP estimated year (2025). See [Appendix A2.2 Estimated NAEP 2025 Values](#) for more details on how we predict 2025 NAEP data.

We recover OL pooled estimates and VCV matrices for states via GLS. For the subject-specific estimates, we perform the following calculations using data for each state-year-subject-scale:

$$\begin{aligned}\hat{\mathbf{B}}_s &= (\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1}(\mathbf{X}'\mathbf{W}^{-1}\mathbf{Y}) \\ \hat{\mathbf{V}}_s &= (\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1}\end{aligned}\tag{9A.1}$$

Where $\mathbf{X}$ is a matrix of the predictors in the model including an intercept, centered grade term ($G - 5.5$), centered cohort term (where cohort is defined by year minus grade or $C = Y - G$ and is centered at mc , the middle cohort), and the interaction of the centered grade and a centered year term $((Y - my)(G - 5.5))$; $\mathbf{Y}$ is the vector of NAEP means in the given state and subject, $\hat{\mu}_{gys}$; and $\mathbf{W}$ is a diagonal matrix of their variances, $var(\hat{\mu}_{gys})$. The middle year (my) is 2014 for the 2009 to 2019 period and 2023.5 for the 2022 to 2025 period; the middle cohort (mc) is 2008.5 for the 2009 to 2019 period and 2018 for the 2022 to 2025 period.

Using the same data, we also fit an HLM model by subject-scale with state-grade-year observations, nested in states to get the subject-specific pooled EB estimates:

$$\hat{\mu}_{gys} = \alpha_{gys} + \epsilon_{gys}\tag{9A.2}$$

$$\gamma_{3s} = \delta_{30} + u_{3s}$$

$$\epsilon_{gys} \sim (0, \zeta_{gys}); e_{gys} \sim (0, \sigma); u_s \sim (0, \tau)$$

Where all terms are defined as above, ϵ_{gys} is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean. The residual variances σ and τ are estimated.

Because there are only unique values of grade (4 and 8), we cannot include a squared centered grade term as a predictor in the above models. Our preferred specification for other units (where we include additional values of grade) will have this term. So, we take a few additional computational steps to transform the output to be parallel to what we will do for those units. We first define:

$$\mathbf{D} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (9A.3)$$

Then compute:

$$\begin{aligned} \hat{\mathbf{B}}_s &= \mathbf{D}\hat{\mathbf{G}}_s \\ \hat{\mathbf{V}}_s &= \mathbf{D}\hat{\mathbf{V}}_s\mathbf{D}' \\ \hat{\mathbf{B}}_s^* &= \mathbf{D}\hat{\mathbf{G}}_s^* \\ \hat{\mathbf{V}}_s^* &= \mathbf{D}\hat{\mathbf{V}}_s^*\mathbf{D}' \end{aligned} \quad (9A.4)$$

Where $\hat{\mathbf{G}}_s$ is a matrix of the $\hat{\gamma}_{.s}$, from 9A.2 and $\hat{\mathbf{V}}_s$ is the variance-covariance matrix of $\hat{\mathbf{G}}_s$; $\hat{\mathbf{G}}_s^*$ and $\hat{\mathbf{V}}_s^*$ are the Empirical Bayes shrunk coefficients and their variances. At the end of this step, we have state-subject pooled estimates for each of five parameters: (a) intercept $\hat{\gamma}_{0s}$ (average), (b) centered grade $\hat{\gamma}_{1s}$ (learning rate), (c) centered cohort $\hat{\gamma}_{2s}$ (trend in average), (d) interaction of centered grade and centered year $\hat{\gamma}_{3s}$ (trend in learning rate), and (e) squared centered grade term.

9B. Other Unit All-Student Pooled Estimates

We then use unit-subject-grade-year data for all other units, along with the state pooled-subject estimates from **Step 9A**, to construct unit pooled-subject estimates for all students. Note that we use the unit data *without* the adjusted standard errors that account for the error in

NAEP linking from **Step 5B** as input for these models. We will create a second adjusted standard error after model estimation.

We get OL estimates and VCV matrices ($\hat{\mathbf{B}}_u$ and $\hat{\mathbf{V}}_u$) for unit-year-subjects via GLS. The process is similar to that for states (see Equation 9A.1) except that we include five predictors: intercept, centered grade term, centered cohort term, interaction of centered grade and centered year, and a squared centered grade term. For all units except schools, grade is centered at $mg = \frac{3+8}{2} = 5.5$, the middle grade of our sample. For schools, $mg_n = \frac{\max(\text{grade})_n + \min(\text{grade})_n}{2}$, the middle grade of the school for which we have test score estimates, regardless of whether the school serves additional grades or tested in other grades for which we could not produce estimates. For reference, the grade spans of schools are shown in [Table 14](#). C is defined as above and centered at mc . For all units except schools, cohort is centered at $mc = (my - 5.5)$, the middle cohort of our sample. For schools, we define $mc_n = (my - mg_n)$. Note that we use the same middle year for cohort centering regardless of whether the school was observed over the whole or a subset of the time period. We then construct an adjusted VCV matrix by adding the VCV matrix from the state-year-subject model (created in Step 9A, Equations 9A.2 and 9A.4) to the unit-year-subject VCV matrix:

$$\hat{\mathbf{V}}_u^{adj} = \hat{\mathbf{V}}_u + \hat{\mathbf{V}}_s \quad (9B.1)$$

To construct the EB estimates, we fit the following HLM model (separately within each state):

$$\begin{aligned} \hat{\mu}_{(s)gyu} = & \beta_{(s)0u} + \beta_{(s)1u}(G - mg) + \beta_{(s)2u}(C - mc) \\ & + \beta_{(s)3u}((G - mg)(Y - my)) + \beta_{(s)4u}((G - mg)^2) \\ & + \epsilon_{(s)gyu} \end{aligned} \quad (9B.2)$$

$$\beta_{(s)0u} = \gamma_{(s)00} + r_{(s)0u}$$

$$\beta_{(s)1u} = \gamma_{(s)10} + r_{(s)1u}$$

$$\beta_{(s)2u} = \gamma_{(s)20} + r_{(s)2u}$$

$$\beta_{(s)3u} = \gamma_{(s)30} + r_{(s)3u}$$

$$\beta_{(s)4u} = \gamma_{(s)40} + r_{(s)4u}$$

$$\epsilon_{(s)gyu} \sim (0, \zeta_{(s)gyu}); \quad \mathbf{r}_{(s)u} \sim (0, \boldsymbol{\eta}_{(s)})$$

Where all terms are defined as above, $\epsilon_{(s)gyu}$ is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean. The residual variance $\eta_{(s)}$ is estimated.

From this model, we get the EB posterior means and variances: $\mathbf{B}_{(s)u}^*$ and $\hat{\mathbf{V}}_{(s)u}^*$. However, we want to adjust the posterior means to be centered at the state EB posterior mean from Step 9A to account for shrinkage of the state estimate to the national mean. To do this, we subtract the state OL estimate and add the state EB estimate. We also construct an adjusted posterior variance matrix by adding the posterior VCV matrix from the state-year model (created in **Step 9A**, Equations 9A.2 and 9A.4) to the district-year posterior VCV matrix.

$$\begin{aligned}\hat{\mathbf{B}}_u^* &= (\hat{\mathbf{B}}_{(s)u}^* - \hat{\mathbf{B}}_s) + (\hat{\mathbf{B}}_s^*) \\ \hat{\mathbf{V}}_u^* &= \hat{\mathbf{V}}_{(s)u}^* \\ \hat{\mathbf{V}}_u^{*adj} &= \hat{\mathbf{V}}_{(s)u}^* + \hat{\mathbf{V}}_s^*\end{aligned}\tag{9B.3}$$

9C. Subgroup Pooled Estimates for All Units

To construct unit pooled-subject OL estimates for each subgroup, we follow the process described in **Step 9B**. This returns: $\hat{\mathbf{B}}_u^h$ and $\hat{\mathbf{V}}_u^h$, and $\hat{\mathbf{V}}_u^{hadj}$.

To get EB estimates, we first compute the difference in mean scores between group h and the *all* mean in each unit-grade-year (where p_{gyu}^h is the proportion of group h in the unit-grade-year) as done in 8C.1. We will use these differences as the outcome in our HLM models, such that the resulting estimates will be for the *differences* between the subgroup and the all group.

We then fit an HLM model, separately for each subgroup-subject:

$$\Delta\hat{\mu}_{gyu}^h = \beta_{0u} + \beta_{1u}(G - mg) + \beta_{2u}(C - mc) + \beta_{3u}((G - mg)(Y - my))\tag{9C.1}$$

$$\epsilon_{gyu} \sim (0, \zeta_{gyu}); \quad \mathbf{r}_u \sim (0, \boldsymbol{\tau})$$

Where all parameters are defined as above, ϵ_{gyu} is a normally distributed error term with mean zero and known variance equal to the sampling variance of $\Delta\hat{\mu}_{gyu}^h$ from 8C.1. The residual variance $\boldsymbol{\tau}$ is estimated.

From this model, we get EB estimates of the difference between the subgroup h and the all group, $\hat{\mathbf{B}}_u^{\Delta h*}$ and $\hat{\mathbf{V}}_u^{\Delta h*}$. To get the unit-subject subgroup estimates, we need to add the resulting EB estimates from the model above to those for all students:

$$\begin{aligned} \hat{\mathbf{B}}_u^{h*} &= \hat{\mathbf{B}}_u^{\Delta h*} + \hat{\mathbf{B}}_u^{all*} \\ \hat{\mathbf{V}}_u^{h*} &= \hat{\mathbf{V}}_u^{\Delta h*} + \hat{\mathbf{V}}_u^{all*} \\ \hat{\mathbf{V}}_u^{h*adj} &= \hat{\mathbf{V}}_u^{\Delta h*} + \hat{\mathbf{V}}_u^{*adj} \end{aligned} \tag{9C.2}$$

9D. National Variance and Reliability Parameters

To get national variance and reliability estimates, we estimate a single model for each subgroup-subject-scale of the form:

$$\hat{\mu}_{gyu} = \beta_{0u} + \beta_{1u}(G - mg) + \beta_{2u}(C - mc) + \beta_{3u}(G - mg)(Y - my) + \beta_{4u}((G - mg)^2) + \epsilon_{gyu} \quad (9D.1)$$

$$\beta_{0u} = \gamma_{00s} + r_{0u}$$

$$\beta_{1u} = \gamma_{10s} + r_{1u}$$

$$\beta_{2u} = \gamma_{20s} + r_{2u}$$

$$\beta_{3u} = \gamma_{30s} + r_{3u}$$

$$\beta_{4u} = \gamma_{40s} + r_{4u}$$

$$\gamma_{00s} = \delta_{000} + u_{00s}$$

$$\gamma_{10s} = \delta_{100} + u_{10s}$$

$$\gamma_{20s} = \delta_{200} + u_{20s}$$

$$\gamma_{30s} = \delta_{300} + u_{30s}$$

$$\gamma_{40s} = \delta_{400} + u_{40s}$$

$$\epsilon_{(y)gu} \sim (0, \zeta_{gyu}); \mathbf{r}_{(y)us} \sim (0, \boldsymbol{\eta}); \mathbf{u}_s \sim (0, \boldsymbol{\tau})$$

Where all terms are defined as above, ϵ_{gyu} is a normally distributed error term with mean zero and known variance equal to the sampling variance of the mean. The residual variances $\boldsymbol{\eta}$ and $\boldsymbol{\tau}$ are estimated.

We take the $\boldsymbol{\tau}$ from Model 9A.2 (the state-level variance from the NAEP data) and add that to the $\boldsymbol{\eta}$ from 9D.1 (the within-state, unit-level variance from the state accountability data). We report this sum as the national tau for that unit-subgroup-subject-year-scale case. The national reliabilities reported come from model 9D.1.

We use national taus for the all subgroup to determine suppression based on imprecision. For the intercept, we suppress $\hat{\beta}_{ku}$ if the $se(\hat{\beta}_{ku}) < \sqrt{\frac{3}{7}\tau_{kk}}$. We do all suppression based on the reliabilities in the CS scale to ensure consistency across scales. We use the standard error threshold determined for all students to censor estimates for every subgroup rather than calculate a subgroup-specific threshold.

Tables 15a-h report the variance and covariance terms from the $\hat{\boldsymbol{\tau}}$ matrices and **Tables 16a-d** report the estimated reliabilities of estimates from the pooling models for all units.

Step 10. Aggregating Cross-State Unit Components

For the subset of units where we split the unit into components for pooled HETOP estimation, we need to “re-aggregate” the components into complete unit estimates. (This only affects units from 2009-2019.) The following summary is written for metropolitan areas that cross state lines; however, the same logic can be applied to units serving BIE schools or commuting zones that cross state lines. Suppose there are a set of M metropolitan areas that cross state lines (e.g., have two or more metropolitan area-by-state components). The metropolitan area mean is then estimated as the weighted average of metropolitan area-by-state means across all D_m metropolitan area components in metropolitan area m , computed as

$$\hat{\mu}_{mrygb}^{state} = \sum_{d=1}^{D_m} p_{dm} \hat{\mu}_{mrygb}^{state}, \quad (10.1)$$

where p_{dm} is the proportion of metropolitan area m represented by metropolitan area-by-state component d . The estimated metropolitan area standard deviation is estimated as the square root of the estimated total variance between and within metropolitan area-by-state components in the metropolitan area,

$$\hat{\sigma}_{mrygb}^{state} = \sqrt{\hat{\sigma}_{B_m}^2 + \hat{\sigma}_{W_m}^2} \quad (10.2)$$

where $\hat{\sigma}_{B_m}^2$ is the estimated variance between metropolitan area-by-state components in metropolitan area m and $\hat{\sigma}_{W_m}^2$ is the estimated variance within metropolitan area-by-state components in metropolitan area m . The formulas used to estimate $\hat{\sigma}_{B_m}^2$ and $\hat{\sigma}_{W_m}^2$ are based on equations in Reardon et al. (2017). These formulas and formulas for estimating the standard errors of the metropolitan area means and standard deviations, $\hat{\mu}_{mrygb}^{state}$ and $\hat{\sigma}_{mrygb}^{state}$, are included in [Appendix A1](#).

Step 11. Calculating Achievement Gaps

SEDA 2025.1 does not include any achievement gap estimates. We plan to release these estimates in future iterations. However, interested users can construct these from the subgroup

estimates from any file. The gap will be the difference in the mean estimates; the standard error will be the square root of the sum of the squared standard errors for each group.

Below we show the example calculation for the White-Black gap in a given unit, year, grade, and subject using the long form (grade-subject-year) data; however, the same process can be applied to all subgroup pairs of interest (e.g., male-female, nec-ecd) and types of estimates (e.g., annual or pooled):

$$\widehat{wb}g_{uygb}^x = \hat{\mu}_{u(r=wht)ygb}^x - \hat{\mu}_{u(r=blk)ygb}^x \quad (11.1)$$

The standard errors of the gap are given by:

$$\begin{aligned} se(\widehat{wb}g_{uygb}^x) &= \sqrt{se(\hat{\mu}_{u(r=wht)ygb}^x)^2 + se(\hat{\mu}_{u(r=blk)ygb}^x)^2} \\ se^{adj}(\widehat{wb}g_{uygb}^x) &= \sqrt{se^{adj}(\hat{\mu}_{u(r=wht)ygb}^x)^2 + se^{adj}(\hat{\mu}_{u(r=blk)ygb}^x)^2} \end{aligned} \quad (11.2)$$

Step 12. Suppressing and Flagging Data for Release

Long Form Files. Our agreement with the U.S. Department of Education requires (1) that all reported unit-subgroup-subject-grade-year estimates reflect at least 20 unique students; and (2) that a small amount of random noise is added to each estimate in proportion to the sampling variance of the respective estimate. The added noise is roughly equivalent to randomly removing one student's score from each unit-subgroup-subject-grade-year estimate. These measures are taken to ensure that the raw counts of students in each proficiency category cannot be recovered from published estimates. The random error added to each unit-subgroup estimate is drawn from a normal distribution $\mathcal{N}(0, (1/n) * \hat{\omega}^2)$ where $\hat{\omega}^2$ is the squared estimated standard error of the estimate and n is the number of student assessment outcomes to which the estimate applies. The SEs of the mean are adjusted to account for the additional error. For 2022-2025 estimates based on the public EDC data, we enforce the minimum number of unique students, but we do not add noise to the estimates.

In addition, we remove any imprecise individual estimates where the CS scale standard error is greater than 1. We suppress all subgroup estimates if the mean is missing for the "all"

subgroup. Finally, we suppress estimates whenever the mean, standard deviation, or standard error of either is missing. [Table 17](#) summarizes the cases removed in the long files.

Annual and Pooled Files. For all annual files, our agreement with the U.S. Department of Education requires that all reported estimates (1) reflect at least 20 unique students; and (2) are pooled across at least two subject-grades. For all pooled files, our agreement with the U.S. Department of Education requires that all reported mean estimates (1) reflect at least 20 unique students; and (2) are pooled across at least two subject-year-grades. We enforce these same requirements for 2022-2025 estimates based on EDC data.

We make a number of additional suppressions for data quality:

- We additionally suppress annual and pooled estimates if they are based only on 3rd and 4th or only on 7th and 8th grade data. Our models are trying to estimate the mean in grade 5.5, which means that producing estimates in these cases would require considerable extrapolation.
- We suppress 2009-2019 pooled estimates if they are based only on 2009 through 2011 or only on 2017 through 2019 data. Our models are trying to estimate the mean in year 2014, which means that producing estimates in these cases would require considerable extrapolation.
- We suppress subgroup estimates if the mean is missing for the “all” subgroup.
- For a small number of units, there is insufficient data to recover an OLS estimate or SE for a given parameter. While we can recover EB estimates for these parameters, we do not release them. Moreover, SEDA does not report EB or OLS parameter estimates for a unit when the parameter estimate is imprecise or unreliable. The process for reliability suppression is described in Sections 8D (annual estimates) and 9D (pooled estimates).
- We suppress estimates whenever the parameter or its standard error is missing.
- To ensure consistency across the parameters we report, we suppress the trend in average scores, learning rate, and trend in learning rate if the average test score is missing or suppressed for any reason. We also suppress the trend in learning

rates parameter if either the learning rate or trend in average test scores is missing or suppressed for any reason.

Tables 18 and 19 summarize the cases removed in the annual and pooled files.

II.F. Additional Notes

Gender Estimates. Research reported by Reardon, Kalogrides, et al. (2019) suggests that the magnitude of gender achievement gaps can be impacted by the proportion of test items that are multiple-choice versus constructed response. As a result, differences in gender estimates across states (or across time when a state changes the format of its test) may confound true differences in achievement with differences in the format of the state test used to measure achievement. See Reardon, Fahle, et al. (2019) for a description of an analytic strategy that can be used to adjust for these potential effects.

Estimates for Small Schools and Charter Schools. Research reported in Reardon, Papay, et al. (2019) shows that estimates of student learning rates (the coefficient on the “grade” term in the pooling models in Step 9) are generally unbiased and reliable, except when student mobility in and out of schools is high. In the three states’ data they used, student mobility was higher, on average, in small schools and districts, schools with long grade spans, and in charter schools. In addition, in very small schools and charter schools, the estimated grade slope is biased upwards, as a result of differential mobility (more lower-achieving students leave schools than enter). As a result, we recommend that users interpret the school level grade slopes with some caution, particularly for small schools, schools that span 4 or more of the grades from 3 to 8, and charter schools. Moreover, users should be cautious in comparing grade slopes in charter schools to those of traditional public schools, given the evidence of systematic upward bias in the charter sector estimates.

Data Quality and Validation Checks. Inferences regarding changes in the estimated test scores from 2009 to 2025 hinge on the comparability of the data over time. There are a few potential threats to these inferences:

Data discrepancies. Because state reported proficiency data are the source data for *EDFacts*, the data reported in the two sources should be equivalent (save for differences in data suppression). However, as part of the *EDFacts* data collection process, state proficiency data are vetted and cleaned. As such, it is possible that the 2022-2025 state data are of lower or different quality than the 2009-2019 *EDFacts* data. Additionally, there may be differences in how the data are reported, for example, whether alternate assessments are reported in the count data for each district.

To better understand any potential discrepancies in the data, we cleaned the 2019 state reported data using the same rules (described above) as the 2022-2025 state reported data. We then compared the count data and estimates produced from both sources. Differences in estimated means are associated with differences in the underlying count data at the state and/or district levels (i.e., the two sources reported different proportions of students scoring at the same proficiency level). On average, these differences in the estimated means were small. Only in a small percentage of cases (6.6%) did estimates differ by more than .2 grade levels. More than three-quarters of cases with differences larger than .2 grade levels are in district-subject-grades with fewer than 100 test-takers; only 3% are in cases with more than 500. Thus, while there are differences in the data, they would impact the estimates we report in limited ways for relatively few test-takers.

Notably, as a result of this analysis, we determined that the data reported in *EDFacts* for Arkansas RLA is not comparable to the data reported by the state in either Reading or English. As such RLA data for Arkansas was removed from website and data files.

NAEP Linking. We replicated a subset of the analyses from Reardon, Kalogrides, and Ho (2021) to demonstrate that the NAEP linking continues to perform well in 2022 and 2024.

Comparison to percent proficient data. As a face validity check of the data, for each state-subject we correlated our estimated changes in means (pooled over grades) with changes in the probit transformed reported percent proficient (pooled over grades). Correlations suggest that our estimates tell a story that is largely consistent with that

from the reported percent proficient ($M = .79$, $SD = .10$). We do not expect the two to be identical given that means are not a linear function of probit-transformed proportions above a threshold if the standard deviations of the underlying distributions differ.

Because our data use information from all available cut scores simultaneously, they are preferable to changes in percentages proficient or transformations of such changes.

OLS and EB Estimates. SEDA contains two sets of estimates derived from the annual (Section 8) and the pooling models (Section 9): (1) the OLS estimates and (2) the Empirical Bayes (EB) shrunk estimates. The OLS estimates are unbiased estimates generated via generalized least squares, but they may be noisy in small units.

The EB estimates are based on the fitted HLM models, but they include an EB shrunk residual. The EB estimates are biased toward the grand mean but have statistical properties that make them suited for inclusion as predictor variables or when one is interested in identifying outlier units. We report the square root of the posterior variance of the EB estimates as the standard error of the EB estimate.

In general, the EB estimates (marked as “eb” in the data files) should be used for descriptive purposes and as predictor variables on the right-hand side of a regression model; they are the estimates shown on the website (<https://edopportunity.org>). They should not be used as outcome variables in a regression model because they are shrunk estimates. Doing so may lead to biased parameter estimates in fitted regression models. The OLS estimates (marked as “ol” in the data files) are appropriate for use as outcome variables in a regression model. When using the OLS estimates as outcome variables, we recommend fitting precision-weighted models that account for the known error variance of the OLS estimates.

Replicating the Annual and Pooled Estimates. Notably, we used non-noised long-form estimates to create annual and pooled estimates. Users will not be able to identically replicate our annual or pooled estimates given two differences between the public long files and the ones we used to create annual and pooled estimates: added noise and more dropped estimates (see Step 12 below). However, the results should be very similar.

II.G. How to Obtain Values Shown on edopportunity.org

In order to replicate the values shown on the edopportunity.org website from the downloadable data files (csv/Stata), please refer to the steps below:

1. Determine what geography you would like to focus on and download the corresponding GYS Pooled File: `seda_[geography]_pool_gys_[seda_version]`, where **geography** is *school*, *admindist*, or *state* and where **seda_version** is a number, e.g., *2025.1* for version 2025.
2. To get average test score, do the following calculation: `gys_mn_avg_eb` – `gradecenter`.
3. To get the learning rate, do the following calculation: `gys_mn_lrn_eb` – 1.
4. To get the trend in average scores, use `gys_mn_tav_eb`.
5. To get the trend in learning rates, use `gys_mn_tlr_eb`.
6. In order to get the corresponding Margins of Error (we use a 95% confidence interval) on our site, multiply the adjusted Standard Errors on the variable you would like to use by 1.959964 (the inverse cumulative standard normal distribution of .975). For example, the average test score margin of error is `gys_mn_avg_eb_se_adj` * 1.959964.

III. Covariate Data Construction

SEDA contains CCD and ACS data that have been curated for use with the school, administrative school district, and state achievement data.

III.A. ACS Data and SES Composite Construction

For all units except schools we use data from the ACS to construct measures of median family income, proportion of adults with a bachelor's degree or higher, proportion of adults that are unemployed, the household poverty rate, the proportion of households receiving SNAP benefits, and the proportion of households with children that are headed by a single mother. We also combine these measures to construct a single socioeconomic status composite.

ACS data are available as 5-year pooled samples, from which we use samples from 2005-2009 through 2020-2024. The samples we use here reflect data for the total population of residents in each unit. In select years, district-level tabulations are also available for families who live in each school district in the U.S and who have children enrolled in public school. However, the most recent sample of this data that has all of the information we need is the 5-year 2007-

2011 sample. We prefer to use the total population tabulation data from more recent years. We have compared measures constructed using the total population samples and the relevant children enrolled in public schools samples in years where both samples are available and the measures are highly correlated ($r > 0.99$) and not sensitive to which sample we use.

The construction of our measures derived from the ACS data occurs in a variety of steps, which we describe below. Our derivation of these measures is complicated by the fact that we use the ACS-reported margins of error to compute Empirical Bayes shrunken versions of our key ACS measures. The shrunken measures help account for attenuation bias that results from the fact that smaller units' measures include more measurement error due to smaller sample sizes. [Appendix B2](#) describes the problems of measurement error and attenuation bias in detail. Below we describe the steps we take to create our measures derived from the raw ACS data. Note that we do not compute standard errors or Empirical Bayes shrunken versions of state-level measures. State samples are sufficiently large as to not be concerned about measurement error due to small samples. Therefore, many of the steps described below are not relevant for states.

Step 1: We download and clean the raw ACS data for each year and unit, saving the measures of interest along with their margins of error. We use data from the 2005-2009, 2006-2010, 2007-2011, 2008-2012, 2009-2013, 2010-2014, 2011-2015, 2012-2016, 2013-2017, 2014-2018, 2015-2019, and 2020-2024 samples. In [Appendix B1](#) we provide a list of the raw ACS data tables we downloaded and use to compute each derived measure.

Step 2: Some of our derived measures require combining various fields from ACS in order to compute our desired metric. For example, in order to compute the proportion of adults with a bachelor's degree or higher we sum the number of men with a bachelor's degree, a master's degree or a professional degree with the number of women with a bachelor's degree, a master's degree or a professional degree and divide that sum by the total number of adults in the unit. Each of these component measures is reported with its own margin of error in the raw ACS data. We use the margins of error from each component measure to generate a single standard error for the combined bachelor's degree attainment rate variable (and do the same for all 6 socioeconomic measures that make up the SES composite). [Appendix B3](#) describes our methodology for computing the sampling variance of sums of ACS variables in detail.

Step 3: After constructing the 6 SES measures and their standard errors we impute some missing data using Stata’s **—mi impute chained—** routine, which fills in missing values iteratively by using chained equations. We reshape the data from long (one observation for each unit and race group [all, White, Black, Hispanic, Asian, and Native American] in each year) to wide (one observation for each unit and a separate variable for each of the 6 SES by race measures in each year). We use both the 6 SES measures and their standard errors in the imputation model as well as the total population count in each unit. The imputation model, therefore, includes median income, proportion of adults with a bachelor’s degree or higher, child poverty rate, SNAP receipt rate, single mother headed household rate, and unemployment rate for each race group (all, White, Black, Hispanic, Asian, and Native American) in each of the 12 5-year spans for both the estimates and their standard errors.²⁵ We estimate the imputation model 5 times.

Step 4: Next, we use the imputed data to compute the SES composite. This measure is computed as the first principal component score of the following measures (each standardized): log(median income), percent of adults ages 25 and older with a bachelor’s degree or higher, child poverty rate, SNAP receipt rate, single mother headed household rate, and employment rate for adults ages 16-64. We calculate the component loadings by conducting the analysis in 2008-2012 at the geographic school district level and weighting by geographic school district enrollment. We then use the loadings from this principal component analysis to calculate SES composite values for different subgroups, years and units. Note that only observations without any imputed ACS data are used in the computation of the factor weights. The SES composite computed for each imputed data set and then averaged over imputations.

Table 20 shows the component loadings for the socioeconomic status composite as well as the mean and standard deviation of each measure it includes. The “standardized loadings” indicate the coefficients used to compute the overall geographic school district SES composite score from the 6 standardized indicator variables in 2008-2012, resulting in an SES composite that has an enrollment-weighted mean of 0 and standard deviation of 1 across all geographic

²⁵ Our imputation method varies slightly for metropolitan statistical areas. An imputation model that includes a variable for each group and each year for each of the 6 measures is not feasible for metropolitan statistical areas because it results in more variables than units. Therefore, we impute the metropolitan area data separately by year and only use 5 years of surrounding data, e.g., we impute 2013-2017 data with 2011-2015, 2012-2016, 2013-2017, 2014-2018 and 2015-2019.

school districts in 2008-2012 without any imputed data. The “unstandardized loadings” are rescaled versions of the coefficients that are used to construct an SES composite score from the raw (unstandardized) indicator variables, but which is on the same scale as the standardized SES composite scores.

To provide context for interpreting values of the SES composite, [Table 21](#) reports average values of the indicator variables at different values of the SES composite.

Step 5: The next step is to construct a standard error of the SES composite. We discuss our methodology in detail in [Appendix B4](#).

Step 6: The final step is to do the Empirical Bayes shrinkage of the SES composites as well as for each of the 6 SES measures that go into making the composite. The shrinkage is done using a random effects meta-analysis regression model weighted by the standard error of each measure. Note that we use the same suppression rules described on page 36 to suppress SES estimates with large standard errors. We do not report group specific SES if the reliability of the all group estimate for a given geography is below .70. Furthermore, if either the White or Black/Hispanic/Asian/Native American group specific SES is censored by this first rule then we censor the SES gap between the two groups as well. See page 36 for more details of the censoring rules.

III.B. Common Core of Data Imputation

We use data derived from the CCD from the Longitudinal Imputed School Dataset (LISD).²⁶ Detailed documentation for that dataset is available upon request from sedasupport@stanford.edu. The dataset includes school-level data from the CCD from Fall 1991 to Fall 2024. The CCD is published by the National Center for Education Statistics and includes enrollment counts by race/ethnicity and free or reduced-price lunch eligibility. However, CCD enrollment data are missing in some cases and have data quality issues in others. To address these issues, the LISD replaces missing and implausible race/ethnicity and free or reduced-price

²⁶ At the time of the publication of the SEDA 2025.1 data, we were using a pre-release version of the LISD that includes more years than the public version available at www.edopportunity.org/segregation.

lunch enrollment data via multiple imputation. The imputation model included school-level data from 1991-1992 through 2024-2025 school years. The imputation model includes annual school-level total enrollment, enrollments by race, enrollments by free or reduced-price lunch status, as well as several auxiliary variables (school economic disadvantaged rate, child poverty rate in the school's census tract, percent of students direct-certified as free or reduced-price lunch, and Title 1 status) in years they are available. The imputation is run 10 times, and the values are averaged across datasets. More details are provided in the LSDD data documentation.

IV. Website-Specific Variables

Identification of Similar Places. On our website we report a set of “similar places” for each district and school. These similar places chosen based on the following school or district characteristics, averaged over 2022-2025, as reported in the SEDA covariate files:

- (a) the log of the average grade-level enrollment in grades 3 to 8
- (b) the average proportion of students receiving federal subsidized lunches
- (c) the average proportions of students who identify as Asian, Black, Hispanic, and White in the district
- (d) the average proportions of students in urban schools, in suburban schools, and in town schools
- (e) the average socioeconomic status (SES) of families in the district

Note that the average SES is not available for schools; matching is done without that variable for schools.

The set of similar places are those in the same state as the focal district or school that are most similar (closest in Mahalanobis distance) to it on these 5 dimensions.

Chronic absenteeism. In our downloadable reports for the website, there are graphs containing chronic absenteeism data. These data are provided to our team courtesy of Nat Malkus at the American Enterprise Institute. For more information on these data, see:

<https://www.returntolearntogether.net/>.

V. Versioning and Publication

New or revised data will be posted periodically to the SEDA website. SEDA updates that contain substantially new information are labeled as a new version (e.g., 1.0, 2.0, etc.). Updates that make corrections or minor revisions to previously posted data are labeled as a subsidiary of the current version (e.g., 1.1, 1.2, etc.). When citing any SEDA data set for presentation, publication or use in the field, please include the version number in the citation. All versions of the data will remain archived and available on the SEDA website to facilitate data verification and research replication.

SEDA 2025.1 makes the following additions to data contained in SEDA 2024.3:

- Includes state, administrative school district, and school pooled estimates.
- Includes state and administrative school district annual and long-form estimates for 2025.

SEDA 2025.1 makes the following modifications to the procedures used in SEDA 2024.3:

- Draws on more complete historical (2022-2024) EDC data, as well as new 2025 EDC data.
- Uses new shape files and a new crosswalk to improve tracking of districts from 2009-2025.
- Estimates 2025 NAEP state and national average test scores and uses these for linking states' estimates to a common scale.
- Implements a new pooling model (from SEDA 6.0) to recover average test scores, trends in average test scores, learning rates, and trends in learning rates.
- Uses a different HLM model for producing variances, covariances, and reliabilities.

References

- Education Data Center. (2025). *State assessment data repository (Version 3.0): Data documentation*. Retrieved from https://www.zelma.ai/data_documentation/EDC_technical_documentation_v3.0.pdf
- Reardon, S. F., Fahle, E. M., Kalogrides, D., Podolsky, A., & Zárate, R. C. (2019). Gender Achievement Gaps in U.S. School Districts. *American Educational Research Journal*, 000283121984382. <https://doi.org/10.3102/0002831219843824>
- Reardon, S. F., Ho, A. D., & Min, J. (2026). *Two Watches: Measurement Error Models for Estimating Educational Progress from Discrepant Test Score Trends*. Educational Opportunity Project Working Paper. Retrieved from: https://edopportunity.org/papers/two_watches_reardon_ho_min_2026.pdf
- Reardon, S. F., Kalogrides, D., Fahle, E. M., Podolsky, A., & Zárate, R. C. (2018). The relationship between test item format and gender achievement gaps on math and ELA tests in fourth and eighth grades. *Educational Researcher*, 1–11. <https://doi.org/10.3102/0013189X18762105>
- Reardon, S. F., Kalogrides, D., & Ho, A. D. (2021). Validation Methods for Aggregate-Level Test Scale Linking: A Case Study Mapping School District Test Score Distributions to a Common Scale. *Journal of Educational and Behavioral Statistics*, 107699861987408. <http://doi.org/10.3102/1076998619874089>
- Reardon, S.F., Papay, J.P., Kilbride, T., Strunk, K.O., Cowen, J., An, L., & Donohue, K. (2019). Can Repeated Aggregate Cross-Sectional Data Be Used to Measure Average Student Learning Rates? A Validation Study of Learning Rate Measures in the Stanford Education Data Archive. (CEPA Working Paper No.19-08). Retrieved from Stanford Center for Education Policy Analysis: <http://cepa.stanford.edu/wp19-08>
- Reardon, S. F., Shear, B. R., Castellano, K. E., & Ho, A. D. (2017). Using heteroskedastic ordered probit models to recover moments of continuous test score distributions from coarsened data. *Journal of Educational and Behavioral Statistics*, 42(1), 3–45. <https://doi.org/10.3102/1076998616666279>

Shear, B. R., & Reardon, S. F. (2021). Using Pooled Heteroskedastic Ordered Probit Models to Improve Small-Sample Estimates of Latent Test Score Distributions. *Journal of Educational and Behavioral Statistics*, 46(1), 3–33. <https://doi.org/10.3102/1076998620922919>

Tables

Table 1. Test Score Files

File Name	Form	Metric	Unit			Disaggregated by			Subgroups			
			School	Administrative District	State	Year	Grade	Subject	Means			
									All	Race	Gender	ECD
seda_admindist_annualsub_cs_2025.1	Annual by Subject	CS		X		X		X	X	X	X	X
seda_admindist_annualsub_gcs_2025.1	Annual by Subject	GCS		X		X		X	X	X	X	X
seda_admindist_annualsub_gys_2025.1	Annual by Subject	GYS		X		X		X	X	X	X	X
seda_admindist_annualsub_ys_2025.1	Annual by Subject	YS		X		X		X	X	X	X	X
seda_admindist_annual_cs_2025.1	Annual	CS		X		X			X	X	X	X
seda_admindist_annual_gcs_2025.1	Annual	GCS		X		X			X	X	X	X
seda_admindist_annual_gys_2025.1	Annual	GYS		X		X			X	X	X	X
seda_admindist_annual_ys_2025.1	Annual	YS		X		X			X	X	X	X
seda_admindist_long_cs_2025.1	Long	CS		X		X	X	X	X	X	X	X
seda_admindist_long_gcs_2025.1	Long	GCS		X		X	X	X	X	X	X	X
seda_admindist_long_gys_2025.1	Long	GYS		X		X	X	X	X	X	X	X
seda_admindist_long_ys_2025.1	Long	YS		X		X	X	X	X	X	X	X
seda_admindist_poolsb_cs_2025.1	Pooled by subject	CS		X				X	X	X	X	X
seda_admindist_poolsb_gcs_2025.1	Pooled by subject	GCS		X				X	X	X	X	X
seda_admindist_poolsb_gys_2025.1	Pooled by subject	GYS		X				X	X	X	X	X
seda_admindist_poolsb_ys_2025.1	Pooled by subject	YS		X				X	X	X	X	X
seda_state_annualsub_cs_2025.1	Annual by Subject	CS			X	X		X	X	X	X	X
seda_state_annualsub_gcs_2025.1	Annual by Subject	GCS			X	X		X	X	X	X	X
seda_state_annualsub_gys_2025.1	Annual by Subject	GYS			X	X		X	X	X	X	X
seda_state_annualsub_ys_2025.1	Annual by Subject	YS			X	X		X	X	X	X	X
seda_state_annual_cs_2025.1	Annual	CS			X	X			X	X	X	X
seda_state_annual_gcs_2025.1	Annual	GCS			X	X			X	X	X	X
seda_state_annual_gys_2025.1	Annual	GYS			X	X			X	X	X	X
seda_state_annual_ys_2025.1	Annual	YS			X	X			X	X	X	X
seda_state_long_cs_2025.1	Long	CS			X	X	X	X	X	X	X	X
seda_state_long_gcs_2025.1	Long	GCS			X	X	X	X	X	X	X	X
seda_state_long_gys_2025.1	Long	GYS			X	X	X	X	X	X	X	X
seda_state_long_ys_2025.1	Long	YS			X	X	X	X	X	X	X	X
seda_state_pool_cs_2025.1	Pooled	CS			X				X	X	X	X
seda_state_pool_gcs_2025.1	Pooled	GCS			X				X	X	X	X
seda_state_pool_gys_2025.1	Pooled	GYS			X				X	X	X	X
seda_state_pool_ys_2025.1	Pooled	YS			X				X	X	X	X
seda_state_poolsb_cs_2025.1	Pooled by subject	CS			X			X	X	X	X	X
seda_state_poolsb_gcs_2025.1	Pooled by subject	GCS			X			X	X	X	X	X
seda_state_poolsb_gys_2025.1	Pooled by subject	GYS			X			X	X	X	X	X
seda_state_poolsb_ys_2025.1	Pooled by subject	YS			X			X	X	X	X	X
seda_school_pool_cs_2025.1	Pooled	CS	X						X			
seda_school_pool_gcs_2025.1	Pooled	GCS	X						X			
seda_school_pool_gys_2025.1	Pooled	GYS	X						X			
seda_school_pool_ys_2025.1	Pooled	YS	X						X			
seda_school_poolsb_cs_2025.1	Pooled by subject	CS	X					X	X			
seda_school_poolsb_gcs_2025.1	Pooled by subject	GCS	X					X	X			
seda_school_poolsb_gys_2025.1	Pooled by subject	GYS	X					X	X			
seda_school_poolsb_ys_2025.1	Pooled by subject	YS	X					X	X			

Metric: CS = Cohort Scale; YS = Year Scale; GCS = Grade Cohort Scale; GYS = Grade Year Scale

Unit: Admindist = Administrative School District

Academic Years: 2008/09 – 2018/19 & 2022/25

Grades: 3 – 8

Subjects: Math, RLA

Race: Asian, Black, Hispanic, Native American, White

Gender: male, female

ECD: economically disadvantaged, not disadvantaged (as defined by states)

Table 2. Covariate Data Files

Covariate Data				
File Name	Form	Disaggregated by		
		Unit	Year	Grade
seda_cov_admindist_long_2025.1	Long	X	X	X
seda_cov_admindist_annual_2025.1	Annual	X	X	
seda_cov_admindist_pool_2025.1	Pool	X		
seda_cov_state_long_2025.1	Long	X	X	X
seda_cov_state_annual_2025.1	Annual	X	X	
seda_cov_state_pool_2025.1	Pool	X		
seda_cov_school_pool_2025.1	Pool	X		

Table 3. Example *EDFacts* Data Structure

FIPS	NCESSCH	Subgroup	Subject	Grade	Year	Number of students scoring at...			
						Level 1	Level 2	Level 3	Level 4
99	997777755555	All Students	Math	3	2009	26	87	185	32
99	997777755555	All Students	RLA	3	2009	13	102	195	20
99	997777755556	All Students	Math	3	2009	35	238	192	7
99	997777755556	All Students	RLA	3	2009	7	278	187	0

Note. The data shown in this table are not real.

Table 4a. Availability of SEDA Estimates by State-Subject-Year-Grade

[illegible]

Table 4a. Availability of SEDA Estimates by State-Subject-Year-Grade (Continued)

[illegible]

Table 4a. Availability of SEDa Estimates by State-Subject-Year-Grade (Continued)

	2009			2010			2011			2012			2013			2014			2015			2016			2017			2018			2019			2022			2023			2024			2025											
	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8
Michigan																																																						
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Minnesota																																																						
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Mississippi																																																						
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Missouri																																																						
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Montana																																																						
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Nebraska																																																						
Math																																																						
RLA																																																						
Nevada																																																						
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
New Hampshire																																																						

Table 4a. Availability of SEDA Estimates by State-Subject-Year-Grade (Continued)

	2009		2010		2011		2012		2013		2014		2015		2016		2017		2018		2019		2022		2023		2024		2025								
	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	3	4	5	6	7	8	
North Carolina																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
North Dakota																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Ohio																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Oklahoma																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Oregon																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Pennsylvania																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Rhode Island																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
South Carolina																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
South Dakota																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Tennessee																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Texas																																					
Math	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
RLA	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	

Table 4a. Availability of SEDA Estimates by State-Subject-Year-Grade (Continued)

[illegible]

Table 4b. Reason for Unavailability of SEDA Estimates by State-Subject-Year-Grade

[illegible]

Table 4b. Reason for Unavailability of SEDA Estimates by State-Subject-Year-Grade (Continued)

[illegible]

Table 4b. Reason for Unavailability of SEDA Estimates by State-Subject-Year-Grade (Continued)

[illegible]

Table 4b. Reason for Unavailability of SEDA Estimates by State-Subject-Year-Grade (Continued)

[illegible]

Table 5. Double Counting of Bureau of Indian Education Schools

NCESSCH Dropped from ED<i>Facts</i>	School Name	NCESSCH Kept in SEDA	Note
590002500172	TURTLE MOUNTAIN COMMUNITY MIDDLE SCHOOL	380253000751	CCD Task Force
590010600170	TURTLE MOUNTAIN COMMUNITY ELEMENTARY SCHOOL	380253000750	CCD Task Force
590007700173	TWIN BUTTES ELEMENTARY SCHOOL	381860000757	CCD Task Force
590011700174	WHITE SHIELD ELEMENTARY SCHOOL	381968000807	CCD Task Force
590018700086	NAH TAH WAHSH PUBLIC SCHOOL ACADEMY	260010300646	CCD Task Force
590005400080	JOSEPH K LUMSDEN BAHWETING ANISHNABE ACADEMY	260007100492	CCD Task Force
590018900167	MANDAREE ELEMENTARY SCHOOL	381185000747	SEDA Team
590018900167	MANDAREE HIGH SCHOOL	381185000006	SEDA Team
230006400664	INDIAN TOWNSHIP SCHOOL	590004200052	SEDA Team
370015302953	CHEROKEE ELEMENTARY	590006600044	SEDA Team
370015303150	CHEROKEE MIDDLE SCHOOL	590010100045	SEDA Team
230006900630	BEATRICE RAFFERTY SCHOOL	590013700042	SEDA Team
230006600671	INDIAN ISLAND SCHOOL	590016000051	SEDA Team

Note. CCD Task Force indicates the school was listed in the CCD's Documentation (Double Counting of Bureau of Indian Education Schools). SEDA Team indicates that it was determined by looking at school coordinates, assessments received by grade, and school contact information since the CCD task force only began with SY 2009-10.

Table 6a. SBAC States and Years Used to Create BIE Waiver Cutscores

	2014	2015	2016	2017	2018	2019
California		X	X	X	X	X
Connecticut		X	X	X	X	X
Delaware		X	X	X	X	X
Hawaii	X	X	X	X	X	X
Idaho		X	X	X	X	X
Maine		X				
Michigan		X	X	X		
Missouri		X				
Montana		X	X	X	X	X
Nevada		X	X	X		
New Hampshire		X	X	X	X	X
North Dakota		X	X	X		
Oregon		X	X	X	X	X
South Dakota		X	X	X	X	X
Vermont		X	X	X		
Washington		X	X	X	X	X
West Virginia		X	X	X	X	

Table 6b. PARCC States and Years Used to Create BIE Waiver Cutscores

	2014	2015	2016	2017	2018	2019
Colorado		X	X	X		
DC		X	X	X	X	X
Illinois		X	X	X	X	
Maryland		X	X	X	X	X
Mississippi		X				
New Jersey		X	X	X	X	X
New Mexico					X	
Rhode Island		X	X	X		

Table 7. BIE Schools Dropped from SEDA

State Abbreviation	Year	NCES School ID	School Name
MS	2016	590005300056	Standing Pine Elementary School
MS	2017	590005300056	Standing Pine Elementary School
MS	2016	590007800048	Choctaw Central Middle School
MS	2017	590007800048	Choctaw Central Middle School
MS	2016	590010000057	Tucker Elementary School
MS	2017	590010000057	Tucker Elementary School
MS	2016	590011100050	Conehatta Elementary School
MS	2017	590011100050	Conehatta Elementary School
MS	2016	590012300054	Pearl River Elementary School
MS	2017	590012300054	Pearl River Elementary School
MS	2016	590017200055	Red Water Elementary School
MS	2017	590017200055	Red Water Elementary School
MS	2016	590019400043	Bogue Chitto Elementary School
MS	2017	590019400043	Bogue Chitto Elementary School
WI	2012	590010400087	Lac Courte Oreilles Ojibwa School
WI	2012	590011400090	Oneida Nation School
WI	2012	590014400088	Menominee Tribal School

Note. Year indicates the spring of the school year.

Table 8. Subject-Grade-Year-Subgroup Cases Removed Pre-Estimation

2009-2019	sedaadmin		sedafips		sedasch	
No testscores	586,306	4.05%	1,234	2.09%	93,706	1.95%
Only 1 testscore	1,110,915	7.66%	78	0.13%	78,286	1.63%
All participation < 94%	394,277	2.72%			159,734	3.32%
Subgroup participation < 94%	176,084	1.21%	680	1.15%		
Representation < 95%	72,761	0.50%	261	0.44%		
Alternative assessments > 20%	71,289	0.49%			40,571	0.84%
No sufficient cells	212,239	1.46%	295	0.50%	35,255	0.73%
Not dropped	11,870,085	81.90%	56,619	95.69%	4,405,284	91.53%
Total	14,493,956	100.00%	59,167	100.00%	4,812,836	100.00%

2022-2025	sedaadmin		sedafips		sedasch	
No testscores	20,644	0.71%			196	0.01%
Only 1 testscore	14,066	0.49%			915	0.07%
All participation < 94%	66,464	2.29%			34,774	2.48%
Subgroup participation < 94%	35,998	1.24%	192	1.30%		
Representation < 95%	380	0.01%				
Alternative assessments > 20%	66,285	2.29%	310	2.10%	15,488	1.10%
No sufficient cells	63	0.00%				
Not dropped	2,692,692	92.96%	14,243	96.60%	1,353,103	96.34%
Total	2,896,592	100.00%	14,745	100.00%	1,404,476	100.00%

Note: sedafips = State; sedaadmin = Administrative District; sedasch = School

Table 9. NAEP Means and Standard Deviations by Year and Grade.

	Grade	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2022	2023	2024	2025
		Math																
Means	3	227.8	227.7	227.6	228.1	228.6	229.2	229.8	229.3	228.8	228.2	227.7	228.1	228.6	224.7	226.3	227.9	227.9
	4	237.8	238.0	238.1	238.7	239.2	239.8	240.4	239.8	239.1	238.7	238.3	238.6	238.9	234.2	235.4	236.5	236.7
	5	247.9	248.2	248.6	249.2	249.7	250.4	251.0	250.2	249.4	249.2	249.0	249.0	249.1	243.7	244.5	245.2	245.6
	6	257.9	258.5	259.1	259.7	260.3	260.9	261.6	260.7	259.8	259.7	259.6	259.5	259.4	253.3	253.6	253.9	254.5
	7	268.0	268.8	269.6	270.2	270.8	271.5	272.1	271.1	270.1	270.2	270.2	269.9	269.6	262.8	262.7	262.6	263.3
	8	278.0	279.1	280.1	280.8	281.4	282.1	282.7	281.6	280.4	280.6	280.9	280.4	279.9	272.3	271.8	271.3	272.2
SDs	3	27.9	27.9	27.8	27.9	27.9	28.2	28.6	28.7	28.8	29.6	30.5	30.7	31.0	32.5	33.0	33.5	33.7
	4	29.9	29.8	29.8	29.8	29.7	30.0	30.3	30.4	30.5	31.4	32.3	32.6	32.9	33.9	34.6	35.3	35.6
	5	31.9	31.8	31.7	31.6	31.6	31.8	32.0	32.1	32.3	33.2	34.1	34.5	34.8	35.2	36.2	37.1	37.6
	6	33.8	33.8	33.7	33.5	33.4	33.5	33.7	33.8	34.0	35.0	35.9	36.3	36.8	36.6	37.8	38.9	39.5
	7	35.8	35.7	35.6	35.4	35.2	35.3	35.4	35.6	35.8	36.8	37.8	38.2	38.7	38.0	39.4	40.8	41.5
	8	37.8	37.7	37.6	37.3	37.1	37.1	37.1	37.3	37.5	38.5	39.6	40.1	40.6	39.3	41.0	42.6	43.4
		RLA																
Means	3	205.9	206.1	206.2	206.5	206.8	207.3	207.7	208.5	209.3	208.8	208.4	207.9	207.5	204.1	203.3	202.6	202.1
	4	216.3	216.7	217.0	217.4	217.8	218.5	219.1	219.6	220.0	219.8	219.5	218.8	218.1	214.9	214.0	213.2	213.0
	5	226.7	227.3	227.8	228.3	228.8	229.7	230.5	230.6	230.8	230.7	230.6	229.7	228.7	225.6	224.7	223.7	223.9
	6	237.2	237.9	238.6	239.2	239.8	240.9	241.9	241.7	241.5	241.6	241.8	240.6	239.4	236.4	235.4	234.3	234.8
	7	247.6	248.5	249.3	250.0	250.7	252.1	253.4	252.8	252.3	252.6	252.9	251.4	250.0	247.2	246.0	244.9	245.6
	8	258.0	259.1	260.1	260.9	261.7	263.3	264.8	263.9	263.0	263.5	264.0	262.3	260.6	257.9	256.7	255.5	256.5
SDs	3	39.1	38.6	38.1	38.2	38.4	38.8	39.2	38.9	38.6	39.0	39.5	39.7	39.9	41.9	42.2	42.4	42.8
	4	38.7	38.2	37.7	37.8	37.9	38.2	38.4	38.2	38.0	38.5	39.0	39.4	39.8	41.3	41.8	42.2	42.6
	5	38.4	37.9	37.4	37.4	37.4	37.5	37.6	37.5	37.4	38.0	38.5	39.1	39.7	40.7	41.3	42.0	42.3
	6	38.0	37.5	37.0	36.9	36.9	36.9	36.9	36.9	36.9	37.4	38.0	38.8	39.5	40.0	40.9	41.8	42.1
	7	37.6	37.1	36.6	36.5	36.3	36.2	36.1	36.2	36.3	36.9	37.4	38.4	39.4	39.4	40.5	41.6	41.8
	8	37.3	36.8	36.3	36.0	35.8	35.5	35.3	35.5	35.8	36.4	36.9	38.1	39.3	38.8	40.1	41.4	41.6

Note. We use the expanded population estimates, which may differ slightly from those reported publicly on the website.

Table 10. Pooltype-Year-Subgroup Cases Removed Pre-Estimation for Annual Estimates

2009-2019	sedaadmin		sedafips	
Pooltype				
Overall				
Insufficient data	94,917	7.56%	121	2.56%
Not dropped	1,161,272	92.44%	4,600	97.44%
Math				
Insufficient data	78,260	6.38%	30	0.64%
Not dropped	1,148,351	93.62%	4,625	99.36%
RLA				
Insufficient data	72,633	5.87%	29	0.62%
Not dropped	1,164,741	94.13%	4,653	99.38%
2022-2025	sedaadmin		sedafips	
Pooltype				
Overall				
Insufficient data	18,655	6.44%	51	3.90%
Not dropped	271,026	93.56%	1,256	96.10%
Math				
Insufficient data	14,200	5.01%	19	1.46%
Not dropped	269,414	94.99%	1,279	98.54%
RLA				
Insufficient data	12,798	4.51%	3	0.24%
Not dropped	270,807	95.49%	1,264	99.76%

Note: sedafips = State; sedaadmin = Administrative District

Table 11. Administrative District Variances (Annual Estimates)

		2009-2019			2022-2025		
		Overall tau(avg)	Math tau(avg)	RLA tau(avg)	Overall tau(avg)	Math tau(avg)	RLA tau(avg)
CS Scale	all	0.1474	0.1639	0.1384	0.1796	0.2074	0.1622
	asn	0.1741	0.1972	0.1584	0.2560	0.3097	0.2202
	blk	0.1069	0.1135	0.0994	0.1271	0.1397	0.1184
	ecd	0.0900	0.1023	0.0811	0.1030	0.1209	0.0907
	fem	0.1375	0.1488	0.1331	0.1732	0.1966	0.1612
	hsp	0.1015	0.1056	0.0980	0.1301	0.1411	0.1228
	mal	0.1477	0.1665	0.1344	0.1841	0.2173	0.1610
	nam	0.1067	0.1085	0.0977			
	nec	0.1128	0.1281	0.0986	0.1409	0.1652	0.1238
	wht	0.1090	0.1216	0.0990	0.1316	0.1503	0.1210
GCS Scale	all	1.5421	1.6719	1.5645	1.8673	2.1062	1.8335
	asn	1.8224	2.0154	1.7907	2.6603	3.1589	2.4901
	blk	1.1183	1.1595	1.1237	1.3114	1.4206	1.3382
	ecd	0.9401	1.0435	0.9174	1.0648	1.2280	1.0254
	fem	1.4414	1.5189	1.5053	1.8024	1.9997	1.8223
	hsp	1.0636	1.0792	1.1080	1.3490	1.4350	1.3887
	mal	1.5441	1.6982	1.5196	1.9130	2.2091	1.8204
	nam	1.1169	1.1072	1.1061			
	nec	1.1814	1.3105	1.1149	1.4689	1.6845	1.3989
	wht	1.1417	1.2421	1.1192	1.3747	1.5315	1.3679

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 12. Administrative District Reliabilities (Annual Estimates)

		2009-2019			2022-2025		
		Overall rel(avg)	Math rel(avg)	RLA rel(avg)	Overall rel(avg)	Math rel(avg)	RLA rel(avg)
CS Scale	all	0.963	0.941	0.933	0.975	0.957	0.953
	asn	0.803	0.729	0.703	0.931	0.894	0.876
	blk	0.806	0.720	0.709	0.909	0.846	0.850
	ecd	0.908	0.860	0.838	0.930	0.887	0.874
	fem	0.942	0.905	0.897	0.961	0.933	0.927
	hsp	0.807	0.717	0.710	0.921	0.870	0.865
	mal	0.942	0.908	0.895	0.961	0.933	0.926
	nam	0.700	0.582	0.569			
	nec	0.917	0.877	0.852	0.945	0.912	0.896
	wht	0.927	0.890	0.873	0.953	0.922	0.914
GCS Scale	all	0.963	0.941	0.933	0.975	0.958	0.953
	asn	0.802	0.730	0.703	0.931	0.896	0.877
	blk	0.805	0.722	0.710	0.909	0.847	0.850
	ecd	0.907	0.861	0.839	0.929	0.888	0.875
	fem	0.942	0.906	0.897	0.961	0.934	0.927
	hsp	0.807	0.717	0.711	0.921	0.870	0.866
	mal	0.942	0.908	0.895	0.961	0.934	0.926
	nam	0.699	0.583	0.569			
	nec	0.917	0.878	0.852	0.945	0.913	0.897
	wht	0.927	0.891	0.873	0.953	0.923	0.914

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 13. Pooltype-Subgroup Cases Removed Pre-Estimation for Pooled Estimates

2009-2019	sedaadmin		sedafips		sedasch	
Pooltype						
Overall						
Insufficient data	10,512	7.54%			13,438	16.28%
Not dropped	128,839	92.46%	459	100.00%	69,107	83.72%
Math						
Insufficient data	9,020	6.64%	9	1.96%	14,677	18.18%
Not dropped	126,842	93.36%	450	98.04%	66,070	81.82%
RLA						
Insufficient data	7,900	5.79%			16,880	20.68%
Not dropped	128,613	94.21%	459	100.00%	64,744	79.32%
2022-2025	sedaadmin		sedafips		sedasch	
Pooltype						
Overall						
Insufficient data	5,382	6.76%	14	3.99%	12,173	19.16%
Not dropped	74,223	93.24%	337	96.01%	51,357	80.84%
Math						
Insufficient data	6,688	8.53%	14	3.99%	15,776	25.31%
Not dropped	71,682	91.47%	337	96.01%	46,545	74.69%
RLA						
Insufficient data	5,157	6.61%	1	0.30%	13,837	22.64%
Not dropped	72,864	93.39%	337	99.70%	47,288	77.36%

Note: sedafips = State; sedaadmin = Administrative District; sedasch = School

Table 14. Grade Spans of Test Score Estimates by School

2009 -	High Grade					
2019	3	4	5	6	7	8
Low Grade						
3	1,376	3,375	27,136	13,665	841	11,075
4		25	304	450	78	704
5			66	521	125	2,748
6				202	157	13,396
7					84	5,318
8						900
2022 -	High Grade					
2025	3	4	5	6	7	8
Low Grade						
3	1,982	3,383	24,867	8,070	603	6,575
4		42	344	342	46	276
5			54	500	220	1,494
6				120	534	10,253
7					399	3,177
8						249

Table 15a. Administrative District Variances and Covariances (Pooled Estimates – CS Scale – 2009-2019)

	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	cov (avg,lrn)	cov (avg,tav)	cov (avg,tlr)	cov (lrn,tav)	cov (lrn,tlr)	cov (tav,tlr)
Overall										
all	0.15908	0.00226	0.00086	0.00006	0.00133	0.00147	0.00010	0.00069	-0.00001	-0.00003
asn	0.16005	0.00210	0.00138	0.00009	0.00240	0.00115	0.00036	0.00087	0.00003	0.00000
blk	0.10353	0.00240	0.00123	0.00009	0.00139	0.00101	0.00026	0.00090	-0.00006	-0.00004
ecd	0.09427	0.00220	0.00087	0.00006	0.00024	0.00053	0.00013	0.00068	-0.00002	-0.00004
fem	0.14424	0.00200	0.00080	0.00006	0.00165	0.00132	0.00015	0.00062	-0.00001	-0.00003
hsp	0.09711	0.00227	0.00103	0.00008	0.00060	0.00021	0.00025	0.00074	-0.00005	-0.00003
mal	0.15622	0.00222	0.00086	0.00006	0.00135	0.00168	0.00011	0.00068	0.00000	-0.00003
nam	0.10429	0.00267	0.00161	0.00014	0.00092	0.00131	0.00054	0.00103	-0.00002	-0.00004
nec	0.12732	0.00212	0.00080	0.00006	0.00150	0.00095	0.00015	0.00066	0.00001	-0.00003
wht	0.11359	0.00192	0.00068	0.00005	0.00171	0.00103	0.00009	0.00055	0.00001	-0.00003
Math										
all	0.17284	0.00316	0.00111	0.00009	0.00256	0.00207	0.00010	0.00094	0.00003	-0.00003
asn	0.18021	0.00237	0.00135	0.00008	0.00346	0.00190	0.00031	0.00094	0.00004	0.00001
blk	0.11050	0.00276	0.00133	0.00009	0.00204	0.00122	0.00024	0.00103	-0.00003	-0.00004
ecd	0.10578	0.00292	0.00109	0.00008	0.00074	0.00068	0.00015	0.00087	0.00001	-0.00003
fem	0.15299	0.00282	0.00103	0.00008	0.00277	0.00178	0.00017	0.00084	0.00003	-0.00003
hsp	0.10126	0.00293	0.00117	0.00009	0.00120	0.00060	0.00027	0.00092	-0.00002	-0.00002
mal	0.17448	0.00295	0.00108	0.00008	0.00281	0.00256	0.00012	0.00089	0.00003	-0.00003
nam	0.10839	0.00295	0.00155	0.00013	0.00156	0.00177	0.00035	0.00110	-0.00002	-0.00004
nec	0.14204	0.00291	0.00100	0.00008	0.00236	0.00140	0.00017	0.00086	0.00005	-0.00002
wht	0.12559	0.00272	0.00089	0.00008	0.00202	0.00125	0.00011	0.00076	0.00005	-0.00003
RLA										
all	0.15262	0.00197	0.00068	0.00005	0.00060	0.00104	0.00015	0.00056	0.00001	-0.00001
asn	0.14740	0.00171	0.00113	0.00005	0.00158	0.00024	0.00025	0.00074	0.00004	0.00003
blk	0.10016	0.00197	0.00093	0.00005	0.00113	0.00092	0.00018	0.00071	-0.00004	-0.00003
ecd	0.08741	0.00193	0.00068	0.00005	-0.00021	0.00040	0.00016	0.00055	-0.00001	-0.00002
fem	0.14376	0.00167	0.00063	0.00004	0.00094	0.00095	0.00016	0.00050	0.00001	-0.00001
hsp	0.09721	0.00192	0.00080	0.00005	-0.00002	-0.00011	0.00021	0.00056	-0.00001	-0.00001
mal	0.14483	0.00193	0.00068	0.00005	0.00063	0.00106	0.00016	0.00055	0.00001	-0.00001
nam	0.10316	0.00194	0.00121	0.00008	0.00020	0.00092	0.00056	0.00074	-0.00001	0.00001
nec	0.11562	0.00175	0.00064	0.00004	0.00138	0.00073	0.00016	0.00054	0.00003	-0.00001
wht	0.10692	0.00161	0.00055	0.00004	0.00163	0.00090	0.00012	0.00046	0.00002	-0.00001

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 15b. Administrative District Variances and Covariances (Pooled Estimates – GCS Scale – 2009-2019)

					cov	cov	cov	cov	cov	cov
	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	(avg,lrn)	(avg,tav)	(avg,tlr)	(lrn,tav)	(lrn,tlr)	(tav,tlr)
Overall										
all	1.69346	0.02580	0.00898	0.00062	0.05851	0.01528	0.00084	0.00757	0.00014	-0.00002
asn	1.70311	0.02466	0.01448	0.00100	0.06724	0.01269	0.00408	0.00927	0.00068	0.00057
blk	1.10218	0.02691	0.01283	0.00088	0.04335	0.01064	0.00260	0.00955	-0.00028	0.00006
ecd	1.00403	0.02415	0.00911	0.00064	0.02943	0.00549	0.00123	0.00716	0.00001	-0.00006
fem	1.53656	0.02328	0.00830	0.00060	0.05782	0.01371	0.00139	0.00675	0.00016	-0.00002
hsp	1.03696	0.02490	0.01074	0.00084	0.03335	0.00238	0.00225	0.00761	-0.00022	0.00010
mal	1.66253	0.02537	0.00897	0.00064	0.05743	0.01777	0.00100	0.00748	0.00022	0.00002
nam	1.11259	0.02931	0.01683	0.00147	0.03719	0.01445	0.00595	0.01105	0.00030	0.00032
nec	1.35505	0.02441	0.00838	0.00059	0.05290	0.01012	0.00147	0.00718	0.00038	0.00002
wht	1.20937	0.02223	0.00711	0.00055	0.05104	0.01079	0.00098	0.00600	0.00030	-0.00004
Math										
all	1.75230	0.04294	0.01121	0.00088	0.13172	0.02105	0.00253	0.01084	0.00122	0.00062
asn	1.82182	0.03617	0.01378	0.00084	0.14759	0.01946	0.00424	0.01069	0.00136	0.00115
blk	1.12175	0.03584	0.01343	0.00087	0.08962	0.01286	0.00338	0.01114	0.00063	0.00070
ecd	1.07449	0.03560	0.01094	0.00081	0.07141	0.00713	0.00208	0.00921	0.00095	0.00053
fem	1.55110	0.03905	0.01036	0.00080	0.12204	0.01829	0.00304	0.00965	0.00115	0.00058
hsp	1.02715	0.03637	0.01181	0.00085	0.07462	0.00634	0.00309	0.00962	0.00072	0.00071
mal	1.76975	0.04104	0.01086	0.00079	0.13522	0.02603	0.00302	0.01059	0.00116	0.00057
nam	1.10109	0.03717	0.01569	0.00125	0.08163	0.01852	0.00487	0.01228	0.00100	0.00095
nec	1.44179	0.03896	0.01012	0.00079	0.11072	0.01433	0.00272	0.00968	0.00132	0.00058
wht	1.27520	0.03590	0.00897	0.00076	0.09670	0.01272	0.00206	0.00847	0.00118	0.00044
RLA										
all	1.71223	0.02239	0.00771	0.00052	-0.00872	0.01164	0.00153	0.00621	0.00003	-0.00022
asn	1.65357	0.01915	0.01273	0.00057	0.00273	0.00248	0.00278	0.00831	0.00028	0.00017
blk	1.12527	0.02225	0.01057	0.00062	0.00264	0.01029	0.00196	0.00793	-0.00061	-0.00042
ecd	0.98172	0.02203	0.00773	0.00053	-0.01110	0.00443	0.00169	0.00612	-0.00015	-0.00027
fem	1.61287	0.01886	0.00713	0.00047	-0.00400	0.01064	0.00174	0.00555	0.00007	-0.00019
hsp	1.09198	0.02189	0.00905	0.00060	-0.01008	-0.00137	0.00243	0.00632	-0.00021	-0.00022
mal	1.62465	0.02189	0.00772	0.00052	-0.00758	0.01181	0.00172	0.00607	0.00006	-0.00021
nam	1.15743	0.02214	0.01374	0.00093	-0.00788	0.01015	0.00620	0.00824	-0.00026	-0.00005
nec	1.29808	0.01965	0.00725	0.00047	0.00390	0.00813	0.00171	0.00601	0.00021	-0.00015
wht	1.20059	0.01797	0.00620	0.00047	0.00769	0.01000	0.00120	0.00506	0.00017	-0.00017

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 15c. Administrative District Variances and Covariances (Pooled Estimates – CS Scale – 2022-2025)

	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	cov (avg,lrn)	cov (avg,tav)	cov (avg,tlr)	cov (lrn,tav)	cov (lrn,tlr)	cov (tav,tlr)
Overall										
all	0.18724	0.00437	0.00216	0.00042	0.00094	0.00281	0.00278	0.00164	0.00009	-0.00001
asn	0.26155	0.00338	0.00224	0.00041	0.00637	0.00608	0.00409	0.00155	0.00014	0.00007
blk	0.13258	0.00340	0.00198	0.00038	0.00043	0.00143	0.00229	0.00142	-0.00006	-0.00012
ecd	0.10878	0.00433	0.00241	0.00042	-0.00003	0.00180	0.00202	0.00185	0.00005	-0.00004
fem	0.17987	0.00375	0.00199	0.00044	0.00181	0.00303	0.00295	0.00138	0.00010	-0.00003
hsp	0.13533	0.00351	0.00195	0.00039	0.00146	0.00287	0.00257	0.00134	0.00009	-0.00005
mal	0.19257	0.00400	0.00204	0.00046	0.00106	0.00259	0.00291	0.00142	0.00011	0.00003
nec	0.15123	0.00461	0.00257	0.00042	0.00316	0.00352	0.00244	0.00200	0.00014	-0.00001
wht	0.13783	0.00389	0.00197	0.00042	0.00283	0.00310	0.00222	0.00138	0.00015	0.00002
Math										
all	0.21647	0.00777	0.00259	0.00056	-0.00012	0.00280	0.00342	0.00188	0.00029	0.00010
asn	0.31721	0.00519	0.00207	0.00034	0.00845	0.00704	0.00449	0.00134	0.00026	0.00012
blk	0.14847	0.00447	0.00200	0.00043	-0.00091	0.00124	0.00268	0.00114	-0.00003	-0.00006
ecd	0.12975	0.00678	0.00260	0.00049	-0.00125	0.00164	0.00260	0.00185	0.00016	0.00003
fem	0.20408	0.00628	0.00226	0.00051	0.00133	0.00300	0.00370	0.00147	0.00025	0.00005
hsp	0.14848	0.00636	0.00209	0.00040	0.00025	0.00222	0.00280	0.00115	0.00012	0.00001
mal	0.22772	0.00629	0.00220	0.00050	-0.00017	0.00291	0.00357	0.00133	0.00020	0.00009
nec	0.17712	0.00702	0.00275	0.00048	0.00255	0.00357	0.00298	0.00197	0.00025	0.00005
wht	0.15849	0.00656	0.00226	0.00051	0.00185	0.00301	0.00263	0.00158	0.00033	0.00010
RLA										
all	0.16927	0.00410	0.00190	0.00031	0.00202	0.00277	0.00211	0.00148	0.00015	0.00005
asn	0.22983	0.00300	0.00173	0.00027	0.00498	0.00555	0.00353	0.00126	0.00014	0.00011
blk	0.12797	0.00310	0.00171	0.00023	0.00113	0.00144	0.00198	0.00127	0.00002	-0.00001
ecd	0.09590	0.00387	0.00204	0.00028	0.00067	0.00170	0.00146	0.00156	0.00010	0.00003
fem	0.16711	0.00346	0.00169	0.00030	0.00234	0.00283	0.00227	0.00120	0.00014	0.00004
hsp	0.12957	0.00348	0.00169	0.00026	0.00238	0.00344	0.00202	0.00115	0.00018	0.00003
mal	0.16793	0.00374	0.00173	0.00032	0.00230	0.00256	0.00209	0.00128	0.00016	0.00007
nec	0.13456	0.00422	0.00221	0.00028	0.00449	0.00376	0.00183	0.00183	0.00019	0.00009
wht	0.12874	0.00351	0.00168	0.00030	0.00390	0.00326	0.00175	0.00125	0.00017	0.00009

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 15d. Administrative District Variances and Covariances (Pooled Estimates – GCS Scale – 2022-2025)

	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	cov (avg,lrn)	cov (avg,tav)	cov (avg,tlr)	cov (lrn,tav)	cov (lrn,tlr)	cov (tav,tlr)
Overall										
all	1.98824	0.04709	0.02200	0.00434	0.05880	0.03070	0.02931	0.01727	0.00227	0.00081
asn	2.76585	0.04046	0.02295	0.00429	0.13509	0.06666	0.04316	0.01712	0.00289	0.00171
blk	1.40768	0.03552	0.01976	0.00381	0.03364	0.01601	0.02425	0.01427	0.00027	-0.00042
ecd	1.15351	0.04522	0.02443	0.00428	0.02761	0.01984	0.02150	0.01906	0.00165	0.00059
fem	1.90861	0.04105	0.02013	0.00450	0.06623	0.03301	0.03121	0.01459	0.00221	0.00063
hsp	1.44237	0.03741	0.01948	0.00390	0.05123	0.03187	0.02779	0.01407	0.00196	0.00046
mal	2.04290	0.04315	0.02076	0.00473	0.06066	0.02857	0.03039	0.01486	0.00236	0.00120
nec	1.60497	0.05087	0.02637	0.00429	0.07551	0.03849	0.02607	0.02131	0.00274	0.00096
wht	1.46418	0.04296	0.02010	0.00436	0.06681	0.03359	0.02349	0.01475	0.00261	0.00112
Math										
all	2.18896	0.09191	0.02671	0.00551	0.13487	0.03347	0.03566	0.02151	0.00645	0.00345
asn	3.20125	0.08160	0.02151	0.00345	0.29065	0.07829	0.04875	0.01878	0.00633	0.00306
blk	1.50119	0.05410	0.02030	0.00409	0.08611	0.01672	0.02736	0.01239	0.00227	0.00142
ecd	1.31306	0.07698	0.02667	0.00474	0.06881	0.02062	0.02693	0.02021	0.00460	0.00268
fem	2.06288	0.07852	0.02325	0.00502	0.14275	0.03571	0.03845	0.01747	0.00587	0.00274
hsp	1.50365	0.07539	0.02142	0.00383	0.09831	0.02626	0.02877	0.01337	0.00371	0.00203
mal	2.30300	0.07762	0.02284	0.00499	0.14088	0.03488	0.03714	0.01590	0.00521	0.00313
nec	1.79198	0.08617	0.02812	0.00472	0.13663	0.04056	0.03182	0.02267	0.00568	0.00298
wht	1.60334	0.08024	0.02333	0.00499	0.11731	0.03432	0.02805	0.01862	0.00613	0.00321
RLA										
all	1.89870	0.04619	0.02141	0.00352	0.00522	0.03061	0.02350	0.01644	0.00131	0.00030
asn	2.57493	0.03308	0.01949	0.00305	0.03200	0.06140	0.03906	0.01361	0.00110	0.00096
blk	1.43653	0.03504	0.01932	0.00267	-0.00022	0.01563	0.02217	0.01423	-0.00005	-0.00042
ecd	1.07660	0.04387	0.02301	0.00317	-0.00244	0.01879	0.01632	0.01742	0.00086	0.00003
fem	1.87443	0.03897	0.01904	0.00343	0.00891	0.03123	0.02537	0.01324	0.00118	0.00016
hsp	1.45393	0.03902	0.01908	0.00300	0.01287	0.03822	0.02238	0.01261	0.00172	0.00005
mal	1.88358	0.04210	0.01956	0.00358	0.00869	0.02829	0.02330	0.01412	0.00150	0.00052
nec	1.50968	0.04702	0.02495	0.00321	0.03698	0.04177	0.02031	0.02019	0.00184	0.00070
wht	1.44466	0.03910	0.01901	0.00341	0.03091	0.03617	0.01944	0.01370	0.00162	0.00075

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 15e. School Variances and Covariances (Pooled Estimates – CS Scale – 2009-2019)

					cov	cov	cov	cov	cov	cov
	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	(avg,lrn)	(avg,tav)	(avg,tlr)	(lrn,tav)	(lrn,tlr)	(tav,tlr)
Overall										
all	0.21014	0.00414	0.00120	0.00016	0.00337	0.00314	0.00013	0.00093	0.00003	-0.00002
Math										
all	0.22295	0.00627	0.00155	0.00025	0.00567	0.00418	0.00028	0.00127	0.00014	0.00001
RLA										
all	0.20786	0.00309	0.00098	0.00010	0.00176	0.00230	0.00019	0.00077	0.00002	-0.00001

Table 15f. School Variances and Covariances (Pooled Estimates – GCS Scale – 2009-2019)

					cov	cov	cov	cov	cov	cov
	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	(avg,lrn)	(avg,tav)	(avg,tlr)	(lrn,tav)	(lrn,tlr)	(tav,tlr)
Overall										
all	3.19578	0.04615	0.01229	0.00160	0.10069	0.02759	0.00171	0.01036	0.00064	0.00021
Math										
all	3.19889	0.07415	0.01472	0.00240	0.20231	0.03577	0.00811	0.01441	0.00253	0.00115
RLA										
all	3.29911	0.03449	0.01120	0.00117	0.00960	0.02218	0.00125	0.00853	0.00012	-0.00024

Table 15g. School Variances and Covariances (Pooled Estimates – CS Scale – 2022-2025)

					cov	cov	cov	cov	cov	cov
	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	(avg,lrn)	(avg,tav)	(avg,tlr)	(lrn,tav)	(lrn,tlr)	(tav,tlr)
Overall										
all	0.27468	0.00831	0.00401	0.00179	0.00071	0.00248	0.00395	0.00271	0.00022	-0.00008
Math										
all	0.31656	0.01210	0.00483	0.00206	0.00036	0.00255	0.00465	0.00319	0.00039	0.00005
RLA										
all	0.25448	0.00628	0.00330	0.00113	0.00065	0.00207	0.00330	0.00232	0.00016	0.00001

Table 15h. School Variances and Covariances (Pooled Estimates – GCS Scale – 2022-2025)

					cov	cov	cov	cov	cov	cov
	tau(avg)	tau(lrn)	tau(tav)	tau(tlr)	(avg,lrn)	(avg,tav)	(avg,tlr)	(lrn,tav)	(lrn,tlr)	(tav,tlr)
Overall										
all	3.82011	0.08732	0.04016	0.01817	0.09124	0.00215	0.04460	0.02782	0.00430	0.00099
Math										
all	3.62782	0.12704	0.04469	0.01915	0.19494	0.00975	0.04520	0.03178	0.00876	0.00458
RLA										
all	4.00366	0.07110	0.03756	0.01277	-0.00143	0.02673	0.03840	0.02620	0.00122	-0.00031

Table 16a. Administrative District Reliabilities (Pooled Estimates – 2009-2019)

	CS Scale				GCS Scale			
	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)
Overall								
all	0.967	0.838	0.892	0.787	0.967	0.843	0.891	0.787
asn	0.867	0.511	0.668	0.438	0.867	0.526	0.667	0.442
blk	0.849	0.596	0.712	0.519	0.850	0.602	0.711	0.518
ecd	0.940	0.787	0.852	0.712	0.940	0.789	0.851	0.710
fem	0.957	0.792	0.858	0.725	0.957	0.799	0.857	0.724
hsp	0.876	0.628	0.730	0.547	0.876	0.633	0.729	0.544
mal	0.955	0.794	0.857	0.723	0.955	0.800	0.857	0.723
nam	0.783	0.477	0.623	0.422	0.784	0.481	0.622	0.423
nec	0.944	0.782	0.841	0.702	0.944	0.788	0.841	0.703
wht	0.949	0.811	0.859	0.742	0.949	0.817	0.859	0.743
Math								
all	0.954	0.807	0.864	0.745	0.954	0.825	0.864	0.745
asn	0.810	0.396	0.538	0.288	0.811	0.452	0.538	0.295
blk	0.788	0.499	0.619	0.404	0.789	0.524	0.617	0.405
ecd	0.918	0.738	0.809	0.643	0.918	0.751	0.807	0.641
fem	0.939	0.750	0.820	0.663	0.939	0.775	0.818	0.664
hsp	0.821	0.550	0.647	0.434	0.822	0.573	0.645	0.431
mal	0.940	0.744	0.815	0.649	0.940	0.770	0.814	0.649
nam	0.690	0.364	0.490	0.281	0.691	0.388	0.489	0.283
nec	0.925	0.737	0.799	0.644	0.926	0.759	0.798	0.645
wht	0.934	0.773	0.823	0.691	0.934	0.792	0.821	0.690
RLA								
all	0.951	0.754	0.833	0.670	0.951	0.754	0.833	0.671
asn	0.791	0.359	0.531	0.252	0.790	0.356	0.531	0.251
blk	0.780	0.454	0.583	0.356	0.780	0.453	0.583	0.358
ecd	0.908	0.678	0.768	0.565	0.908	0.678	0.768	0.566
fem	0.936	0.675	0.775	0.571	0.936	0.674	0.775	0.572
hsp	0.816	0.473	0.597	0.366	0.815	0.472	0.597	0.366
mal	0.933	0.690	0.780	0.583	0.933	0.690	0.780	0.583
nam	0.686	0.302	0.462	0.238	0.686	0.302	0.462	0.238
nec	0.914	0.664	0.757	0.557	0.914	0.662	0.757	0.558
wht	0.925	0.702	0.782	0.610	0.925	0.700	0.782	0.611

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 16b. Administrative District Reliabilities (Pooled Estimates – 2022-2025)

	CS Scale				GCS Scale			
	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)
Overall								
all	0.978	0.787	0.768	0.691	0.978	0.790	0.768	0.691
asn	0.940	0.526	0.536	0.428	0.940	0.551	0.536	0.430
blk	0.923	0.583	0.573	0.475	0.924	0.584	0.570	0.473
ecd	0.946	0.697	0.683	0.569	0.947	0.697	0.682	0.569
fem	0.967	0.703	0.685	0.615	0.968	0.710	0.685	0.615
hsp	0.934	0.613	0.597	0.507	0.935	0.618	0.595	0.505
mal	0.967	0.702	0.677	0.609	0.967	0.706	0.677	0.610
nec	0.957	0.709	0.698	0.579	0.957	0.716	0.699	0.580
wht	0.961	0.719	0.694	0.621	0.962	0.727	0.695	0.622
Math								
all	0.969	0.746	0.690	0.605	0.969	0.762	0.693	0.608
asn	0.927	0.446	0.387	0.260	0.928	0.516	0.391	0.268
blk	0.897	0.463	0.433	0.345	0.898	0.488	0.434	0.344
ecd	0.927	0.621	0.568	0.448	0.928	0.635	0.570	0.450
fem	0.954	0.640	0.581	0.497	0.955	0.669	0.584	0.500
hsp	0.909	0.557	0.480	0.373	0.910	0.578	0.482	0.372
mal	0.954	0.621	0.554	0.472	0.955	0.648	0.559	0.477
nec	0.941	0.636	0.593	0.464	0.942	0.661	0.595	0.467
wht	0.946	0.660	0.595	0.508	0.947	0.683	0.599	0.513
RLA								
all	0.963	0.671	0.641	0.529	0.963	0.671	0.641	0.529
asn	0.907	0.382	0.372	0.264	0.906	0.378	0.371	0.263
blk	0.892	0.441	0.427	0.296	0.892	0.441	0.427	0.297
ecd	0.911	0.549	0.531	0.384	0.910	0.550	0.531	0.384
fem	0.946	0.555	0.524	0.424	0.946	0.554	0.524	0.424
hsp	0.899	0.469	0.444	0.332	0.899	0.468	0.444	0.332
mal	0.946	0.566	0.525	0.428	0.946	0.565	0.525	0.428
nec	0.927	0.569	0.555	0.400	0.927	0.567	0.554	0.400
wht	0.937	0.577	0.544	0.446	0.937	0.575	0.544	0.446

Note: all = all students; asn = Asian; blk = Black; hsp = Hispanic; nam = Native American; wht = White; fem = female; mal = male; ecd = economically disadvantaged; nec = not economically disadvantaged

Table 16c. School Reliabilities (Pooled Estimates – 2009-2019)

	CS Scale				GCS Scale			
	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)
Overall								
all	0.971	0.796	0.894	0.715	0.977	0.804	0.893	0.717
Math								
all	0.957	0.757	0.861	0.670	0.967	0.777	0.858	0.673
RLA								
all	0.955	0.674	0.832	0.538	0.963	0.670	0.832	0.537

Table 16d. School Reliabilities (Pooled Estimates – 2022-2025)

	CS Scale				GCS Scale			
	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)	rel(avg)	rel(lrn)	rel(tav)	rel(tlr)
Overall								
all	0.985	0.778	0.804	0.658	0.989	0.785	0.806	0.661
Math								
all	0.977	0.727	0.726	0.546	0.981	0.746	0.727	0.551
RLA								
all	0.974	0.616	0.663	0.437	0.981	0.614	0.662	0.436

Table 17. Suppressed Estimates by Unit Post-Estimation, Long Form Data

2009-2025	sedaadmin		sedafips	
NCES privacy restrictions	4,430,430	30.43%	176	0.26%
Estimation failed	192	0.00%		
Standard error 1 or higher	142	0.00%		
Missing or suppressed mean for All subgroup	1,877	0.01%		
Not suppressed	10,127,055	69.56%	67,061	99.74%
Total	14,559,696	100.00%	67,237	100.00%

Note: sedafips = State; sedaadmin = Administrative District

Table 18. Suppressed Estimates by Unit Post-Estimation, Annual Data

2009-2025	sedaadmin		sedafips	
Overall				
NCES privacy restrictions	141,917	9.91%	5	0.09%
Extreme grade distribution	15,164	1.06%	13	0.22%
Estimation failed	3,073	0.21%	1	0.02%
Low reliability	9,858	0.69%		
Missing or suppressed mean for All subgroup	27	0.00%		
Not suppressed	1,262,274	88.13%	5,837	99.68%
Math				
NCES privacy restrictions	151,429	10.68%	7	0.12%
Extreme grade distribution	28,302	2.00%	127	2.15%
Estimation failed	3,356	0.24%	4	0.07%
Low reliability	27,975	1.97%		
Missing or suppressed mean for All subgroup	123	0.01%		
Not suppressed	1,206,595	85.10%	5,766	97.66%
RLA				
NCES privacy restrictions	158,810	11.06%	7	0.12%
Extreme grade distribution	17,703	1.23%	16	0.27%
Estimation failed	3,316	0.23%		
Low reliability	29,911	2.08%	1	0.02%
Missing or suppressed mean for All subgroup	91	0.01%		
Not suppressed	1,225,732	85.38%	5,893	99.59%

Note: sedafips = State; sedaadmin = Administrative District

Table 19. Suppressed Estimates by Unit Post-Estimation, Pooled Data

2009-2025	sedaadmin		sedafips		sedasch	
Overall						
NCES privacy restrictions	7,739	3.81%			379	0.31%
Extreme year distribution	2,406	1.18%			3,741	3.11%
Estimation failed	617	0.30%			446	0.37%
Low reliability	4,731	2.33%			583	0.48%
Missing or suppressed mean for All subgroup	11	0.01%				
Not suppressed	187,560	92.36%	796	100.00%	115,315	95.73%
Math						
NCES privacy restrictions	8,164	4.11%			363	0.32%
Extreme year distribution	1,931	0.97%			2,693	2.39%
Estimation failed	272	0.14%	1	0.13%	177	0.16%
Low reliability	7,187	3.62%	1	0.13%	1,058	0.94%
Missing or suppressed mean for All subgroup	16	0.01%				
Not suppressed	180,956	91.15%	785	99.75%	108,324	96.19%
RLA						
NCES privacy restrictions	8,642	4.29%			449	0.40%
Extreme year distribution	2,030	1.01%			2,694	2.40%
Estimation failed	267	0.13%			171	0.15%
Low reliability	7,294	3.62%			1,038	0.93%
Missing or suppressed mean for All subgroup	18	0.01%				
Not suppressed	183,228	90.94%	796	100.00%	107,680	96.12%

Note: sedafips = State; sedaadmin = Administrative District; sedasch = School

Table 20. Component Loadings and Summary Statistics for Socioeconomic Status Composite Construction.

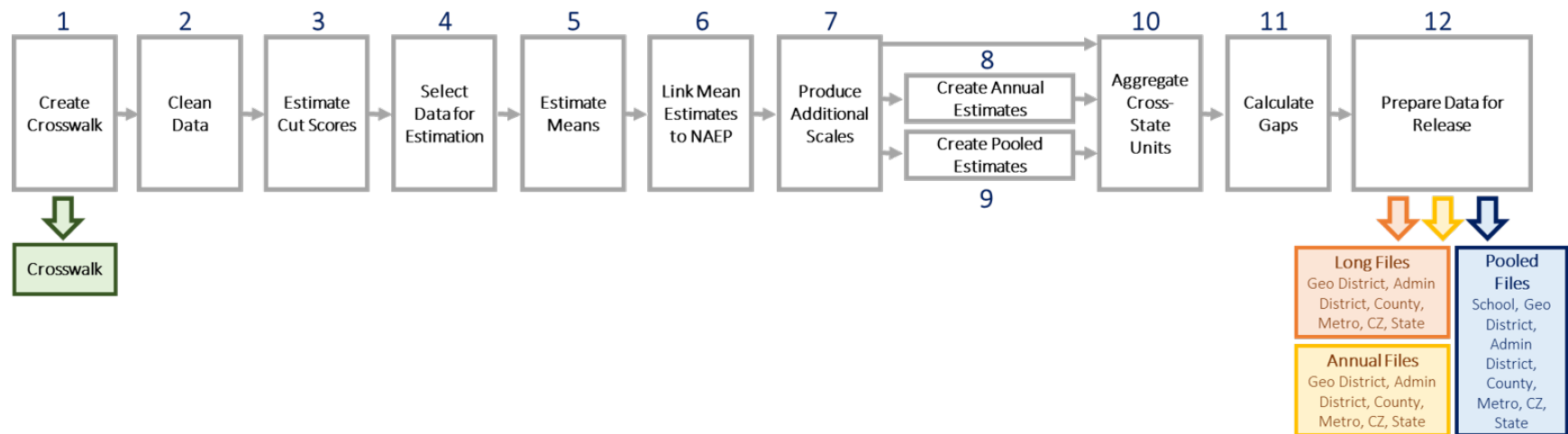
	Standardized Loadings	Unstandardized Loadings	Mean	SD
log(Median Family Income)	0.914	0.638	10.886	0.335
% with BA or Higher	0.674	1.170	0.285	0.135
Poverty Rate	-0.925	-2.664	0.151	0.081
SNAP Eligibility Rate	-0.931	-2.958	0.115	0.074
Unemployment Rate	-0.791	-5.221	0.097	0.035
Single Mother Headed Household Rate	-0.799	-2.315	0.199	0.081

Table 21. Summary Statistics at Different Values of the Socioeconomic Status Composite.

	SES Composite						
	below -2.5	-2.5 to -1.5	-1.5 to -.5	-.5 to .5	.5 to 1.5	1.5 to 2.5	above 2.5
log(Median Family Income)	10.17	10.31	10.51	10.74	11.03	11.55	12.11
% with BA or Higher	0.10	0.12	0.14	0.18	0.27	0.54	0.79
Poverty Rate	0.35	0.29	0.21	0.14	0.08	0.04	0.03
SNAP Eligibility Rate	0.32	0.26	0.18	0.11	0.06	0.02	0.01
Unemployment Rate	0.14	0.10	0.08	0.06	0.05	0.04	0.04
Single Mother Headed Household Rate	0.39	0.30	0.22	0.16	0.12	0.10	0.07

Figures

Figure 1. SEDA Construction Process



Appendices

Appendix A: Additional Detail on Statistical Methods

1. Estimating Means and Standard Deviations for Units that Cross State Lines

This section briefly describes how means, standard deviations, and standard errors are estimated for units serving BIE schools or units that cross state lines. As described above, we first estimate unit “component” means and standard deviations. We then estimate the overall unit mean as weighted averages of the component means and the unit standard deviations as estimates of total variance within a unit based on the component means and standard deviations.

Let $\hat{\mu}_d$ and $\hat{\sigma}_d$ be the estimated means and standard deviations for the D components $d = 1, \dots$, that will be aggregated for a given unit. We also have estimates of the standard errors for each mean and standard deviation, $se(\hat{\mu}_d)$ and $se(\hat{\sigma}_d)$. We do not include grade, subject, year, or state subscripts here for clarity.

To estimate the aggregate parameters, we make the simplifying assumption that $cov(\hat{\mu}_i, \hat{\mu}_j) = cov(\hat{\sigma}_i, \hat{\sigma}_j) = cov(\hat{\mu}_i, \hat{\sigma}_i) = 0$ for $i \neq j$. The derivations for these expressions are based on the formulas in the appendix of Reardon et al. (2017) used to estimate overall mean and variance of a set of groups in the HETOP model. Let p_d be the proportion of all students in the aggregate unit c that are in component d :

$$p_d = \frac{n_d}{\sum_{d=1}^D n_d} = \frac{n_d}{N_c} \quad (A1.1)$$

We estimate the aggregate mean for aggregate unit c as the weighted average of the component estimated means and its estimated standard error as:

$$\hat{\mu}_c = \sum_{d=1}^D p_d \hat{\mu}_d \quad se(\hat{\mu}_c) = \sqrt{\sum_{d=1}^D [p_d^2 \cdot se(\hat{\mu}_d)^2]} \quad (A1.2)$$

We estimate the standard deviation for aggregate unit c as the square root of the sum of the estimated between and within-unit variance and its associated estimated standard error as:

$$\hat{\sigma}_c = \sqrt{\sum_{d=1}^D [p_d(\hat{\mu}_d - \hat{\mu}_c)^2 + q_d \hat{\sigma}_d^2]} \quad (\text{A1.3})$$

$$se(\hat{\sigma}_c) = \sqrt{z_c * \left(\frac{1}{\hat{\sigma}_c}\right)}$$

In these expressions we define

$$q_d = \left(\frac{p_d + (n_d - 1)}{n_d}\right) \left(\frac{p_d}{1 + 2\left(\frac{1}{2\tilde{n}_c}\right)}\right) \quad (\text{A1.4})$$

$$\tilde{n}_c = \left[\left(\frac{1}{D}\right) \sum_{d=1}^D \left(\frac{1}{n_d - 1}\right)\right]^{-1}$$

$$z_c = \sum_{d=1}^D [(p_d^2(\hat{\mu}_d - \hat{\mu}_c)^2 se(\hat{\mu}_d)^2) + (q_d^2 \cdot \hat{\sigma}_d^2 \cdot se(\hat{\sigma}_d)^2)]$$

2. NAEP 2024 Adjustment and Estimated NAEP 2025 Values

2.1 NAEP 2024 EPE Adjustments

Because we do not have the NAEP Expanded Population Estimates (EPEs) for 2024 at the time of this release, we created adjustments based on 2019 and 2022 NAEP EPE and regular data, which we then apply to the 2024 regular NAEP means and SDs. Specifically, we regress the difference between the EPE and regular NAEP estimates (separately for means and SDs, pooling data from 2019 and 2022) on four grade-subject dummy variables. We use the coefficients on those dummy variables to adjust the NAEP regular estimates in 2024.

The adjustments we use are shown in the table below. Each value is added to the corresponding 2024 NAEP regular means and SDs to recover an estimated 2024 NAEP EPE mean and SD.

Table A2.1. EPE Adjustments Applied to 2024 NAEP Means and Standard Deviations

Subject	Grade	Mean Adjustment	SD Adjustment
Math	4	- .7579991	+ .5540244
Math	8	- .9166223	+ .5968
Reading	4	- 1.120848	+ .7713522
Reading	8	- 1.186662	+ 1.005793

2.2 Estimated NAEP 2025 State Means and Standard Deviations

Because NAEP data are not available for 2025 (it is an untested year) and because the 2026 NAEP data are not yet available to allow for interpolation, we estimate NAEP 2025 state means and standard deviations using a multi-step procedure based on the historical relationship between state test score changes and NAEP score changes.

Step 1. Estimating the NAEP-State Test Relationship. To estimate the historical relationship between changes in state test scores and changes in NAEP scores, we use paired NAEP and state-test 2-year changes from 2015-2024, excluding the change from 2019-2022 because that spans three years and reflects unusually large pandemic-era changes. We start in 2015 because that is the first operational year for PARCC and Smarter Balanced, the two primary

Common Core assessment consortia. We use the NAEP EPE data for consistency with the rest of SEDA construction; for 2024, we use the EPE NAEP estimates produced in **Appendix A2.1**.

For each state-grade-subject, we divide data into test eras (periods with no assessment change) and fit era-specific Heteroskedastic Ordered Probit (HETOP) models. From these models, we construct standardized changes (deltas), then pair these state test changes with the corresponding changes in NAEP scores from the same two-year periods.

We stack the paired NAEP and state-test changes and fit a two-level Hierarchical Linear Model (HLM). To account for the negative covariance structure of the observed changes within each state-era, we apply a Cholesky transformation. Let $\mathbf{F}$ be the Cholesky factor of the covariance matrix, we premultiply the model terms (delta, NAEP indicator, and state test indicator) by $\mathbf{F}^{-1}$. $\mathbf{N}$ and $\mathbf{T}$ are dummy variables indicating whether the observed change $\hat{\delta}$ corresponds to the NAEP or state test, respectively.

Level-1: (A2.1)

$$\mathbf{F}^{-1}\hat{\delta} = \mathbf{F}^{-1}\mathbf{N} \cdot \delta_N + \mathbf{F}^{-1}\mathbf{T} \cdot \delta_T + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim (0, \mathbf{I})$$

Level-2:

$$\begin{aligned}\delta_N &= \gamma_N + \Lambda_{Nbg y} + u_N \\ \delta_T &= \gamma_T + \Lambda_{Tbg y} + u_T\end{aligned}$$

At level 2, δ_N and δ_T are modeled as random effects around their respective grand means, with year-grade-subject fixed effects (the $\Lambda_{bg y}$'s, where b denotes subject, g grade, and y year). The level-1 variance is fixed at 1 (reflecting the Cholesky transformation). From this model, we obtain estimates of γ_N , γ_T , and the variance-covariance matrix $\boldsymbol{\tau} = \begin{pmatrix} \tau_0 & \tau_{01} \\ \tau_{01} & \tau_1 \end{pmatrix}$. τ_0 is the within-year-grade-subject variance of the true NAEP changes, τ_1 is the within-year-grade-subject variance of the true state-test changes, and τ_{01} is their covariance.

Step 2. Estimating Standardized Changes from State Assessments.

A. Estimating changes for each state-subject-grade-year. We use state test data from EDC for the 2023-24 and 2024-25 school years to estimate standardized changes in mean achievement and in the ratio of standard deviations between 2024 and 2025 for each state-

grade-subject. We first identify the set of states with no test change between the two years using EDC test-change flags. For states without a test change, we fit HETOP models jointly to 2024 and 2025 proficiency counts (2024 as reference), separately by state-grade-subject for tested grades 3-8. From each model, we extract the standardized change in means (delta, $\hat{\delta}_{Tsgb}$) and the ratio of standard deviations ($\hat{P}_{Tsgb}$), along with their standard errors. These standardized changes are adjusted for test score reliability.

B. Estimating the average state change for each subject-grade. For each grade-subject, we estimate the pooled average state test change ($\bar{\delta}_{gb}$) using precision-weighted meta-regression across states using the data from **Step 2A**.

Step 3. State-Level NAEP 2025 Changes. Using the model parameters from **Step 1** and the estimated 2024-2025 state test changes from **Step 2**, we predict each state's NAEP change from 2024 to 2025 via Empirical Bayes shrinkage. Based on the analyses from Reardon et al. (2026), we compute the posterior mean of the NAEP score change for state s , grade g , and subject b , given the observed state test change and the parameters from Model A2.1 above:

$$\begin{aligned}\hat{\delta}_{Nsgb}^* &= E[\delta_{Nsgb} | \hat{\delta}_{Tsgb}, \tau, \gamma_N, \gamma_T, \sigma_{Tsgb}, \bar{\delta}_{gb}] \\ &= \left(\bar{\delta}_{gb} - \frac{\gamma_T - \gamma_N}{2} \right) + \frac{\tau_{01}}{\tau_1 + 2\sigma_{Tsgb}} (\hat{\delta}_{Tsgb} - \bar{\delta}_{gb})\end{aligned}\tag{A2.2}$$

The posterior variance of the true NAEP mean change is:

$$v_{\mu,sgb}^* = \frac{1}{2} \left(\tau_0 - \frac{\tau_{01}^2}{\tau_1 + 2\sigma_{Tsgb}} \right)\tag{A2.3}$$

$\hat{P}_{sgb}^{naep}$ (predicted ratio of the 2025 to 2024 standard deviation) and $v_{\sigma,sgb}^*$ (variance of the predicted SD ratio) are computed analogously to the mean change using the parameters from the standard deviation model from **Step 1** combined with HETOP-estimated ratio of standard deviations from **Step 2**.

Because the HETOP-estimated changes are not available in both grades 4 and 8 for all state-subjects, we then fit weighted linear regression of predicted NAEP changes on grade within each state-subject (separately for mean changes and SD ratios), using inverse prediction variance weights. We use the regression predicted NAEP change in grades 4 and 8 for all states in all

subsequent steps. The predicted variance is updated to the squared standard error of the fitted value from this regression.

Step 4. Predicting National NAEP 2025 Changes. For the national estimates, we compute enrollment-weighted averages of the state-level predicted NAEP mean changes within each grade-subject. For the national SD ratio, we compute the square root of the enrollment-weighted average of the squared state predicted SD ratios within each grade-subject. These 2025 national NAEP estimates are based on 41 states in math and 39 states in reading for which we were able to estimate NAEP changes from their 2024 and 2025 data.

Step 5. Constructing Predicted NAEP 2025 Means and Standard Deviations. We add the predicted NAEP changes to the EPE-adjusted 2024 NAEP state means and standard deviations to produce predicted 2025 NAEP values and their standard errors, as follows:

$$\begin{aligned}
 \hat{\mu}_{2025sgb}^{naep} &= \hat{\mu}_{2024sgb}^{naep} + \hat{\delta}_{sgb}^{naep} \cdot \hat{\sigma}_{2024sgb}^{naep} \\
 \hat{\sigma}_{2025sgb}^{naep} &= \hat{\sigma}_{2024sgb}^{naep} \cdot \hat{p}_{sgb}^{naep} \\
 se(\hat{\mu}_{2025sgb}^{naep}) &= \sqrt{se(\hat{\mu}_{2024sgb}^{naep})^2 + v_{\mu,sgb}^* \cdot (\hat{\sigma}_{2024sgb}^{naep})^2} \\
 se(\hat{\sigma}_{2025sgb}^{naep}) &= \sqrt{(P_{sgb})^2 \cdot se(\hat{\sigma}_{2024sgb}^{naep})^2 + v_{\sigma,sgb}^* \cdot (\hat{\sigma}_{2024sgb}^{naep})^2 + v_{\sigma,sgb}^* \cdot se(\hat{\sigma}_{2024sgb}^{naep})^2}
 \end{aligned} \tag{A2.4}$$

The state-level 2025 predicted NAEP means, SDs, and their SEs are shown in Table A2.2.

Table A2.2. Estimated NAEP 2025 State Means, by Subject and Grade

State	Math								RLA							
	Grade 4				Grade 8				Grade 4				Grade 8			
	Mean	SE(Mean)	SD	SE(SD)	Mean	SE(Mean)	SD	SE(SD)	Mean	SE(Mean)	SD	SE(SD)	Mean	SE(Mean)	SD	SE(SD)
Alabama	235.9	1.7	33.8	0.8	261.6	1.6	41.1	0.9	212.6	2.0	40.8	1.0	249.6	1.6	39.8	1.0
Arizona	231.9	1.7	36.1	0.9	268.9	1.9	45.4	1.1	206.3	2.0	45.5	1.3	254.4	1.7	39.7	0.9
Arkansas	230.0	1.4	36.1	0.8	265.9	1.6	40.2	1.0								
California	232.4	1.3	37.2	0.8	268.9	1.6	46.7	1.1	210.8	1.7	42.6	1.0	254.3	1.8	43.8	1.1
Connecticut	238.5	1.2	37.2	0.7	276.7	1.6	44.3	1.0	218.1	1.3	44.0	0.9	262.9	1.6	42.3	1.1
Delaware	232.4	1.0	37.5	0.8	263.0	1.3	41.4	0.9	208.2	1.3	43.2	1.0	249.3	1.2	42.7	1.0
District of Columbia	230.5	1.1	40.8	0.7	262.3	1.0	46.7	0.9	208.4	1.3	49.2	1.1	251.8	1.0	44.9	1.0
Florida	243.0	1.2	32.8	0.7	267.6	2.0	40.2	1.0	217.4	1.4	40.7	0.7	253.4	1.1	41.5	0.8
Georgia	235.4	1.4	35.1	0.9	268.7	1.4	42.6	0.9	212.5	1.7	42.9	0.8	258.0	1.5	41.0	0.9
Hawaii	238.6	1.0	36.1	0.9	270.1	1.0	40.0	0.8	214.6	1.1	41.9	0.9	257.3	1.0	40.4	0.7
Idaho	237.7	1.1	35.2	0.6	277.6	1.4	40.0	0.8	214.3	1.4	41.5	0.9	260.3	1.1	39.1	0.9
Indiana	239.5	1.4	35.4	0.8	278.1	1.4	40.0	1.0	218.7	1.4	40.5	0.9	260.5	1.4	40.2	0.9
Iowa	236.4	1.1	33.8	0.7	274.7	1.2	38.6	0.8	213.8	1.5	39.9	0.8	260.0	1.4	38.3	0.8
Kentucky	237.1	1.2	32.4	0.7	270.8	1.2	41.0	0.7	217.1	1.7	41.5	0.9	257.9	1.4	40.3	0.8
Louisiana	234.0	1.3	32.5	0.8	266.8	1.6	40.0	0.9	214.7	1.6	42.6	0.9	255.8	1.6	39.1	0.9
Maine	232.5	1.1	33.7	0.8	272.5	1.4	39.8	1.0	208.1	1.3	41.8	0.9	253.8	1.3	39.6	0.9
Maryland	233.7	1.5	40.0	0.9	269.0	1.8	45.5	1.2	214.8	1.6	46.0	1.0	258.3	1.5	45.3	0.9
Massachusetts	245.5	1.4	36.3	0.7	282.8	1.5	44.3	0.8	223.5	1.4	42.6	1.1	268.0	1.6	40.4	0.9
Michigan	234.4	1.2	35.1	0.9	269.5	1.3	40.6	0.9	207.7	1.3	40.5	0.9	254.6	1.4	37.7	0.8
Minnesota	239.6	1.2	36.9	0.7	281.7	1.6	39.7	0.8	212.8	1.5	41.3	0.9	259.5	1.2	35.2	0.8
Mississippi	238.3	1.1	29.2	0.5	267.5	1.3	38.9	0.8	216.7	1.4	37.5	0.9	253.2	1.5	37.6	0.9
Missouri	234.1	1.2	34.4	0.7	270.4	1.3	40.0	0.9	210.5	1.3	41.4	0.9	254.9	1.2	39.8	0.9
Nebraska	237.2	1.1	33.4	0.7	279.3	1.2	40.2	0.8	210.0	1.6	41.6	1.0	255.7	1.2	39.9	0.8
Nevada	233.0	1.3	37.0	0.7	265.5	1.1	41.1	0.8	212.7	1.6	42.7	0.9	254.0	1.4	42.1	1.1
New Hampshire	241.0	1.0	32.5	0.8	279.6	1.3	39.9	0.8	220.3	1.4	39.8	1.1	263.6	1.3	37.3	0.8
New Jersey	239.8	1.9	36.9	1.3	282.5	1.6	46.8	1.2	220.4	2.3	45.2	1.7	266.1	1.4	43.1	1.1
New Mexico	223.4	1.1	34.4	0.8	256.6	1.4	39.6	0.8	200.4	1.4	44.0	0.8	244.8	1.4	40.7	1.0
North Carolina	238.3	1.4	35.2	0.8	275.9	1.5	45.0	1.1	212.3	1.7	43.7	0.9	255.2	1.4	41.5	0.9
Ohio	239.2	1.4	35.1	0.9	278.5	2.0	42.6	1.0	214.5	1.6	42.2	1.1	260.6	1.5	41.1	1.2
Pennsylvania	237.7	1.3	36.4	0.6	275.9	1.5	45.9	0.9	213.3	1.6	42.8	1.1	258.3	1.8	41.6	1.3
Rhode Island	236.5	1.1	35.5	0.7	269.7	1.0	45.3	0.8	215.1	1.4	43.2	1.0	258.5	0.9	42.9	0.9
South Carolina	237.1	1.2	35.3	0.7	268.1	1.4	42.2	1.0								
South Dakota	239.0	1.3	33.2	0.7	280.8	1.3	38.8	0.9	212.7	1.4	39.3	1.0	259.7	1.0	36.4	0.7
Tennessee	239.0	1.2	34.6	0.8	276.8	2.1	43.4	0.9	213.9	1.6	43.5	1.1	258.8	1.6	40.8	1.0
Texas	240.0	1.3	33.4	0.7	268.9	2.1	41.5	1.0	210.7	1.5	41.3	1.0	252.1	1.5	41.9	0.9
Utah	240.5	1.1	34.7	0.6	280.6	1.8	40.2	0.7								
Vermont	234.2	0.9	33.9	0.7	275.4	1.1	39.9	0.7	212.0	1.4	43.4	1.1	256.8	1.3	40.0	1.0
Virginia									212.7	1.8	43.0	1.0	256.4	1.3	42.9	0.9
Washington	236.8	1.4	36.3	0.7	273.6	1.5	45.6	1.0	214.4	1.3	42.8	0.9	259.1	1.6	41.3	0.9
West Virginia	231.0	1.0	33.8	0.7	261.1	1.2	41.0	0.8	204.9	1.4	43.8	0.9	247.5	1.4	41.5	1.1
Wisconsin	238.5	1.2	36.4	0.7	282.5	1.4	41.6	0.8	213.8	1.3	42.0	1.0	259.8	1.5	39.2	1.1
Wyoming	242.9	0.8	31.0	0.6	277.8	1.0	38.3	0.8	220.9	1.1	37.9	0.9	259.9	0.9	36.9	0.8

Step 6. National NAEP Estimates. We follow the same procedure as in Step 5 to construct the national 2025 NAEP estimates but adding the estimated national change to the 2024 national NAEP estimates. The estimates are shown in Table A2.3.

Table A2.3. Estimated National Means and SDs, by Subject and Grade

Subject	Grade	Mean	SD
Math	4	236.7	35.6
Math	8	272.2	43.4
Reading	4	213.0	42.6
Reading	8	256.5	41.6

Appendix B: Covariates

1. List of Raw ACS Tables Used for SES Composite

Table Description	Table ID	Universe	Description	Usage	Derived Construct
Median household income	B19013	Households	median family income in the past 12 months	we adjust the reported median income for inflation (2012 constant dollars)	Median Income
Median household income	B19013B	Families with a householder who is Black or African American alone	median family income in the past 12 months	we adjust the reported median income for inflation (2012 constant dollars)	White Median Income
Median household income	B19013H	Families with a householder who is white alone (not Hispanic or Latino)	median family income in the past 12 months	we adjust the reported median income for inflation (2012 constant dollars)	Hispanic Median Income
Median household income	B19013I	Families with a householder who is Hispanic or Latino	median family income in the past 12 months	we adjust the reported median income for inflation (2012 constant dollars)	Black Median Income
Sex by Educational Attainment for the Population 25 and Older	B15002	Population 25 years and over	counts of number of individuals that fall into each of 16 educational attainment categories, by sex	we use the counts of men and women with a bachelor's degree or higher along with the total count to generate the BA+ rate	Bachelor's Degree Rate
Sex by Educational Attainment for the Population 25 and Older	C15002B	Black or African American alone population 25 years and over	counts of number of individuals that fall into each of 4 educational attainment categories, by sex	we use the counts of men and women with a bachelor's degree or higher along with the total count to generate the BA+ rate	Black Bachelor's Degree Rate
Sex by Educational Attainment for the Population 25 and Older	C15002H	White alone, not Hispanic or Latino population 25 years and over	counts of number of individuals that fall into each of 4 educational attainment categories, by sex	we use the counts of men and women with a bachelor's degree or higher along with the total count to generate the BA+ rate	White Bachelor's Degree Rate
Sex by Educational Attainment for the Population 25 and Older	C15002I	Hispanic or Latino population 25 years and over	counts of number of individuals that fall into each of 4 educational attainment categories, by sex	we use the counts of men and women with a bachelor's degree or higher along with the total count to generate the BA+ rate	Hispanic Bachelor's Degree Rate

Poverty Status in the Last 12 Months by Age	B17020	Population for whom poverty status is determined	counts of number of individuals living in households above and below the poverty line in various age bins	we use the counts of those living in poverty that are school aged (6-17 years old)	Poverty Rate, 6-17 Year Olds
Poverty Status in the Last 12 Months by Age	B17020B	Black or African American alone population for whom poverty status is determined	counts of number of individuals living in households above and below the poverty line in various age bins	we use the counts of those living in poverty that are school aged (6-17 years old)	Black Poverty Rate, 6-17 Year Olds
Poverty Status in the Last 12 Months by Age	B17020H	White alone, not Hispanic or Latino population for whom poverty status is determined	counts of number of individuals living in households above and below the poverty line in various age bins	we use the counts of those living in poverty that are school aged (6-17 years old)	White Poverty Rate, 6-17 Year Olds
Poverty Status in the Last 12 Months by Age	B17020I	Hispanic or Latino population for whom poverty status is determined	counts of number of individuals living in households above and below the poverty line in various age bins	we use the counts of those living in poverty that are school aged (6-17 years old)	Hispanic Poverty Rate, 6-17 Year Olds
Sex by Age by Employment Status for the Population 16 and Over	B23001	Population 25 to 64 years	counts of individuals by age, labor market status and employment status	we use the count of those employed divided by the count of those in the labor market for civilians ages 16-64 to compute an unemployment rate	Unemployment Rate
Sex by Age by Employment Status for the Population 16 and Over	C23002B	Black or African American alone, not Hispanic or Latino population 16 years and over	counts of individuals by age, labor market status and employment status	we use the count of those employed divided by the count of those in the labor market for civilians ages 16-64 to compute an unemployment rate	Black Unemployment Rate
Sex by Age by Employment Status for the Population 16 and Over	C23002H	White alone, not Hispanic or Latino population 16 years and over	counts of individuals by age, labor market status and employment status	we use the count of those employed divided by the count of those in the labor market for civilians ages 16-64 to compute an unemployment rate	White Unemployment Rate
Sex by Age by Employment Status for the Population 16 and Over	C23002I	Hispanic or Latino population 16 years and over	counts of individuals by age, labor market status and employment status	we use the count of those employed divided by the count of those in the labor market for civilians ages 16-64 to compute an unemployment rate	Hispanic Unemployment Rate

Receipt of Food Stamps/SNAP in the past 12 months by poverty status in the past 12 months for households	B22003	Households	counts of households receiving food stamps/SNAP benefits by poverty status	we use the counts of households receiving SNAP divided by the total number of households to compute the SNAP rate	SNAP Rate
Receipt of Food Stamps/SNAP in the past 12 months by poverty status in the past 12 months for households	B22005B	Households with a householder who is Black or African American alone	counts of households receiving food stamps/SNAP benefits by poverty status	we use the counts of households receiving SNAP divided by the total number of households to compute the SNAP rate	Black SNAP Rate
Receipt of Food Stamps/SNAP in the past 12 months by poverty status in the past 12 months for households	B22005H	Households with a householder who is White alone, not Hispanic or Latino	counts of households receiving food stamps/SNAP benefits by poverty status	we use the counts of households receiving SNAP divided by the total number of households to compute the SNAP rate	White SNAP Rate
Receipt of Food Stamps/SNAP in the past 12 months by poverty status in the past 12 months for households	B22005I	Households with a householder who is Hispanic or Latino	counts of households receiving food stamps/SNAP benefits by poverty status	we use the counts of households receiving SNAP divided by the total number of households to compute the SNAP rate	Hispanic SNAP Rate
Household Type	B11001	Households	counts of different types of households	we use the count of family households with a female householder, no husband present divided by the total number of family households	Female Headed Household Rate
Household Type	B11001B	Households with a householder who is Black or African American alone, not Hispanic or Latino	counts of different types of households	we use the count of family households with a female householder, no husband present divided by the total number of family households	Black Female Headed Household Rate
Household Type	B11001H	Households with a householder who is White alone, not Hispanic or Latino	counts of different types of households	we use the count of family households with a female householder, no husband present divided by the total number of family households	White Female Headed Household Rate
Household Type	B11001I	Households with a householder who is Hispanic or Latino	counts of different types of households	we use the count of family households with a female householder, no husband present divided by the total number of family households	Hispanic Female Headed Household Rate

2. Measurement Error, Attenuation Bias and Solutions

Formally, attenuation bias can be specified as follows. As an example, consider the true relationship between race-specific achievement and socioeconomic status we would like to estimate:

$$Y_g = \beta_{0g} + \beta_{1g}(SES_g) + \varepsilon_g \quad (\text{B-2.1})$$

Where Y is White or non-White minority achievement in a unit (district, county, or metropolitan area) (g indexes group), and SES is the average socioeconomic status of the group. Race specific SES is measured with error and measurement error will be larger in units with relatively smaller sample sizes of non-White minorities. Thus, the data we observe are $W_g = SES_g + \varepsilon_g$. In this case, the bias in β_{1g} is known as attenuation bias. This bias can be quantified by multiplying by the variable's reliability $\lambda = \frac{\text{var}(SES_g)}{\text{var}(SES_g) + \sigma_1^2}$, i.e., the true variance of the variable SES_g relative to the true variance plus the variance of the measurement error.

To address attenuation bias, we use regression calibration, which makes use of the fact that the measurement error in SES_g (and consequently $SESGap$) are known from Census data.²⁷ Regression calibration is a method that replaces the error-prone variable W with its best linear prediction (blp). The best linear predictor of $SESGap$ can be defined as:

$$\begin{aligned} SESp_g^{blp} &= E(SES_g) + \frac{\text{cov}(SES_g, W_g)}{\text{var}(W_g)} (W_g - E(W_g)) \\ &= \mu + \frac{\text{cov}(SES_g, SES_g + \varepsilon_g)}{\sigma_{SES_g}^2 + \sigma_g^2} (W_g - \mu) \end{aligned}$$

²⁷ Specifically, the ACS reports margins of error which can be easily converted standard errors for each Census variable. Appendix B3: Computing the sampling variance of sums of ACS variables provides a full description of how standard errors for cross-tabulated Census data are constructed.

$$= \mu + \lambda(W_g - \mu) \quad (\text{B-2.2})$$

Note that SES_g^{blp} is “shrunk” towards the mean value of SES_g as a function of λ which, recall, is equal to the reliability of the variable SES_g and can be estimated as a random effect (or empirical Bayes estimate) from a generalized linear model.

Now, we show that regressing Y_g on SES_g^{blp} results in consistent estimates of β_{1g} .

$$\begin{aligned} \frac{cov(Y_g, \mu + \lambda(W_g - \mu))}{var(\mu + \lambda(W_g - \mu))} &= \frac{cov(Y_g, \lambda W_g)}{\lambda^2 (\sigma_{SES_g}^2 + \sigma_g^2)} \\ &= \frac{cov(Y_g, SES_g)}{\lambda (\sigma_{SES_g}^2 + \sigma_g^2)} \\ &= \frac{cov(Y_g, SES_g)}{\sigma_{SES_g}^2} = \beta_{1g} \end{aligned}$$

(B-2.3)

3. Computing the Sampling Variance of Sums of ACS Variables

In each unit we are given counts in K cells: $\widehat{n1}_d, \widehat{n2}_d, \dots, \widehat{nK}_d$; we also know total counts t_d ; we also have margins of error of the counts:

$$MoE(\widehat{n1}_d), MoE(\widehat{n2}_d), \dots, MoE(\widehat{nK}_d).$$

We then compute the sampling variances of the

$$var(\widehat{nk}_d) = \left[\frac{MOE(\widehat{nk}_d)}{1.645} \right]^2$$

from these we compute

$$\widehat{pk}_d = \frac{\widehat{nk}_d}{t_d}$$

and

$$var(\widehat{pk}_d) = \frac{var(\widehat{nk}_d)}{t_d^2}.$$

We do not know the sampling rate in unit d ; let's call it r_d . If the estimates come from a simple random sample, we would have:

$$var(\widehat{pk}_d)^* = \frac{pk_d(1 - pk_d)}{r_d t_d}$$

The estimated design effect in district d for variable k is then:

$$\widehat{Dk}_d = \frac{var(\widehat{pk}_d)}{var(\widehat{pk}_d)^*}$$

We can compute the average design effect in unit d as

$$D_d = \frac{1}{K} \sum_{k=1}^K \widehat{Dk}_d$$

Now we compute:

$$\hat{P}_d = \frac{1}{t_d} \sum_{k=1}^K \widehat{nk}_d = \sum_{k=1}^K \widehat{pk}_d$$

We want to know $var(\hat{P}_d)$. If we had a simple random sample, we would have:

$$var(\hat{P}_d)^* = \frac{P_d(1 - P_d)}{r_d t_d}$$

Given the design effect in unit d , however, we would expect this to be inflated by a factor D_d .

So, we have:

$$\begin{aligned} var(\hat{P}_d) &= D_d var(\hat{P}_d)^* \\ &= D_d \frac{P_d(1 - P_d)}{r_d t_d} \\ &= \left[\frac{1}{K} \sum_{k=1}^K \widehat{Dk}_d \right] \frac{P_d(1 - P_d)}{r_d t_d} \\ &= \left[\frac{1}{K} \sum_{k=1}^K \frac{var(\widehat{pk}_d)}{var(\widehat{pk}_d)^*} \right] \frac{P_d(1 - P_d)}{r_d t_d} \\ &= \left[\frac{1}{K} \sum_{k=1}^K \frac{r_d t_d var(\widehat{pk}_d)}{pk_d(1 - pk_d)} \right] \frac{P_d(1 - P_d)}{r_d t_d} \\ &= \left[\frac{1}{K} \sum_{k=1}^K \frac{var(\widehat{pk}_d)}{pk_d(1 - pk_d)} \right] P_d(1 - P_d) \\ &= \left[\frac{1}{K} \sum_{k=1}^K \frac{1}{nk_d} \right] P_d(1 - P_d) \\ &= \frac{1}{\tilde{n}_d} P_d(1 - P_d) \end{aligned}$$

where $nk_d = \frac{pk_d(1-pk_d)}{var(\widehat{pk}_d)}$ is the effective sample size in cell k in unit d (the sample size nk_d such

that $\frac{pk_d(1-pk_d)}{nk_d} = var(\widehat{pk}_d)$), and $\tilde{n}_d = \left(\frac{1}{K} \sum_{k=1}^K \frac{1}{nk_d} \right)^{-1}$ is the harmonic mean of the effective

sample sizes across cells within unit d . Note that $\frac{\tilde{n}_d}{t_d} = \tilde{r}_d$ is the harmonic mean of the effective sampling rate across cells within d .

An alternate approach is to assume a common design effect across units:

$$\begin{aligned} \text{var}(\hat{P}_d) &= D_d \text{var}(\hat{P}_d)^* \\ &= D_d \frac{P_d(1 - P_d)}{r_d t_d} \\ &= D \frac{P_d(1 - P_d)}{r_d t_d} \end{aligned}$$

where $D = \frac{1}{T} \sum_{j=1}^J t_j D_j$ is the average design effect across units (weighted by unit size to increase precision). We can write:

$$\begin{aligned} D &= \frac{1}{T} \sum_{j=1}^J t_j D_j \\ &= \frac{1}{T} \sum_{j=1}^J t_j \left[\frac{1}{K} \sum_{k=1}^K \frac{r_j t_j}{n k_j} \right] \\ &= \sum_{j=1}^J \frac{t_j r_j}{T \tilde{r}_j} \end{aligned}$$

So then,

$$\begin{aligned} \text{var}(\hat{P}_d) &= D_d \text{var}(\hat{P}_d)^* \\ &= D_d \frac{P_d(1 - P_d)}{r_d t_d} \\ &= D \frac{P_d(1 - P_d)}{r_d t_d} \end{aligned}$$

$$\begin{aligned}
&= \left[\sum_{j=1}^J \frac{t_j r_j}{T \tilde{r}_j} \right] \frac{P_d(1 - P_d)}{r_d t_d} \\
&= \left[\sum_{j=1}^J \frac{t_j r_j t_d}{T \tilde{r}_j t_d} \right] \frac{P_d(1 - P_d)}{r_d t_d}
\end{aligned}$$

Assume r_j is constant across units and assume the effective sampling rate in unit j is independent of the unit size t_j ; then this simplifies to:

$$\text{var}(\hat{P}_d) = \frac{P_d(1 - P_d)}{t_d \tilde{r}},$$

where:

$$\tilde{r} = \left[\sum_{j=1}^J \frac{t_j}{T} \frac{1}{\tilde{r}_j} \right]^{-1}$$

is the (weighted) harmonic mean of the effective sampling rates. We can compute $\tilde{r}$ without knowing the actual sampling rates:

$$\begin{aligned}
\tilde{r} &= \left[\sum_{j=1}^J \frac{t_j}{T} \frac{1}{\frac{1}{t_j} \left(\frac{1}{K} \sum_{k=1}^K \frac{\text{var}(\widehat{p}k_j)}{pk_d(1 - pk_j)} \right)^{-1}} \right]^{-1} \\
&= \left[\sum_{j=1}^J \frac{t_j^2}{T} \left(\frac{1}{K} \sum_{k=1}^K \frac{\text{var}(\widehat{p}k_j)}{pk_d(1 - pk_j)} \right) \right]^{-1}
\end{aligned}$$

To recap, we have two approaches to compute the sampling variance of $\hat{P}_d$:

1. For each unit, compute the harmonic mean of the effective sample size:

$$\tilde{n}_d = \left(\frac{1}{K} \sum_{k=1}^K \frac{\text{var}(\widehat{p}k_d)}{pk_d(1 - pk_d)} \right)^{-1}$$

then

$$Var(\hat{P}_d) = \frac{P_d(1 - P_d)}{\tilde{n}_d}.$$

Or:

2. Compute the weighted harmonic mean of the effective sampling rate across units (using any of these formulas, all identical):

$$\begin{aligned}\tilde{r} &= \left[\sum_{j=1}^J \frac{t_j}{T} \frac{1}{\tilde{r}_j} \right]^{-1} \\ &= \left[\sum_{d=1}^D \frac{t_d^2}{T} \left(\frac{1}{K} \sum_{k=1}^K \frac{var(\widehat{pk}_d)}{pk_d(1 - pk_d)} \right) \right]^{-1} \\ &= \left[\frac{1}{(1.645^2)TK} \sum_{d=1}^J \sum_{k=1}^K \frac{MoE(\widehat{nk}_d)^2}{pk_d(1 - pk_d)} \right]^{-1}\end{aligned}$$

then

$$Var(\hat{P}_d) = \frac{P_d(1 - P_d)}{\tilde{r}t_d}.$$

The first approach allows a different design effect in each unit, but the design effect is probably noisily estimated, so will have more noise in the estimated sampling variances. The second assumes a common design effect across units. Our decision criteria for generating sampling variances is as follows:

1. When $K = 1$ and $P_d > 0$, use the sampling variance provided by ACS, i.e., $var(\hat{p}_d) =$

$$\frac{var(\hat{n}_d)}{t_d^2}$$

2. When $K = 1$ and $P_d = 0$, use the sampling variance method 2, i.e., $Var(\hat{P}_d) = \frac{P_d(1-P_d)}{\tilde{r}t_d}$,

where $P_d = \frac{1}{t_d}$.

3. When $K > 1$ and $P_d > 0$, use the sampling variance method 2, i.e., $Var(\hat{P}_d) = \frac{P_d(1-P_d)}{\tilde{r}t_d}$

4. When $K > 1$ and $P_d = 0$, use the sampling variance method 2, i.e., $Var(\hat{P}_d) = \frac{P_d(1-P_d)}{\tilde{r}t_d}$,

where $P_d = \frac{1}{t_d}$.

4. Estimating Sampling Variance of Composite SES Measures

Let $\bar{\mathbf{X}}_d$ be the vector of 6 variables we use to construct the SES composite in unit d . Let

$\mathbf{W}_d$ be the diagonal matrix containing the standard errors of $\hat{\mathbf{X}}_d$.²⁸

Our estimated SES composite (S) in unit d is:

$$\hat{S}_d = \bar{\mathbf{X}}_d \mathbf{B},$$

where $\mathbf{B}$ is a 6×1 vector of unstandardized coefficients. The sampling variance of $\hat{S}_d$ is:

$$var(\hat{S}_d) = \mathbf{B}' \mathbf{V}_d \mathbf{B},$$

where $\mathbf{V}_d$ is the covariance matrix of $\hat{\mathbf{X}}_d$. We know the diagonal elements of $\mathbf{V}_d$ ($\mathbf{W}_d$); but not the off-diagonals. We need to know $\mathbf{V}_d$ to get the standard error of $\hat{S}_d$. How can we compute $\mathbf{V}_d$?

Define $\mathbf{R}_d$, the correlation matrix describing the correlations of the estimates $\hat{\mathbf{X}}_d$. If we knew $\mathbf{R}_d$, then we can get:

$$\mathbf{V}_d = \mathbf{W}_d \mathbf{R}_d \mathbf{W}_d.$$

The key is getting an estimate of $\mathbf{R}_d$. We can use PUMS data to estimate $\mathbf{R}$ empirically (via bootstrapped samples). We do this as follows:

- a. Set $N = 5,000$, and $J = 1,000$ (or some other values)
- b. Pick PUMA k .
- c. From all families in PUMA k , draw a random sample of N families.

²⁸ Note that we get the standard errors of these variables from ACS. The exception is $\ln(\text{median income})$, as we get a standard error for median income. Let $\hat{M}_d$ be the estimated median income in unit d . The Delta method gives us:

$$se[\ln(\hat{M}_d)] \approx \frac{1}{\hat{M}_d} se(\hat{M}_d).$$

- d. Compute $\hat{\mathbf{X}}_k$ from the micro-data (so if $\mathbf{X}$ includes $\ln(\text{median income})$, then estimate $\ln(\text{median income})$ in PUMA k from the sample, and likewise for the 6 variables we include in $\mathbf{X}$).
- e. Repeat (c) and (d) J times for PUMA k .
- f. Estimate $\hat{\mathbf{R}}_k^B$ from the J samples.
- g. Repeat (b)-(f) for all PUMAs $k = 1, \dots, K$.
- h. Repeat (b)-(g) for each race/ethnic group r to get $\hat{\mathbf{R}}_{kr}^B$. We might need to set $N = 1,000$ for race-ethnic groups, because race samples are smaller in each PUMA.

Next, we examine how $\hat{\mathbf{R}}_k$ and $\hat{\mathbf{R}}_{kr}$ vary across PUMAs and race/ethnic groups. If $\hat{\mathbf{R}}_k$ and $\hat{\mathbf{R}}_{kr}$ are relatively constant across PUMAs and subgroups, we can just use a single common value of $\hat{\mathbf{R}}$ for all units and subgroups. We find that they are generally similar, so we use a common $\hat{\mathbf{R}}$ in all PUMAs.